

Dynamic College Admissions

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Dynamic considerations are central to the design of centralized assignment mechanisms, as students learn, update beliefs, and may reapply. Through a stylized model, we show that dynamics create a tradeoff between flexibility and preference elicitation; balancing these forces can mitigate inefficiencies from initial mismatches and learning. Using administrative and survey data, we document the importance of both channels. We estimate a structural model of college progression in which students learn about preferences and ability, hold biased beliefs, and reapply. Counterfactuals show that front-loading learning and admission-rule changes reduce switching; the latter also raise welfare by eliciting preference intensity. Keywords: college admissions, dynamic matching, college retention

Choosing a college program is rarely a one-time decision. Students enter higher education with imperfect information about their academic abilities, their tastes for different fields, their admission chances, and the returns to alternative paths. As they progress through college, they can learn about their abilities and opportunities through their academic performance and experiences within their programs, updating their beliefs and expectations accordingly. These updates can lead students to persist in their original program, switch to a different one, reapply to the admissions system, or leave higher education altogether. As a result, tertiary education is inherently a dynamic decision-making process in which students continuously learn and adjust their choices over time, resulting in trajectories that often diverge from their initial assignment. In Chile, for example, roughly 30% of students switch programs and another 30% drop out, while only 16% of students complete their degree on time (OECD, 2019). These patterns—common across many other higher-education systems—suggest that learning and reassessment are central features of the college experience and potentially important determinants of students’ outcomes and welfare.

Because these decisions unfold over time, the design of admissions systems and the information available to students can have consequences that extend

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well beyond the initial assignment. Centralized admissions systems are now used in dozens of countries—including Turkey, Taiwan, Tunisia, Hungary, and Chile—around the world (Kapor, Karnani and Neilson, 2024), providing policy-makers with a powerful tool to shape how students sort across programs. Yet most discussions in market design focus on the assignment generated at the time of admission, abstracting from the adjustments that occur after enrollment. Accounting for learning and subsequent decision-making may improve outcomes by reducing inefficient mismatches, facilitating productive experimentation, and helping students discover programs that better fit their abilities and interests. Understanding these dynamic forces is therefore essential for designing admissions systems that improve welfare not only at the point of assignment, but throughout students’ educational trajectories.

In this paper, we study how admissions systems should be designed when college choice is a dynamic process rather than a one-time allocation problem, allowing students to learn about their abilities and preferences through experience, possess private information that affects future outcomes, and hold potentially biased beliefs about their prospects. We analyze how the information environment and the assignment mechanism affect both initial placements and subsequent choices, thereby influencing outcomes such as academic performance, retention, and on-time graduation. Our analysis highlights the importance of dynamic incentives and the elicitation of preference intensity for both student welfare and the efficiency of the higher-education system.

Using a stylized two-period model with learning, reapplication, and switching costs, we show how market design interventions shape application behavior and outcomes through two key channels. The first—the initial mismatch channel—arises when students enroll in a less-preferred program and switch later. The second—the learning channel—emerges as students acquire information about their preferences and abilities after enrollment and seek to revise their assignment to avoid *ex post* mismatches. Once these dynamic incentives are taken into account, the assignment mechanism itself influences application behavior and must balance a fundamental design trade-off between flexibility and elicitation of preference intensity. We show that mechanisms that allow unconstrained reapplication and switching (i.e., unconstrained DA) preserve flexibility while partially eliciting preference intensity, as students with sufficiently strong preferences and high switching costs may strategically truncate their first-period lists. These mechanisms can improve welfare by enabling students to correct initial mismatches and unfavorable learning realizations, thereby reducing inefficient *ex post* mismatches, particularly when congestion and switching costs are low. By contrast, mechanisms that impose static or dynamic barriers to reapplication and switching (i.e., constrained DA or no-switch rules) sacrifice flexibility but more effectively elicit preference intensity by forcing students to internalize the opportunity cost of targeting a risky program or getting locked into fallback options. In addition, these mechanisms mitigate the crowd-out externality generated by

reapplications: when a student occupies a seat in an over-demanded program and later vacates it, the seat may remain unfilled while displacing another student to a worse assignment. Limiting switching can therefore improve welfare when learning is limited, preferences are relatively stable, and congestion and switching costs are high. Which force dominates depends on the relative importance of learning and these externalities, and is therefore ultimately an empirical question.

To answer it, we combine administrative data from the Chilean college admissions system with multiple nationwide surveys that we designed and conducted, allowing us to quantify the prevalence of both channels and assess their implications for student and institutional outcomes. We show that students assigned to lower-ranked reported preferences are substantially more likely to reapply and switch, consistent with the initial mismatch channel. We also document changes in choices that are consistent with students learning about both their preferences and abilities during college, affecting switching and dropout decisions. Finally, we show that these adjustments have important consequences for colleges. About a fifth of first-period enrollments are vacated within the first year—mostly at programs with excess demand—and, because vacated seats are neither reallocated nor added to the following year’s offer, the resulting loss of capacity generates substantial foregone tuition revenue for schools.

While these findings establish that both the mismatch and learning channels are empirically important, assessing their welfare consequences and the effects of alternative admissions policies requires a model of students’ dynamic decisions. We therefore introduce a structural model that captures the application behavior of students, as well as their decisions to enroll, retake the admission tests, reapply, switch, and drop out, allowing students to learn about their unknown abilities and their preferences during their academic progression. In particular, we assume that students base their application and enrollment decisions on both (i) the value of studying each program, (ii) the continuation value of retaking the admission tests and reapplying to the system, and (iii) their labor market prospects. Students have heterogeneous preferences—which evolve to capture persistent idiosyncratic shocks that occur each period—and potentially biased beliefs over their future college grades and labor market prospects—which they update as they observe noisy signals of their unknown ability from their grades. Based on their updated continuation values, students decide whether to continue in their current program, reapply to the system, or drop out and choose their outside option. Finally, students face graduation probabilities, and then enter the labor force and receive pecuniary and non-pecuniary values from the labor market.

The main challenge in estimating our model is separately identifying the learning and mismatch channels. To identify the former, we use the longitudinal structure of the data, focusing on how students update their beliefs and change their preferences over time. We isolate learning about preferences by examining the composition of initial applications and reapplications following non-informative grade signals, where any preference change likely reflects evolving tastes rather

than updated beliefs about ability. In turn, we identify learning about unknown abilities by using elicited beliefs about academic performance (e.g., expected grades) and their correlation with (i) changes in the composition of preferences and (ii) decisions (e.g., switching or dropping out) after receiving first-year grades. The correlation between these belief updates and switching behavior—and its heterogeneity across majors and subjects—helps us distinguish between belief-based learning and preference shifts. To identify the initial mismatch channel, we exploit quasi-experimental variation around admission cutoffs, which allows us to compare students who are similar in their initial preferences and abilities, both observed and unobserved, but are initially assigned to different programs. For instance, we show a strong causal effect of not being assigned to the top-reported preference on the probabilities of reapplying and switching (59% and 66% increase, respectively), consistent with the mismatch channel. We formalize the core of this strategy in an extension of the stylized model (Appendix I), where we show that the *direction* of switches around admission cutoffs reveals relevant information, as students who switch *down* from their top choice to their fallback reveal an unfavorable match-quality signal, whereas students who switch *up* reveal a strong preference for their top choice. How these switching rates respond to match-quality signals identifies the primitives behind the two channels: the switching cost, which governs how easily initial mismatches are corrected, and the relative scale of learning. Distance, a persistent preference shifter, then delivers their levels and the distribution of preference intensity.

After estimating the structural model, we conduct a series of counterfactual analyses. First, we show that learning accounts for about one-third of all switching decisions, while initial mismatches and congestion explain the remainder and part of dropout decisions. Next, we examine how changes in (i) the information environment and (ii) the assignment process affect students' outcomes. On the information side, we find that enabling students to front-load their learning process (through a mandatory one-year post-secondary program) can help reduce switching rates. On the mechanism side, we find that penalizing students who switch (as in Turkey); giving a score bonus for all first-year applicants (as in Finland); or allowing students to signal one of their preferences to get a score bonus (as in the economics job market) can significantly reduce switching rates and increase students' welfare. Finally, we find that these effects are robust to changes in the share of strategic participants, unlike policies such as restricting application list lengths, which are widespread but far less effective and sometimes detrimental for students' outcomes.

Our counterfactual experiments highlight the importance of carefully balancing the two key behavioral forces at play: the benefits of learning through experimentation and the crowd-out externality created by initial mismatches. We show that *facilitating learning*, incorporating dynamic incentives, and eliciting students' cardinal preferences—through modifications to reapplication rules and the assignment mechanism—can meaningfully influence both educational out-

comes and overall welfare. More broadly, these findings underscore the relevance of accounting for dynamic behavior and externalities in the design of matching markets, with potential implications for improving efficiency and equity in contexts such as school choice, organ allocation, and entry-level labor markets.

The remainder of the paper is organized as follows. Section I discusses the most closely related literature. Section II presents a stylized model that illustrates the main mechanisms at play. Section III describes the Chilean college admissions system and provides empirical evidence for the two behavioral channels. Section IV presents our model. Section V describes our identification strategy. Section VI describes the estimation approach and its results. Section VII reports our counterfactual results, and Section VIII concludes.

I. Literature

Our paper relates to the literature on the empirical analysis of (i) centralized assignment mechanisms and (ii) education choices under uncertainty.

The first strand focuses on understanding the incentives that centralized assignment mechanisms introduce; how to use the data generated by these mechanisms to identify and estimate students' preferences/beliefs (Agarwal and Somaini, 2019); and measuring the welfare effects of changing assignment mechanisms in different settings (Agarwal and Somaini, 2018; Calsamiglia, Fu and Güell, 2020; He, 2012; Kapor, Neilson and Zimmerman, 2020; Luflade, 2017; Hernandez-Chanto, 2021). Despite progress in this area, the aforementioned studies either consider static settings, assume fixed preferences over time, or simply ignore the effects of assignment mechanisms on downstream outcomes.¹ Ignoring the system's dynamics, however, can lead to biased estimates of the welfare effects of changing the mechanism, as continuation values are themselves affected by changes in the mechanism. Moreover, static approaches cannot evaluate intrinsically dynamic policies that could enhance welfare, such as modifying reapplication rules, as is the case in Finland and Turkey. In response, a recent line of research has begun to examine dynamic settings in which agents interact repeatedly with the assignment process, including the allocation of public housing (Waldinger, 2021), hunting licenses (Reeling and Verdier, 2021), general practitioners (Huitfeldt, Marone and Waldinger, 2025), teachers' labor markets (Ederer, 2023), and kidneys from deceased donors (Agarwal, Hodgson and Somaini, 2024; Agarwal, Somaini and Waldinger, 2024). In educational markets, dynamic models have been recently used to evaluate the impact of middle school placements on high school outcomes (Hahm and Park, 2022), daycare assignments (Kuno, 2023; De Groote and Rho, 2026), and sequential college admissions with off-platform options (De Groote et al., 2025). Among these, the closest to ours is Narita (2018), who analyzes theoretically and empirically the welfare

¹With some exceptions in school reforms (Moriguchi, Narita and Tanaka, 2024), college admissions (Otero, Barahona and Dobbin, 2021), kidney allocation (Agarwal, Hodgson and Somaini, 2024), and the medical match (Friedrich et al., 2024).

effects of centralized school-choice mechanisms with rematching when demand evolves over time. We contribute to this literature by developing a dynamic framework for centralized college admissions that captures how different forms of learning, strategic behavior, and repeated interactions with the mechanism shape system yield, student outcomes, and welfare.

The second strand studies education and occupation choices, focusing on how students gather information and learn about their preferences, abilities and future earnings (Arcidiacono, 2005; Malamud, 2011; Stinebrickner and Stinebrickner, 2012; Stange, 2012; Stinebrickner and Stinebrickner, 2014; Arcidiacono et al., 2025); and how their choices impact outcomes and returns (Altonji, Blom and Meghir, 2012; Altonji, Arcidiacono and Maurel, 2016). Within this strand, the closest paper to ours is Bordon and Fu (2015), who analyze the effects of switching from a system where students choose both college and major simultaneously to one where they choose college first and then major. Our paper complements their work by studying the effects of changing the assignment mechanism and reapplication rules in a dynamic context, allowing for heterogeneous and potentially biased subjective beliefs rather than imposing rational expectations, identifying their effect on students' learning about their abilities and their preferences, and measuring its effect on student outcomes. In the same vein of studying how admission and screening rules affect enrollment success, Declercq and Verboven (2018) estimate a dynamic model of enrollment, reorientation, and dropout in an open-enrollment system with ex post selection (e.g., based on academic performance during higher education), and show that moderate ex ante screening (e.g., through entry exams or admission standards based on high school performance) can reduce unsuccessful enrollment without lowering degree completion.

Finally, within these two strands of the literature, we also contribute to recent papers exploiting survey data for identification of structural models in college admissions (Wiswall and Zafar, 2015a, 2021; Tincani, Kosse and Miglino, 2023) and school-choice (Kapor, Neilson and Zimmerman, 2020). We add to this literature by leveraging students' subjective and potentially biased beliefs over their college outcomes and exploiting this information to identify their learning process.

To the best of our knowledge, this is the first paper to structurally evaluate how the information environment and market design choices, including reapplication rules, affect students' welfare and outcomes—such as retention and on-time graduation—beyond the initial match. We also contribute by revisiting the trade-off between eliciting preference intensity and ensuring *strategy-proofness* in a dynamic framework that incorporates private information, learning about abilities and preferences, and heterogeneous, subjective, and potentially biased beliefs about ability and future outcomes. Accounting for these elements is essential to understand how changes in the assignment mechanism influence students' indirect utilities and welfare, and to evaluate intrinsically dynamic policies designed to improve both student outcomes and the system's yield.

II. A Simple Model

To illustrate how market-design policies shape reapplication incentives and downstream outcomes, we extend the example in [Azevedo and Leshno \(2016\)](#) to a two-period environment with learning, reapplications, and switching costs.

Consider two programs $j \in \{1, 2\}$ with first- and second-period capacities $\mathbf{q} = (q_1, q_2)$ and $\phi \cdot \mathbf{q} = (\phi q_1, \phi q_2)$, where $\phi \in (0, 1)$ is an exogenous parameter. Without loss of generality, we assume that $0 < q_1 < q_2 < 1$. In addition, consider a unit mass of students with half of them preferring program 1 over 2 and half the reverse. Each student has a vector of application scores $\mathbf{s} = (s_1^1, s_2^1, s_1^2, s_2^2)$, where s_j^t is i.i.d. and uniformly distributed in $[0, 1]$. To capture students' intensity of preferences, we assume that the ex ante utility of a student in their top choice is $1 + \rho$, where ρ is privately known and uniformly distributed on $[0, 1]$, while the second choice yields normalized utility 1. Students assigned to their top choice then learn, after one period, a match-quality shock ξ that is uniformly distributed on $[-1, 1]$ and independent of ρ . We denote a student's type by $\theta := (\succ_\theta, \mathbf{s}_\theta, \rho_\theta, \xi_\theta)$, where $\succ_\theta$ is the student's strict preference ranking over the two programs, $\mathbf{s}_\theta$ is the vector of scores, ρ_θ is the student's preference intensity, and ξ_θ is the student's match-quality shock. With a slight abuse of notation, let h_θ and l_θ denote the student's top and second choice, respectively; when clear from the context, we suppress the type subscript θ and use $h, l, \mathbf{s}, \rho$ and ξ .

As discussed in [Azevedo and Leshno \(2016\)](#), the allocation in each period t can be fully described by the vector of cutoffs $(\bar{s}_1^t, \bar{s}_2^t)$, where $\bar{s}_j^t$ captures the minimum score required to be admitted to program j in period t . These cutoffs are determined by market-clearing conditions that equate supply (i.e., the number of available seats in each period) and demand, which in turn depends on students' preferences, their admission probabilities—defined as $p_j^t = \mathbb{P}(s_j^t > \bar{s}_j^t)$ —and, importantly, the application behavior induced by the mechanism used to perform the allocation. We illustrate the implications of the latter by comparing three different mechanisms: (i) DA - unconstrained (DA-U), where students can apply to both programs in each period; (ii) DA-constrained (DA-C), where students can apply to at most one program in each period; and (iii) DA - No Switches (DA-NS), where students who enroll in some program in period 1 cannot switch in period 2. Across all these mechanisms, we assume that students who desire to remain enrolled do not need to re-apply and those who aim to re-apply but do not succeed can keep their current seat. Furthermore, we assume that students who successfully switch incur a switching cost $\tau \geq 0$ that is constant and homogeneous across students.²

Application and re-application behavior. We start characterizing the application

²The switching cost is incurred exclusively by students who switch and is therefore not borne by initially unassigned students who obtain a seat in period 2. Accordingly, τ captures the opportunity cost of switching, including delays in graduation and labor-market entry associated with restarting a degree, as is common in many college-admissions environments.

behavior under each mechanism.

DA - Unconstrained. Lemma 1 characterizes the (re-)application behavior under DA-U. In period 1, since students may apply to both programs in each period, they may strategically truncate their list and skip their second choice if the expected continuation value of remaining unassigned and re-applying is weakly higher than attending their second choice in period 1. At the end of period 1, a student re-applies if one of the following conditions holds: (i) they were initially assigned to their second choice and $\rho > \tau$; (ii) they were initially assigned to their top choice and learn a match-quality shock such that $\rho + \tau < \xi$; or (iii) they were initially unassigned. As students can keep their assignment if their re-application is unsuccessful, initially assigned students who wish to switch apply only to the alternative program, whereas initially unassigned students apply truthfully.

DA - Constrained. Lemma 2 formalizes the (re-)application behavior in this case. As students can apply to at most one program in each period, they must strategically decide which program to list factoring in their preferences, their admission probabilities, and the potential net gains from re-applying. Consequently, the re-application behavior for initially assigned students is the same as in DA-U. However, for initially unassigned students, they must decide whether to apply to $\{h\}$ or $\{l\}$ taking into account their admission probabilities.

In period 1, a student strictly prefers applying to their top choice if and only if the expected continuation value of applying to $\{h\}$ is weakly higher than that of applying to $\{l\}$, taking into account the potential switching decision that may take place in period 2. Since $q_1 < q_2$, we have $p_1^1 < p_2^1$, so students with $h = 2$ will always apply to their top choice. For students with $h = 1$, the application decision depends on the relative magnitudes of the admission probabilities and the expected continuation values, which in turn depend on the switching cost τ ; thus, only those students with sufficiently high preference intensity ρ will apply to their top choice in period 1, while the rest will apply to their second choice.

DA - No Switches. Lemma 3 characterizes the application behavior in this case. When switches are forbidden, re-application behavior is straightforward: initially assigned students do not re-apply, while initially unassigned students re-apply reporting their true preferences. However, the application decision in period 1 is more subtle. Since students cannot switch in period 2, they must factor in the potential loss of utility from being assigned to their second choice in period 1 and being unable to switch to their top choice in period 2. As a result, students may strategically truncate their list and only apply to their top choice in period 1, especially when the intensity of their preferences is sufficiently high.

Comparison of mechanisms. The three mechanisms differ in the dynamic considerations they introduce and, ultimately, in how they balance the trade off between ex post flexibility and the extent to which they elicit preference intensity. Under DA-U, students can re-apply after the first period to correct two sources of ex post mismatch. First, students initially assigned to their second choice may switch to their original top choice, mitigating the *initial mismatch*

channel.³ Second, students initially assigned to their top choice may learn that their match quality is low and switch to their second choice, mitigating the *learning* channel. These reallocations generate gross gains for switchers, but they also introduce inefficiencies to the system. Seats vacated by successful switchers are not reallocated, even though they could have been filled by students with stronger preferences for those programs.⁴ Furthermore, facilitating re-applications creates congestion in the second-period market: switchers compete for the same seats as initially unassigned students, reducing the latter group’s probability of admission.

Importantly, the option to reapply also introduces strategic considerations. In the first period, students with strong preferences for their top choice may strategically truncate their list if preserving the option to reapply yields higher expected value than attending their fallback, especially when switching costs are high. Preference intensity is elicited again at the reapplication stage, as students initially assigned to their second choice reapply only when the expected value of doing so—the utility gain net of the switching cost, weighted by the probability of admission—is positive, which depends directly on how intensely students prefer their top choice. Thus, DA-U can endogenously elicit preference intensity through dynamic considerations—both in the first-period report and in the reapplication decision—departing from standard strategy-proofness in static settings.

DA-C and DA-NS restrict flexibility by introducing static and dynamic considerations, respectively, enabling these mechanisms to elicit more information about preference intensity. Under DA-C, students who face a binding application constraint internalize the opportunity cost of targeting a risky top-choice program, so only those with sufficiently high preference intensity apply to it in period 1. Under DA-NS, students who would otherwise be locked into a fallback seat may truncate their period-1 list, and this truncation is again more attractive for students with stronger preferences for their top choice. In both cases, the mechanism changes the selection of students who obtain first-period seats: high-intensity students are more likely to reach their top choice, and those assigned to their top choice require a more extreme negative match-quality shock to switch later. Consequently, these two mechanisms may decrease inefficient switching in period 2, which is especially important when the reapplication market is thin, i.e., when seats are scarce and the gross gains from ex post reallocation are small.

These forces also shape the welfare ranking across mechanisms, which is ambiguous. As formalized in Propositions 4 and 5 and illustrated in Figures A1 and A2, selection in the first period can dominate the value of flexibility when the market is highly congested, making DA-C or DA-NS preferable to DA-U. By contrast, when congestion is low, the value of correcting initial mismatches and bad learning shocks can be large enough for DA-U to dominate, especially

³Any system with excess demand will inevitably have some students assigned to lower preferences. We refer to this as an initial mismatch, as students may later seek to improve their assignment according to their initial preferences, and these adjustment can generate aggregate inefficiencies.

⁴This assumption mirrors the Chilean system, where seats vacated through switching or dropout are neither refilled during the year nor added to the following year’s admission offer (see Section III.B).

when the mechanism’s dynamic incentives also induce students to avoid low-value fallback assignments. Ultimately, the relative importance of these channels and, thus, the preferable policy lever, is an empirical question that depends on the interplay between persistent preferences, initial assignments, learning, and congestion in the centralized system. The empirical evidence, structural model, and counterfactual analysis that follow are designed to shed light on these trade-offs.

These dynamic choices also make the model’s primitives identifiable, as we formalize in Appendix I. The intuition is that the direction of switching around a cutoff is informative, as students assigned to their top choice who switch *down* to their fallback reveal that they received a sufficiently bad match-quality signal, while students assigned to their fallback who switch *up* reveal a sufficiently strong preference for their top choice. How each of these switching rates responds to the match-quality signal identifies the switching cost and the relative scale of learning. To recover their levels, we exploit a persistent preference shifter—such as distance to each program—that moves the payoff of an option without reshaping the underlying distribution of preference intensities. Section III.B documents these patterns in the data: discontinuities in reapplication and switching around admission cutoffs, and the correlation between students’ dynamic choices—including the direction of switching—and the signals from their first-year grades. These are the sources of variation underlying the moments we match in estimation.

III. College Admissions in Chile

The college admissions process in Chile is semicentralized, with the most selective universities having a centralized system and the remaining institutions conducting their admissions processes independently. This paper’s empirical application follows the cohort of 2014 and focuses on the centralized part of the system, known as *Sistema Único de Admisión* (SUA). This part of the system is organized by the *Consejo de Rectores de las Universidades Chilenas* (CRUCH), and its admissions process is operated by the *Departamento de Evaluación, Medición y Registro Educacional* (DEMRE).

To apply to any of the close to 1,500 academic programs offered by the 41 universities that are part of the centralized system, students must undergo a series of standardized tests (*Prueba de Selección Universitaria* or PSU). These tests include Math, Language, and a choice between Science or History, and provide a score for each. Students’ performance during high school yields two additional scores: one obtained from the average grade during high school (NEM) and a second that depends on the relative position of the student among their cohort (Rank). Admission to any program is solely based on these *admission factors*.

After scores are published, students can submit—at no cost—a list with no more than ten programs, ranked in strict order of preference. We refer to these lists as Rank Order Lists (ROLs). Notice that students directly apply to a program, i.e., they must list university-major pairs on their ROL. In the remainder of the paper, we refer to these pairs as programs.

On the other side of the market, each program announces its vacancies, the weights on each admission factor, and the set of additional requirements they will consider for applications to be valid. For instance, universities may require a minimum application score or a minimum score in some PSU tests, among other less common requirements. Each program’s preference list is defined by first filtering all applicants who do not meet these requirements. Students are then ordered based on their application scores, which are computed as the weighted sum of the applicants’ scores and the predefined weights.

Considering the vacancies and the preferences of the applicants and programs, DEMRE runs an algorithm to match students to programs. The mechanism used is a variant of the student-proposing Deferred Acceptance algorithm, in which all students tied for the last seat in a program are admitted (Rios et al., 2021). A description of the mechanism can be found in Appendix B.B1. As a result of the assignment process, each program is associated with a cutoff, such that all students whose weighted score is above it are granted admission, and all students with scores below the cutoff are waitlisted and thus may have to enroll in a lower-ranked preference. This property is known as the *cutoff structure*.

Enrollment begins with the release of assignment results and unfolds in two rounds. In the first, only assigned students may enroll in their designated program. In the second, programs with remaining seats invite waitlisted students to enroll. Throughout the process, applicants may also choose to enroll in non-centralized programs or enter the workforce.

Students may retake the admissions test and reapply to the centralized system in subsequent years. However, retaking the test is not required for those who last took it in the previous year. For students who take the test in two consecutive years, the system allows them to choose the better of the two score pools—i.e., they cannot combine individual test scores across years—for each program to which they apply (Rios and Sharma, 2026). Although some students switch programs without formally reapplying, the vast majority of switches—approximately 70%—occur through reapplication. Therefore, in the remainder of the paper, we focus on switches that follow a reapplication and abstract from internal transfers, capturing exactly how the centralized system works. Finally, the centralized system allocates only the first-year vacancies that programs announce each year, ranking all applicants—first-time and reapplicants alike—solely by their admission scores: college academic performance plays no role in the allocation, and there are no reserved seats for students switching programs (Rios et al., 2021). Consequently, seats that students vacate when they switch are not reallocated to other students nor included in the offer of the next year, so there may be programs that are under-utilized ex post despite being oversubscribed ex ante. These dynamic features are not specific to Chile: exam re-taking and re-applying in later years is common across centralized systems, such as Turkey (Krishna, Lychagin and Frisanchi, 2018), Finland, and Brazil (Otero, Barahona and Dobbin, 2021).

A. Data

Our dataset includes (i) administrative data provided by DEMRE and the Ministry of Education (MINEDUC), (ii) several surveys designed and conducted in collaboration with CRUCH and DEMRE, and (iii) grade records facilitated by CRUCH. Specifically, we consider the following sources of data:

- **Admissions process:** This includes students’ socioeconomic characteristics—including self-reported family income, parents’ education, and the municipality in which the student lives, among others—; scores; applications; final assignment; and enrollment decisions, spanning from 2007 to 2023. In addition, this includes data on programs characteristics, including their vacancies, weights, tuition, duration, major, location, etc.
- **Labor market:** This includes aggregate information about the labor market outcomes in each program—including average wages four years after graduation and overall employment probabilities one year after graduation—spanning from 2014 to 2018. We also have data at the major level, including average wage for the first five years after graduation; employment probabilities for the first and second year after graduation; and the evolution of average wages from the first ten years after graduation.
- **Grades:** This includes the cumulative GPA for the first three years of college for every student who enrolls in a program that is part of the centralized system in 2014 and 2015.
- **Surveys post-application:** This includes responses to surveys that we designed and conducted in 2019 and 2020 to gather information on students’ preferences—specifically, about their top-true preference—and their beliefs about their admissions probabilities for each program on their ROL and the elicited top-true preference. As many students reapply to the centralized system after a year, these two surveys provide a panel for reapplicants’ preferences and their beliefs (see Appendix B.B2 for more details).
- **Surveys pre and post-application:** To complement the previous surveys, in 2023, we designed and conducted similar surveys before and after the application time window. In the latter, we ask students about their beliefs (about admission chances, grades, wages, among others) for programs both included in their ROL and also outside it, allowing us to capture beliefs on choices and also on counterfactual programs.

To our knowledge, this is the first paper to use the grades data described above and to incorporate nationwide-survey data on students’ true preferences, beliefs about admission probabilities, college progression, and future labor-market outcomes.

Throughout the paper, we focus on a subset of the population to reduce computational complexity. Specifically, we focus on students who graduated from a

high school within the Region Metropolitana (RM) in 2013, participated in the 2014 admissions process, had an average score between Math and Language above 475, and applied to programs located in RM. The latter allows us to reduce the number of programs to less than half (439) without major loss, as close to 80% of applications from students living in RM include only programs located in that region. Finally, we exclude students with average score below 475 (less than 13% of students who can apply) because they do not satisfy loan eligibility requirements. Contrary to previous evidence (Lochner and Monge-Naranjo, 2012, 2016; Solis, 2017), Card and Solis (2020) show that enrollment decisions and college completion rates among students whose average score exceeds 475 (thus, eligible for scholarships) are not highly sensitive to differences in prices. Hence, we can ignore potential effects of credit constraints on students' dynamic choices.

B. Empirical Facts

We now show the prevalence of the two behavioral channels described in Section II: (i) *initial-mismatching*, whereby students assigned to less preferred programs reapply to improve their allocation, and (ii) *learning*, whereby students learn about their match-quality and preferences over time and potentially decide to move to other programs.

Before quantifying the importance of these two channels, we first document the prevalence and economic relevance of the dynamic decisions they generate, namely switching and dropout. In Table 1, we report the average switching and dropout rates in the first four years, separating by income level—high or low—and gender.⁵ First, we observe that close to 21.2% of students switch from their first program of enrollment, and 15.3% drop out within the first four years. Second, comparing outcomes by gender (within an income level), we observe that women are more persistent in their academic progression, since their switching and dropout rates are lower than those for men. Third, comparing these rates by income level (within gender), we observe that low-income students are less likely to switch programs during their academic progression. However, we also observe that low-income students are significantly more likely to drop out. These patterns—suggesting large differences in switching and dropout rates by gender and income—are similar to those observed for first-year outcomes. Therefore, for simplicity, we restrict our analysis to switching and dropout within the first year. This restriction captures the largest share of switching—close to 80% of switching takes place in the first two years, and roughly 70% of those switches occur in the first year—suggesting that focusing on first-year transitions entails little loss of empirical relevance (see Figure B4 in Appendix B.B4).

Furthermore, these first-year transitions have a significant effect on colleges

⁵We refer to *majors* as the fields of education provided by the International Standard Classification of Education (ISCED) (UNESCO (2012)) and adapted for Chile, which classifies programs into: Farming, Art and Architecture, Science, Social Sciences, Law, Humanities, Education, Technology, Health, Management, and Commerce.

yield and revenue. In our sample, 18.3% of first-period enrollments are vacated within the first year (12.7% resulting from switching and 5.6% from dropout), and over 80% of the vacated seats belonged to programs with an active waiting list and could have been filled by waitlisted applicants alone. However, because these seats are neither reallocated nor incorporated into the following year’s offer, they remain permanently unfilled, resulting in forgone tuition revenue that amounts to 49-69 billion Chilean pesos (2014), equivalent to 70-99% of one full year of tuition for the entire cohort in our sample.⁶

Table 1—Switching and Dropout by Gender and Income

		Switches				Dropout
	Income	Program	University	Major	Math type	
Men	Low	0.213 (0.007)	0.108 (0.005)	0.099 (0.005)	0.038 (0.003)	0.232 (0.007)
	High	0.24 (0.006)	0.128 (0.004)	0.132 (0.005)	0.044 (0.003)	0.134 (0.005)
Women	Low	0.173 (0.006)	0.089 (0.005)	0.095 (0.005)	0.042 (0.003)	0.186 (0.007)
	High	0.209 (0.006)	0.102 (0.004)	0.125 (0.005)	0.063 (0.003)	0.096 (0.004)
Overall		0.212 (0.003)	0.109 (0.002)	0.116 (0.002)	0.048 (0.002)	0.153 (0.003)

Note: Standard errors reported in parenthesis. Program denotes any change in the university-major pair; University, Major, and Math-type switches are not mutually exclusive. Sample: students from our analysis sample (Section III.A) who enrolled in a centralized-system program in 2014 (N=17,391); outcomes are measured over their first four years.

MISMATCHING.

A key implication of the initial mismatch channel is that students assigned to lower-ranked choices should be more likely to seek a different program after enrollment. The rank of the assigned program in a student’s submitted preference list therefore provides a natural proxy for mismatch. Figure 1 explores this prediction by plotting first-year switching and dropout rates by assignment rank. We observe that students initially assigned to lower ranked preferences are substantially more likely to switch programs than those assigned to their top choice. For instance, 9.86% of students assigned to their first choice switch at the end of their

⁶For each vacated seat, we multiply the program’s annual tuition (*arancel*) by its remaining duration and sum across seats. The lower bound values each seat over the program’s formal duration (minus the first year already paid); the upper bound instead uses the expected duration, scaling the formal duration by the enrollment-weighted ratio of actual to formal duration among university programs (1.33, from SIES statistics). Tuition is held at its 2014 value without discounting, as posted tuition is fixed in nominal terms; discounting future years at 3% lowers the bounds to 45-62 billion. Seats at programs with an active waiting list account for 84% of the total.

first year, compared with nearly 15% of students assigned to their fourth choice. This pattern provides suggestive evidence that initial mismatch is an important determinant of post-enrollment adjustment decisions.

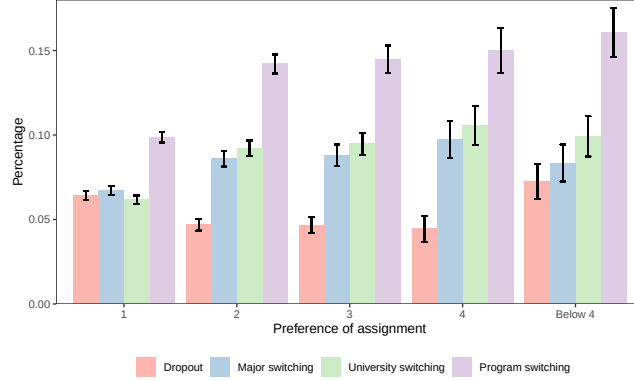


Figure 1. Switchings and dropout

Note: Sample: as in Table 1, restricted to students with a recorded preference of assignment (N=16,790). Error bars denote ± 1 standard error.

To make a causal claim, we use a regression discontinuity design that exploits the algorithm’s cutoff structure to perform the allocation. Because application scores are fixed before students submit their lists and cutoffs are realized only after the assignment is run, students cannot manipulate their position relative to a cutoff. Applicants just above and just below a cutoff are therefore comparable in both observed and unobserved characteristics and differ only in their initial assignment, which allows us to estimate the causal effect of interest. In Figure 2, we display binned means of different outcomes as a function of the distance between cutoffs and students’ scores in their most preferred listed program. Figure 2a shows that students right below the cutoff are 8.4pp more likely to reapply the following year, which corresponds to a relative change of close to 59% compared to those right above the cutoff. Figure 2b shows that students below the cutoff are 7.3pp more likely to switch programs within the centralized system, which corresponds to a relative change of about 66%. Similar results hold when we consider students’ top true preferences instead of their top-reported preferences, as shown in Appendix B.B3. These results confirm our previous findings, i.e., that students assigned to lower preferences are more likely to reapply and switch programs the following year. In Appendix B.B3, we provide more details and additional evidence of the effect of crossing the cutoff of the top-reported preference on students’ decisions. Importantly, we find no causal effect on enrollment, and, contrary to what Figure 1 suggests, we find no causal effect of cutoff crossing on dropout.

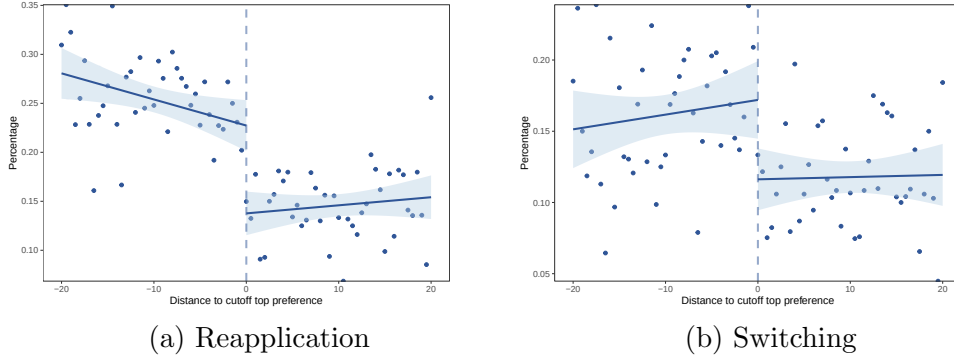


Figure 2. Effect of Cutoff Crossing

Note: Binned means as a function of the distance between the student’s application score and the cutoff of their top-reported program; lines are local-linear fits within ± 20 points. Sample: applicants from our analysis sample who listed at least two programs and are within 20 points of the cutoff ($N=7,797$; switching: $N=6,003$ among those enrolled). See Appendix B.B3 for additional outcomes and diagnostics.

The evidence above establishes that assignment rank has a causal effect on students’ persistence in their initial program. However, this relationship alone does not isolate the mismatch mechanism, since subsequent switching may also reflect learning about abilities, preferences, or other post-enrollment shocks.⁷ To distinguish between these explanations, we use the 2020 survey on students’ preferences and beliefs to examine whether students already anticipate different levels of persistence across assignments before enrollment. We find that a substantial fraction of students report being less likely to remain in a program when it corresponds to a lower-ranked reported preference (see Appendix B.B4 for details).⁸ Because these beliefs are elicited before students begin college, they cannot be driven by subsequent learning or realized academic outcomes. Instead, they indicate that students perceive lower-ranked assignments as less desirable matches and often enroll with the expectation that they may later seek an alternative program. We interpret this as direct evidence of the initial mismatch mechanism.

LEARNING.

We now turn to the learning channel, which arises when students update their

⁷The stylized model in Section II makes this explicit: the switching rates around admission cutoffs depend jointly on switching costs, the persistence of students’ initial preferences—which governs initial mismatch—and post-enrollment learning. Appendix I shows how the direction of switching and its response to first-year signals disentangles these forces.

⁸We test whether there are significant differences in observable characteristics (e.g., gender, income, and average math-verbal score) between students who respond to the survey and those who do not. Differences are modest—respondents are slightly more likely to be female and have somewhat higher scores—and the moments we construct from the surveys condition on these observables (see Appendix B.B2).

preferences and beliefs about their abilities after enrollment. As a result, students may revise their educational choices in response to information that was not available at the time of application. To study these adjustments, we analyze switching decisions and classify them into three categories: (i) *Up*, (ii) *Down*, and (iii) *Out*. Students move *Up* (*Down*) if they switch to a program listed above (below) their initial enrollment on their initial ROL. Students move *Out* if they switch to a program not listed on their initial ROL.

We find that, among students who switch in their first year, 17.3% move *Down*, 14.2% move *Up*, and 67.1% move *Out*. The remaining 1.4% of switchers were initially enrolled in programs outside their original ROL (through special admission channels) and subsequently switched. Moreover, about half of the *Out* switches involve more selective programs, i.e., programs with higher cutoffs compared to their initial enrollment. These results suggest that both channels explain students' switching significantly. Students who move *Down* or *Out* to less selective programs are likely to have received negative signals about their ability or revised their preferences over programs (learning channel), and students who move *Up* or *Out* to more selective programs may be trying to find a better match given their initial preferences.

Table 2—Effect of Signal on Reapplication and Switching

	Reapplied		Changed comp.		Switch		Switch Down		Switch Up	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Signal	−0.980*** (0.033)	−0.092 (0.061)	−0.850*** (0.035)	0.077 (0.073)	−0.790*** (0.031)	0.226*** (0.057)	−0.883*** (0.054)	−0.705*** (0.144)	0.753*** (0.088)	0.997*** (0.107)
Mean	0.119	0.085	0.08	0.057	0.131	0.099	0.022	0.015	0.024	0.026
Test	0.11***	0.086***	0.074***	0.074***	0.124***	0.093***	0.019***	0.017***	0.019***	0.018***
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Passed ($GPA \geq 4.0$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Observations	14,395	12,591	14,395	12,591	14,395	12,591	14,395	12,591	14,395	12,591

Note: Sample includes all students from the cohort that graduated high school in 2014, enrolled in 2014 and 2015 in a program offered by a university within the system, and for whom first-year grade data is available. GPA is measured on a 1–7 scale, with failing grades defined as below 4.0. Test reports the results of t-test assuming no signal (all p-values are below 1×10^{-4}), i.e., testing whether the intercept is different than zero. Significance levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

In Table 2, we analyze the effect of the signal received at the end of the first year—i.e., the difference between the obtained GPA and the expected one—on reapplication and switching decisions. In all of these models, we control for demographics (gender, income); scores (NEM and the average between Language and Math); and the preference of assignment in the initial year. Odd columns include the entire sample, while even columns focus on students with a GPA greater than or equal to 4.0, ruling out forced switches.⁹ We observe that the magnitude of

⁹In Chile, the pass/fail threshold is 4.0 on a scale from 1.0 to 7.0. Since our measure of GPA is

the signal is negatively correlated with students’ likelihood of reapplying, modifying the composition of their rank-ordered list, and switching programs in the full sample. Moreover, the signal is negatively correlated with the probability of switching to less preferred programs (“switching down”) and positively correlated with switching to more preferred programs (“switching up”). These patterns suggest that students who receive more negative signals—indicating they performed worse than expected and thus revise their beliefs about their unobserved ability downward—are less inclined to pursue initially more preferred programs and more inclined to downgrade their choices, consistent with learning about ability.

The evidence above shows that students update their decisions in response to information about their academic ability. However, students may also learn about their preferences through their college experience. To examine this possibility, we focus on students who receive a null signal, i.e., whose grades are as expected, limiting the scope for updating beliefs about ability. Table 2 reports the results of t-tests on the predicted probabilities of reapplying, modifying the rank-ordered list (ROL), and switching behaviors for these students. We find that even in the absence of a signal, students remain significantly likely to reapply, modify the composition of their ROL, and switch programs. This pattern is further supported by the summary statistics in Table 3, which show that students who receive a neutral signal still engage in these behaviors at meaningful rates, generally falling between those observed for students receiving positive or negative signals. For instance, 8.6% of students with a neutral signal reapply, compared to 19.9% of those with a negative signal and 7.6% of those with a positive signal. Similarly, 6.0% of students with a neutral signal change the composition of their ROL, versus 13.5% for negative and 5.3% for positive signals.

Table 3— Switching Statistics by Signal Type

	N	Reapply		Switch		Switch Down		Up
		All	Changed comp.	Program	Major	Program	Major	Program
Negative	5745	0.199 (0.005)	0.135 (0.005)	0.216 (0.005)	0.098 (0.004)	0.039 (0.039)	0.014 (0.002)	0.011 (0.001)
Neutral	1665	0.086 (0.007)	0.060 (0.006)	0.100 (0.007)	0.043 (0.005)	0.019 (0.019)	0.007 (0.002)	0.016 (0.003)
Positive	8325	0.076 (0.003)	0.053 (0.002)	0.105 (0.003)	0.041 (0.002)	0.012 (0.012)	0.003 (0.001)	0.033 (0.002)

Note: Sample includes all students from the cohort that graduated high school in 2014, enrolled in 2014 in a program offered by a university within the system, and for whom first-year grade data is available. Positive / Negative signal considers students who received a sufficiently positive / negative signal, i.e., 0.1 points above or below expected, respectively. Neutral signal includes all students who did not receive a positive or negative signal.

aggregated, it may not capture forced switches due to a single course (e.g., failing a mandatory class multiple times). However, students with high GPA who are forced to quit are uncommon.

Our previous results show that students’ reported preferences may change during their first year in college. To assess whether these changes reflect genuinely evolving preferences rather than students simply becoming more experienced in participating in the system, we analyze changes in true preferences using the surveys conducted in 2019 and 2020. Specifically, we construct a panel that includes students who participated in both surveys and responded to the same questions (close to 1,300 students), and we compare the top true preference reported in each year. In Figure 3, we plot the fraction of students who changed their top true preference (for programs and also for universities) as a function of their initial preference of assignment (in 2019). First, we observe that on average, close to 65% of the students changed their top true preference after their first year in college, and close to 50% changed their most preferred university. Second, we observe that students initially assigned to lower preferences are less likely to change their top true preference, consistent with the existence of both the *mismatch* and *learning* channels. In contrast, students initially assigned to their top-reported preference have a lower probability of being *mismatched*; thus, their reapplication suggests that they learned about their match quality and preferences during their first year in college.

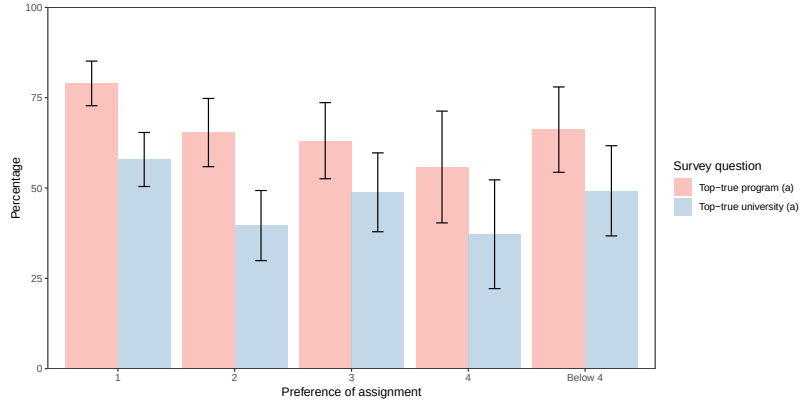


Figure 3. Percentage of reapplicants that change their top-true preference, by preference of assignment in 2019

Note: Sample: respondents to both the 2019 and 2020 surveys with an elicited top-true preference in each wave ($N=1,337$).

IV. Structural Model

In this section, we present a dynamic model of students’ application behavior and academic progression that incorporates how they learn about their abilities and preferences over time. The model adapts and extends the work from [Arcidia-](#)

cono (2004, 2005); Arcidiacono et al. (2025) to (i) capture the empirical patterns documented in the previous section; (ii) quantify the extent to which students' switching behavior is driven by initial mismatch versus subsequent learning; and (iii) assess how alternative learning environments and market design interventions affect students' outcomes in equilibrium.

To accomplish this, we assume that students enter college with imperfect information about two key dimensions: (i) their match-specific abilities—as multidimensional skills relevant to labor market returns—and (ii) their preferences over majors of study. Abilities are partially revealed through college performance, while preferences evolve based on students' experience in their enrolled programs, independently of grades. This learning process shapes students' beliefs about future returns upon graduation and informs their decisions to continue in their current program, reapply to the centralized system, or drop out. Upon graduation, students enter the labor market and receive utility that depends both on a pecuniary component corresponding to their lifetime labor market returns and a non-pecuniary one that captures the alignment between their program of graduation and their preferences.

Timing. The model unfolds over three periods. In period 1, students apply to college, receive an assignment, decide whether to retake the admission exam, and observe first-year grades. In period 2, they may reapply and choose whether to remain in their current program, switch, or drop out. Period 3 aggregates all subsequent academic years through graduation and the labor market, capturing future educational outcomes and discounted earnings.

Notation and Organization. We use $i \in \mathcal{I}$ to refer to students, $j \in \mathcal{J} \cup \{\emptyset\}$ to refer to programs ($\emptyset$ representing no enrollment), and $t \in \mathcal{T} = \{1, \dots, T\}$ to refer to periods. In addition, let $\mathcal{M}$ be the set of majors and $m : \mathcal{J} \rightarrow \mathcal{M}$ be the function that maps programs to majors, i.e., $m(j)$ represents the major of program j . Finally, with a slight abuse of notation, we use functions to make explicit the underlying arguments and structure from which key elements of the model are derived (e.g., $G(\cdot)$ for grades); subscripts to denote their realized values (e.g., $G_{i,j,t}$); tilde notation to denote beliefs over these values (e.g., $\tilde{G}_{i,j,t}$); and hat notation to represent their estimated values (e.g., $\hat{G}_{i,j,t}$).

The remainder of this section is organized as follows. Section IV.A describes the learning process during college. Section IV.B specifies the flow utility from enrollment and the workforce utility obtained upon graduation. Finally, Section IV.C discusses the main decisions modeled in each period.

A. Learning

Learning about preferences. We allow students to learn about their preferences for majors. In particular, we assume that student i 's idiosyncratic preferences for majors in period t are captured by a set of random coefficients $\{\alpha_{i,m,t}\}_{m \in \mathcal{M}}$, whose initial values are normally distributed with mean $\mu_{i,m}^\alpha$ and standard deviation $\sigma_{i,m}^\alpha$. Furthermore, we assume that students only learn about their preferences for

their major of enrollment.¹⁰ Thus, student i 's idiosyncratic preferences for major m , conditional on being enrolled in program j , evolves according to $\alpha_{i,m,t+1} = \alpha_{i,m,t} + \mathbb{1}_{\{m=m(j)\}} \cdot \varepsilon_{i,m,t}^\alpha$, where $\varepsilon_{i,m,t}^\alpha \sim N\left(0, \left(\zeta_{i,m}^\alpha\right)^2\right)$. Finally, as we later discuss in Section IV.B, students observe preference shocks (related to their flow utility) for each degree program that are not persistent over time and i.i.d. across programs and periods, so we do not consider these as part of the (systematic) learning process.

Learning about abilities. Beyond learning about their preferences, we allow students to learn about their abilities. Specifically, let $\mathcal{S}$ be the set of subjects that are mandatory for all students in the standardized exam (i.e., $\mathcal{S} = \{s_m, s_v\}$, with s_m and s_v representing the math and verbal parts of the standardized test, respectively),¹¹ and let $\omega_{j,s}$ be the weight that program j places on subject $s \in \mathcal{S}$. Then, we assume that the ability $A_{i,j}$ of student i in program j corresponds to the sum of three elements: (i) $A_{i,j}^o := \sum_{s \in \mathcal{S}} \omega_{j,s} A_{i,s}^o$, an observable component that is known by the student and the econometrician; (ii) $A_{i,j}^p := A_{i,m(j)}^p + \sum_{s \in \mathcal{S}} \omega_{j,s} A_{i,s}^p$, a private component only known by the student; and (iii) $A_{i,j}^u := A_{i,m(j)}^u + \sum_{s \in \mathcal{S}} \omega_{j,s} A_{i,s}^u$, unknown by both the student and the econometrician but learned from the grades obtained during college. Note that the private and unobserved components comprise two elements: (i) $A_{i,m(j)}^l$, which is major-specific, and (ii) $A_{i,s}^l$, which are subject-specific, with $l \in \{p, u\}$. For private abilities, we assume that these are randomly drawn at the beginning of the academic progression, with $A_{i,k}^p \sim N(\mu_{i,k}^p, \varsigma_{i,k}^2)$ for $k \in \mathcal{M} \cup \mathcal{S}$, where $\mu_{i,k}^p$ is the mean—normalized to zero for majors—¹² and $\varsigma_{i,k}$ is the standard deviation of student i 's private ability in program j and component k . For unobserved abilities, we assume that the true distribution of $A_{i,k}^u$ is normal with mean zero and variance $\sigma_{i,k}^2$ for $k \in \mathcal{M} \cup \mathcal{S}$, but we allow students to have subjective, heterogeneous and potentially biased prior beliefs about these elements; namely, we assume that student i 's prior beliefs about $A_{i,k}^u$ in period t is normally distributed with mean $\tilde{\mu}_{i,k,t}$ and variance $\tilde{\sigma}_{i,k,t}^2$, with initial values $\tilde{\mu}_{i,k}$ and $\tilde{\sigma}_{i,k}^2$. Consequently, student i 's prior mean and variance about their unobserved ability in program j in period t are given by $\tilde{\mu}_{i,j,t} = \tilde{\mu}_{i,m(j),t} + \sum_{s \in \mathcal{S}} \omega_{j,s} \cdot \tilde{\mu}_{i,s,t}$ and $\tilde{\sigma}_{i,j,t}^2 = \tilde{\sigma}_{i,m(j),t}^2 + \sum_{s \in \mathcal{S}} \omega_{j,s}^2 \cdot \tilde{\sigma}_{i,s,t}^2$, respectively.

Furthermore, we assume that students' grades are a function of a set of time

¹⁰University-specific tastes enter the flow utility through college fixed effects. We restrict preference learning to majors both for tractability—each additional learning dimension adds persistent state variables to the continuation-value computation—and because the data show little scope for university-level taste updating: switching consistent with it (changing university while keeping the same major) accounts for about one in ten first-year switches and does not respond to first-year grade signals.

¹¹Elective test scores (Science/History) and high school performance measures (NEM/Rank) are excluded from the ability term because they are not mandatory. However, all five admission factors are used when estimating admission probabilities, and elective scores are updated for exam retakers.

¹²More precisely, the major-specific intercepts of the private-ability means are normalized to zero, as any common major-level shift in abilities is absorbed by the major fixed effects in the grade equation.

invariant observable characteristics $X_{i,j}^G$ — which include characteristics of student i (e.g., their observed ability) and program j (e.g., major)—, their private and unknown abilities $A_{i,j}^p, A_{i,j}^u$, and a noise term $\epsilon_{i,j,t}^G \sim N(0, \sigma_G^2)$,¹³ i.e.,

$$(1) \quad G_{i,j,t} = G\left(X_{i,j}^G, A_{i,j}^p, A_{i,j}^u, \epsilon_{i,j,t}^G\right).$$

Consequently, based on their prior beliefs about their unobserved ability, students construct beliefs about their grades, which we denote by $\tilde{G}_{i,j,t} := \mathbb{E}[G_{i,j,t}]$ —with the expectation taken over $\tilde{A}_{i,j}^u \sim N(\tilde{\mu}_{i,j,t}, \tilde{\sigma}_{i,j,t}^2)$ and $\epsilon_{i,j,t}^G \sim N(0, \sigma_G^2)$ —and, upon observing their actual grades, they receive a noisy signal that they use to update their prior beliefs about their unknown abilities according to Bayes rule, as shown in Proposition 6 in Appendix C.

All learning and belief parameters are student-type specific: we parametrize their means and variances as functions of gender, income, and observable ability (see Appendix C.C1). This heterogeneity allows the model to rationalize the marked differences in switching and dropout across these groups documented in Section III.B—tracing them to differences in preferences, beliefs, and learning—and, in the counterfactuals, to assess the distributional incidence of policies across gender and income groups alongside their aggregate effects.

B. Utilities

Flow Utility. Let $U_{i,j,t}$ be the flow utility that student i receives for attending program j in period t . We assume that this utility can be modeled as:

$$(2) \quad U_{i,j,t} = U\left(X_{i,j}^U, \alpha_{i,m(j),t}, \epsilon_{i,j,t}^U\right),$$

where $X_{i,j}^U$ is a set of time-invariant observable characteristics that include program characteristics and student-program interactions, among them the distance between student i 's and program j 's municipalities and the net cost that student i pays for program j (Section IV.C lists the full set of controls); $\alpha_{i,m(j),t}$ is student i 's idiosyncratic preference for major $m(j)$ in period t ; and $\epsilon_{i,j,t}^U$ is an idiosyncratic preference shock that is distributed i.i.d across programs and periods and follows a type I extreme value distribution with a scale parameter κ . We specify a location normalization and set the systematic flow value of the outside option to $U_{i,\emptyset,t} = 0$ during the college periods ($t \in \{1, 2\}$) and set the distance coefficient to minus one, allowing the scale of the shocks to vary. The value of the outside option from the third period onward—the absorbing state entered upon dropping out—is instead a function of students' observable characteristics, as described in

¹³Grades in our model reflect students' abilities rather than an endogenous effort choice; separately identifying effort would require additional sources of variation, such as direct effort measures (see De Groote (2025) for a dynamic model of student effort and its identification challenges). Following the literature on learning in college, we instead capture the determinants of academic performance through rich ability heterogeneity: observed, private, and unknown.

Appendix E.E4. The exact linear specifications of the flow utility, grade, and wage functions are reported in Appendix C.C1.

Labor Market. Upon graduating from program $j \in \mathcal{J}$ in period $t' \in \mathcal{T}$, we assume that student i enters the work force and obtains a utility $V_{i,j,t'}$ consisting of two elements: (i) a non-pecuniary utility $V_{i,j,t'}^{NP}$, capturing the utility derived from their preferences during their time in college; and (ii) a pecuniary utility $V_{i,j,t'}^P$, capturing the present value of their lifetime wages. Given a retirement period T and a common discount factor δ , we model this as:

$$(3) \quad V_{i,j,t'} = \underbrace{\beta_{m(j)}^V + \beta_{\alpha}^V \alpha_{i,m(j),t'} + \beta_A^V A_{i,j}^p + \beta_{PU}^V \cdot (A_{i,j}^p + A_{i,j}^u)}_{:= V_{i,j,t'}^{NP}} + \underbrace{\beta_W^V \log \left(\mathbb{E} \left[\sum_{t=t'}^T \delta^{t-t'} \cdot P_{m(j),t}^W \cdot W_{i,j,t} \right] \right)}_{:= V_{i,j,t'}^P}.$$

The first term captures the non-pecuniary utility, which depends on student i 's preferences for major $m(j)$ encoded in the random coefficient $\alpha_{i,m(j),t'}$ and on their abilities. The value of the outside state $V_{i,\emptyset,t'}$, entered upon dropping out, is instead a function of observable characteristics, as described in Appendix E.E4. The second term captures the pecuniary utility, which is separable over time and depends on the (major-specific) employment probability $P_{m(j),t}^W$ and the wage $W_{i,j,t}$, which we model as

$$(4) \quad W_{i,j,t} = W(X_{i,j}^W, G_{i,j,t'}, \Lambda_{j,t-t'}),$$

where $X_{i,j}^W$ is a set of time-invariant observable characteristics of student i ; program j and their interaction; $G_{i,j,t'}$ is student i 's GPA at the time of graduation; and $\Lambda_{j,t-t'} = \Lambda(X_{i,j}^A, t-t')$ captures the (potentially heterogeneous across observable characteristics in $X_{i,j}^A$) effect of tenure on program j 's wage. As we discuss in Appendix C, we assume that the effect of grades on wages is heterogeneous across students, and we model it as a random coefficient that is log-normally distributed with mean $\log(\mu_{W \sim G})$ and variance $\sigma_{W \sim G}^2$. Finally, note that student i receives this continuation value only if they graduate from their program. If, instead, student i drops out, we assume that their continuation value is equal to $V_{i,\emptyset,t'}$.

C. Decisions.

Throughout their first two years in college, the key decisions we model are (i) enrollment (in period $t = 2$),¹⁴ i.e., conditional on the assignment, whether to enroll in the newly assigned program, stay in the current one, or dropout; (ii)

¹⁴We treat take-up of a newly assigned program as exogenous for tractability: it occurs with an estimated student-specific probability, while the subsequent choice between remaining in the current program and dropping out is endogenous. Appendix H develops an endogenous-take-up variant in which enrollment increases with the surplus of the assignment relative to the fallback option. This variant fits comparably, although it amplifies the re-application discontinuity by no longer averaging assignment values through a student-specific take-up probability, and leaves our counterfactual conclusions qualitatively unchanged.

exam re-taking (in period $t = 1$), i.e., whether or not to re-take the standardized test; and (iii) application (in periods $t \in \{1, 2\}$), i.e., conditional on the scores received after (re)taking the test, the current beliefs about abilities and the updated preferences, which programs to include in the ROL (if any). For the latter, we assume that students differ in their level of sophistication (Pathak and Sönmez, 2008), with each student's application strategy being exogenously given and following one of two types: (i) weak truth-telling (w.p. ρ),¹⁵ i.e., students report their true preferences; and (ii) strategic (w.p. $1 - \rho$), i.e., students submit a ROL that maximizes their expected value (as in Chade and Smith (2006)) assuming correct beliefs about their marginal admission probabilities. Specifically, a strategic student i solves:

$$(5) \quad R_i^* \in \operatorname{argmax}_{R \subseteq \mathcal{R}_i} \left\{ \sum_{l=1}^{|R|} p_{i,R(l),t} \cdot \bar{p}_{i,R(l),t} \cdot \Psi_{i,R(l)}^* - C_{i,t}^A(R) \right\},$$

where $p_{i,j,t}$ is the probability of getting assigned to program j in period t ; $\bar{p}_{i,R(l),t} = \prod_{l'=1}^{l-1} (1 - p_{i,R(l'),t})$ is the probability of not being assigned to any of the top $l - 1$ programs listed in R , with $\bar{p}_{i,R(1),t} = 1$; $C_{i,t}^A(R) = c|R|$ is the cost of applying to ROL R ;¹⁶ and $\mathcal{R}_i := \{R \in \mathcal{R}(\mathcal{J}) : |R| \leq K, p_{i,j',t} > 0, \forall j' \in R\}$ is the set of ROLs involving at most K programs with positive admission probability. Note that students include programs in their ROL only if it is strictly profitable, so they will skip programs for which their admission chances are zero. This assumption is in line with the findings in (Larroucau et al., 2025), and rules out multiplicity of optimal ROLs involving zero-chance programs (He, 2012).

The key element in (5) is the term $\Psi_{i,R(l)}^*$, which represents the expected continuation value associated with being assigned to program $j = R(l)$ in period t . This value integrates all subsequent dynamic decisions that follow, including whether to retake the admissions exam, re-apply to new programs, switch programs or dropout and join the labor force (see Appendix C.C2 for the explicit form of these continuation values). It also embeds the evolution of beliefs about abilities and the learning processes that unfold conditional on enrollment: students update their expectations based on outcomes and experiences in the program enrollment. In particular, systematic preferences evolve only for the enrolled major, and grades update beliefs about major- and subject-specific abilities according to that program's subject composition; because the subject-specific components are common across programs, these updates propagate to beliefs about ability in all degree programs, including those in other majors. The continuation value also captures the probabilities of graduation and dropout, which we model as

¹⁵Throughout, ρ denotes the probability of weak truth-telling; it is distinct from the preference-intensity parameter in Section II and from the learning-resolution parameters in Section VII.

¹⁶Since submitting an ROL is free of charge (Section III), we set $c = 0$, thus, the length of submitted lists is limited by the cap K and the restriction to programs with positive admission probability.

functions of observable characteristics from period three (see Appendix E.E4), as well as labor market outcomes upon graduation. Finally, it also captures the first-time enrollment cost $C_{i,t}^S$, incurred upon initial enrollment and upon switching alike, which operates as a switching cost. Note that, beyond this direct cost, a substantial portion of the cost of switching is microfounded within the model: switchers restart their academic progression, delaying graduation and the arrival of labor-market payoffs, an opportunity cost that the continuation values capture through discounting.

Computing these continuation values is particularly challenging, as it requires taking expectations over a high-dimensional state space that encompasses the multiple sources of uncertainty previously mentioned. To address this complexity, we assume a large matching market, so students take the distribution of cutoffs as given (Azevedo and Leshno, 2016), and we adapt the results of Fack, Grenet and He (2019) to our dynamic setting, showing that students’ continuation values over their optimal portfolios can be represented by their most preferred ex post feasible programs, integrated over the empirical distribution of future cutoffs. Exploiting the assumption that preference shocks are i.i.d. Type-I extreme value, we then approximate the corresponding expectations using a log-sum expression (see Appendix C.C3). This approach allows us to evaluate continuation values without integrating over all preference shocks, greatly reducing computational burden. Finally, these continuation values can be directly influenced by policy interventions; for example, by altering the information environment, which shapes the learning process, or through market design changes that modify the incentives and feasibility of re-application or switching. We discuss such interventions in greater detail in Section VII.

Controls. The grade, flow-utility, and log-wage functions are linear in their arguments (see Appendix C.C1 for the exact specifications), and each controls for a set of time-invariant observable characteristics—encoded in $X_{i,j}^G$, $X_{i,j}^U$, and $X_{i,j}^W$ —that include characteristics of the student, the program, and their interaction. In the grade equation, we control for broad major fixed effects and students’ observed ability. The flow-utility equation includes program fixed effects—parametrized as a combination of major fixed effects, college fixed effects, a quality index, and dummies for the top 20 programs in terms of market shares—observed ability, distance to the program’s municipality, distance to the program’s mean quality, and net costs (tuition minus scholarships). Finally, the wage equation includes broad major fixed effects, expected grades at graduation, average posted wages at enrollment, and fixed effects for gender and low-income status. See Appendix C.C4 for additional details.

V. Identification

We outline our identification strategy by focusing on the sources of variation that identify the main parameters. While full identification relies on all moment conditions, several parameters can be isolated using targeted moments. For clar-

ity, we present the arguments constructively and conditional on observables.

Preferences. We follow Agarwal and Somaini (2018), who show that students' preferences can be non-parametrically identified from their reported ROLs assuming correct beliefs about their (marginal) admission probabilities and exploiting distance as a special regressor.

Dynamic costs. The dynamic choices described in Section II and the corresponding switching and reapplication moments identify two dynamic costs—the first-time enrollment cost $C_{i,t}^S$ incurred when starting a new program and the entrance-exam retaking cost $C_{i,t}^T$ incurred when retaking¹⁷—separately from the preference distribution identified above. As we formalize in Appendix I, students just above and just below an admission cutoff are comparable in their preferences but start their tertiary education in different programs. Students above the cutoff switch down, and students below switch up, by weighing the same preferences against what they learn in their respective programs, net of the switching cost. Comparing the two switching rates therefore cancels the common preference component and reveals the switching cost relative to the scale of learning; distance, the persistent preference shifter invoked above, then delivers the levels.¹⁸

Application. To estimate the probability of being strategic in the baseline ($1 - \rho$), we use the share of applicants who exclude their top-true preference (elicited in the survey) from their ROLs.

Tuition and welfare effects. We follow Kapor, Karnani and Neilson (2024) and exploit the discontinuous change in tuition generated by the scholarship *Beca Vocación de Profesor* and its heterogeneous effect on enrollment to identify the effect of the net cost that students pay.¹⁹

Private abilities and preferences. Students know their private abilities and the draw of their initial preferences when making their initial choices (Arcidiacono, 2005). Thus, we exploit the correlation in students' initial reported preferences—i.e., before learning about their unknown abilities—across majors and subjects to identify the role of private information on students' choices, and rely on the correlation between students' ROL composition and first-year grades to identify the effect of private abilities on grades using a control function approach (Heckman, 1979; Kline and Walters, 2016). If private abilities largely shape preferences through grades, we should observe a strong correlation between the two.

Variance of abilities and grades. Since variation in students' grades can only be explained by their private abilities (already identified), their unknown abilities,

¹⁷The retaking cost is separated by variation in the value of reapplying, exploiting quasi-random variation in the initial assignment: students just below the cutoff of their top preference are about six percentage points more likely to retake the exam than those just above (Figure B1e in Appendix B.B3). Since about 80% of reapplicants retake the exam, the retake and reapplication decisions are tightly linked.

¹⁸While the argument is most transparent when stated in terms of upward versus downward switching, any two independent combinations of the directional margins carry the same information. In estimation, we target the discontinuities in overall switching (the net of the two margins) and in upward switching (up or out to ex ante infeasible programs), together with the level of downward switching and its correlation with grade signals (see Appendix E.E3).

¹⁹Students with average scores above 600 can enroll in Education programs tuition-free, creating a discontinuity in applications to Education around this cutoff (see Figure D1 in Appendix D.D2).

and the grade shock, we separately identify the variance of the latter two by constructing a system of equations using the variances and covariance of the first-year and second-year signals among non-switchers (see Appendix D.D1).

Biased unknown abilities. Assuming that beliefs about grades follow a similar structure to that in Equation (C2a), considering students' (potentially biased) prior about their unknown ability and an additional measurement error term $\epsilon_{i,j,t}^{\tilde{G}} \sim N(0, \sigma_G^2)$, we identify the initial mean bias in students' beliefs about their first-year grade, $\tilde{\mu}_{i,j}$, by averaging the difference between students' beliefs about their first-year grades and their actual grades.

Learning about preferences and abilities. We directly identify subjective prior variances by exploiting the probability that students assign to their grades falling between their expected grades plus or minus 10 percentiles conditional on attending their *top-reported program*, which reflects students' uncertainty about their GPA given their subjective beliefs. We also use the correlation between the signal and the choices that students make after the first period (including their decisions to switch and dropout) to further separate the effect of subjective beliefs from grade noise; Appendix I formalizes how the response of directional switching to these signals identifies the scale of learning. Intuitively, a high signal-to-noise ratio implies a low variance of the grade shock relative to the subjective variance of unknown abilities, which means students' choices respond more strongly to their observed grades. This translates into stronger correlations between first-year grades and switching decisions. Moreover, if students are more likely to switch majors than subjects when they receive negative signals (lower grades than expected), then the variances related to majors outweigh those for subjects.

Finally, we identify the standard deviation of the evolution of random coefficients, $\zeta_{i,m}^\alpha$, by exploiting variation in preference changes across majors among students who reapply after receiving a non-informative signal, allowing us to isolate the effect of changing preferences from learning about unknown abilities. Because initial reports are submitted before any learning occurs, initial application patterns—the correlation across majors listed within applications and the distribution of top-true preferences elicited in the surveys—identify students' initial preferences over programs, conditional on their observable types and beliefs. Preference learning is then separated using students assigned to their top-true choice: conditional on a non-informative signal, their switching—if it implies a systematic change in their ROL composition—cannot reflect initial mismatch or ability updating, so its incidence reveals the magnitude of preference changes relative to the switching cost.

Labor market. To identify the (subjective) wage equation, we rely on the correlation between students' beliefs about their wages and grades at the individual level and across programs, including students' reported preferences, their top-true program, and also a random program outside their reported ROL—ruling out potential selection concerns. This type of variation allows us to estimate the marginal effect of expected grades on wages. To separately identify the wage

equation parameters from measurement error, we exploit the fact that we elicited students’ beliefs about their admission chances to their top-reported program in two ways: (i) directly, and (ii) indirectly, by asking them to report the probability that their application score will be above the cutoff. The difference between these two responses should be independent of students’ unknown ability, providing a proxy of the measurement error in the survey. Then, to separate the variance of the effect of grades on wages from the measurement error—i.e., $\sigma_{W \sim G}^2$ from $\sigma_{W \sim M}^2$ —we leverage the elicited beliefs on wages for different programs (both included and excluded from the reported preferences)—same as we do for beliefs in grades—to construct a system of equations involving the residuals of the subjective wage regression, as discussed in Appendix D. Finally, we identify the parameter scaling the wage component by leveraging the correlation between students’ subjective wage beliefs and their preferences—specifically, by comparing top true and random programs.

Initial mismatch and counterfactual outcomes. The strength of the initial mismatch channel is not a separate parameter: it is jointly determined by the primitives identified above, including the preference distribution, unobserved heterogeneity, and the first-time enrollment cost. The RDD variation in initial assignments described in Section III.B disciplines this combination: the levels of the directional switching rates introduced above pin down the relative strength of the initial-mismatch and learning channels, while their net measures the causal effect of the initial assignment. In addition, these variations are key to identifying the distribution of outcomes in the counterfactual analysis.²⁰ Indeed, because the primary effect of our counterfactuals is to reassign students around admission cutoffs, these local variations provide a credible basis for predicting counterfactual outcomes. We then use the full model’s structure to extrapolate beyond the cutoffs and account for equilibrium effects that may alter application behavior and initial assignments.

VI. Estimation

We estimate the model using a random sample of 4,000 students drawn from the population described in Section III.A. We next describe the estimation procedure (Section VI.A), present the estimated parameters (Section VI.B), and assess the model fit (Section VI.C).

A. Estimation Procedure

Indirect Inference. We estimate the model parameters via Indirect Inference (II), which relies on building a statistical model that yields a rich description of the data patterns to identify the model parameters (Bruins et al., 2018). This statistical model—also known as the *auxiliary* model—is estimated on both the data and on simulated data from the structural model. The II estimator—also

²⁰See Agarwal, Hodgson and Somaini (2024) for a thorough discussion of this identification challenge.

known as Wald approach to II—minimizes a function of the distance between the estimated data parameters and those estimated from the simulated data. In this sense, the Simulated Method of Moments is a particular case of II, in which the *auxiliary* model is a vector of moments. In Appendices E.E1, E.E2 and E.E3, we formalize the estimator, describe the estimation algorithm, and discuss the moment conditions, auxiliary models and target parameters, respectively.

Estimation of Exogenous Components. As previously discussed, we estimate the distributions of admission probabilities $\hat{p}_{j,t}$ using the bootstrap procedure detailed in Appendix E.E4. Similarly, we estimate the probabilities of enrollment $\hat{P}_{i,t}^e$, graduation $\hat{P}_{i,j,t}^g$ and dropout $\hat{P}_{i,j,t}^d$ (from periods three onward) as discussed in Appendix E.E4. All these probabilities are estimated outside the main estimation procedure and serve as inputs.

Marginal Improvement Algorithm. An important part of the estimation procedure is to solve for the optimal portfolio that a strategic student would choose, given the estimated parameters. To do this, we use the Marginal Improvement Algorithm (MIA) introduced by Chade and Smith (2006), which sequentially adds the program that yields the highest marginal utility gain, stopping when the list-length constraint binds or no further improvement is possible. We then assume that a student’s reported ROL corresponds to the MIA-selected programs, ordered by decreasing utility. See Appendix E.E5 for details.

B. Estimated parameters

Table 4 presents a summary of the key estimated parameters. Full estimation results are presented in Tables E3, E4, E5, E6, and E7 in Appendix E.E7.

Table 4—Estimation Results: Summary of Key Parameters

Parameter	Unknown Abilities							
	Beliefs biases				True		Preferences	
	Mean ($\tilde{\mu}$)		Log-Var ($\tilde{\sigma}^2$)		Log-Var (σ^2)		Log-Var (ζ_m^2)	
	Value	S.E.	Value	S.E.	Value	S.E.	Value	S.E.
Science	0.63	(-)	-1.28	0.21	-0.27	0.19	0.57	0.09
Social Science	0.66	(-)	-2.03	0.13	-0.20	0.22	1.14	0.01
Education and Humanities	0.36	(-)	-0.74	0.16	-0.58	0.26	0.48	< 0.01
Health	0.84	(-)	-1.99	0.09	-1.44	0.18	2.69	0.02
Low income	0.08	(-)	-0.19	0.21	0.12	0.09	2.47	0.02
Female	-0.19	(-)	0.20	0.25	-0.37	0.10	-0.04	0.02
Avg. obs. ability	-0.21	(-)	-1.90	0.08	-0.80	0.16	1.20	0.03

Note: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator. Parameters from external regressions are indicated by (-).

Students’ beliefs about their academic abilities display systematic biases. Prior

means are upward-biased across all major categories, with subjective mean biases ($\tilde{\mu}_m$) ranging from 0.36 to 0.84, consistent with pervasive overconfidence. Students also underestimate the uncertainty in their abilities, as indicated by negative biases in log-variances reflecting overprecision in their beliefs, i.e., $\log(\tilde{\sigma}_m^2) < \log(\sigma_m^2)$. These distortions have important behavioral implications: overconfidence leads students to overestimate their chances of success in competitive programs, generating ex post mismatches once grade signals reveal true abilities; overprecision dampens belief updating in response to new information, slowing learning and discouraging optimal switching. Biases also vary systematically across demographics: low-income students tend to be more overconfident and overprecise than high-income students; female students are generally less overconfident and overprecise than males; and higher-ability students are less overconfident but more overprecise. Beyond learning about their abilities, students also learn about their preferences for majors through direct experience. The estimated log-variance parameters (ζ^2) reveal substantial heterogeneity across majors and student characteristics. Low-income students, for example, display higher variances and thus greater initial uncertainty about their preferences.

A few additional parameters are worth highlighting (see Appendix E.E7 for details). The estimated share of strategic students is 0.88, suggesting that most applicants account for admission probabilities when choosing their rank-ordered lists, underscoring the importance of explicitly modeling strategic behavior. Regarding preference heterogeneity, the estimated scale of idiosyncratic shocks ($\kappa = 17.78$) and the standard deviation of major-specific random coefficients ($\sigma_\alpha = 33.7$) indicate substantial variation across students, with systematic differences in major preferences dominating idiosyncratic tastes. The exam retaking cost ($C^T = 21.84$) is a fraction of the first-time enrollment cost ($C^S = 206.55$), which, spread over the expected duration of a degree, is comparable to the dispersion of the idiosyncratic preference shocks (about 1.4 times their per-period standard deviation). Switching thus requires substantial expected gains beyond the delayed-graduation cost already embedded in the continuation values (Section IV.C). Finally, the variance of the grades shock ($\sigma_g^2 = 0.11$) and the cross-sectional variance of the realized grade signal ($\mathbb{V}(a_{i,j,1}) \approx 0.26$) imply that the grade signals are highly informative and, thus, students can meaningfully learn about their abilities from their performance in college.

Overall, these findings reveal that students exhibit substantial and heterogeneous belief biases—typically overestimating their abilities and underestimating uncertainty—yet the grade signal remains highly informative about their unknown abilities. At the same time, students’ choices are shaped by strategic application behavior, substantial preference heterogeneity, and sizable dynamic adjustment costs. These patterns can have far-reaching implications for application strategies, learning dynamics, and the welfare effects of policy interventions, highlighting the importance of accounting for both bias and heterogeneity in students’ beliefs and learning parameters.

C. Model fit

Table 5—Goodness of Fit: Additional Moments

(a) RDD Moments			(b) Beliefs on Prior Log-Variance						(c) Signal Correlations		
Targets	Model	Data	Belief Variance				Learning Pref.		Targets	Model	Data
<i>RDD Switching</i>			Coefficient	Subjective		Real		Model	Data	<i>Grades $\times$ Switch</i>	
Constant	0.114	0.167		Model	Data	Model	Data			Dropout	-0.227 -0.281
Discontinuity	-0.079	-0.051	Science	-1.592	-1.768	-0.254	-0.191	0.111	0.089	Program	-0.161 -0.218
<i>RDD Re-Application</i>			Social Science	-1.706	-1.746	-0.227	-0.139	0.051	0.029	Broad Major	-0.142 -0.158
Constant	0.289	0.234	Educ. and Human.	-1.517	-1.770	-0.479	-0.548	0.102	0.048	Major	-0.170 -0.213
Discontinuity	-0.158	-0.087	Health	-1.834	-1.749	-0.965	-0.825	0.047	0.070	Math Type	-0.111 -0.139
<i>RDD Switch Up/Out</i>			Female	-0.032	-0.070	-0.248	-0.245			Switch Up	-0.024 0.065
Constant	0.072	0.048	Low-income	-0.004	0.003	0.084	0.068			Switch Down	-0.046 -0.133
Discontinuity	-0.066	-0.034	Observed ability	-0.310	-0.302	-0.521	-0.503			Out (feas.)	-0.162 -0.225
			Verbal weight ²	-0.080	-0.175	0.076	0.351			Out (unfeas.)	-0.023 -0.011
			Signal					-0.058	-0.120		

Notes: Goodness of fit for key moments used in estimation. Panel (a) shows RDD discontinuity coefficients. Panel (b) shows regression coefficients of log total variance of prior beliefs (Belief Variance columns show both Model and Data for Subjective and Real beliefs) and learning preference coefficients (showing both Model and Data). Panel (c) shows correlations between grades and switching behaviors.

To illustrate the key identifying variations discussed in Section V, Table 5 summarizes the moments that help identify the two main channels: (i) initial mismatches and (ii) learning about abilities and preferences. Panel 5a reports the estimated causal effects from the RDD models around admission cutoffs for students' top-reported preferences. These variations are crucial for identifying initial mismatches and accurately predicting switching rates in counterfactuals, since students may shift around the cutoffs in equilibrium. The model matches these moments reasonably well, though it slightly overpredicts the magnitudes of the discontinuities. Panel 5b shows the fit of regressions of the log total variance of students' prior beliefs on demographics and ability measures. Along with the correlation structure of grade signals, these moments identify the degree of learning about unknown abilities and their effects on outcomes. The model replicates well the observed heterogeneity across broad majors and demographic groups (see Table C2 in Appendix C.C5). The same panel also reports the fit for regressions of changes in the norm of reported preferences on students' grade signals and major. Preference changes are negatively correlated with grade signals, but the estimated intercepts show that students also revise their preferences when their grades carry no news, evidence that students learn about their match quality (preferences) outside the grade production function. See Appendix E.E3 for detailed specifications of these auxiliary models. Finally, Panel 5c presents the correlations between grade signals and switching outcomes. The model reproduces these patterns broadly: switching *Up* is only weakly correlated with grades in both the model and the data, while switching *Out* to ex ante feasible programs is strongly negatively correlated, consistent with both channels.

VII. Counterfactuals

We now turn to our counterfactual analysis. As outlined in the introduction, we consider two families of counterfactual policies: (i) changes to the information and learning environment, aimed at reducing belief biases and front-loading learning about abilities and preferences; and (ii) market design interventions—including changes in the assignment mechanism and the reapplication rules—that reduce initial mismatches and capture students’ dynamic incentives. We evaluate each of these policies in equilibrium, accounting for the strategic incentives they may generate; that is, we assume that strategic students respond to changes in the environment by updating their beliefs about the distribution of cutoffs and adjusting their behavior accordingly. Algorithm 2 in Appendix F.F3, which extends Kapor, Neilson and Zimmerman (2020) to account for the mixture between first-time and re-applicants, describes how we estimate students’ equilibrium beliefs over the stationary cutoff distributions.

The welfare gains from these policies stem from their ability to reduce the aggregate inefficiencies created by initial mismatches. These inefficiencies take three forms. First, vacated seats that are neither reallocated nor added to subsequent admissions offers leave institutional capacity unused, generating revenue losses through foregone tuition (Section III.B). Second, displaced students—those who would otherwise have occupied these seats—are pushed into less-preferred options or remain unassigned, making them more likely to reapply and switch and thus incur additional reapplication and switching costs. Third, students who switch restart their academic progression, delaying graduation and postponing labor-market entry. The policies examined below differ in how they trade off reducing these costs against preserving the benefits of learning through enrollment.

Before proceeding to our primary counterfactual analysis, we assess how switching and dropout decisions are shaped by the two behavioral channels (see Table F1 in Appendix F.F5). First, eliminating biases about mean unknown abilities—by setting $\tilde{\mu} = 0$ —decreases switching by roughly 15%, while increasing unassignment by close to 19%. Second, eliminating biases in the mean and variance of unknown abilities—by additionally setting $\tilde{\sigma}_k^2 = \sigma_k^2$ for $k \in \mathcal{M} \cup \mathcal{S}$ —increases the number of switches and dropouts by close to 81% and 23%, respectively. The intuition behind this result is that students are generally overoptimistic (i.e., positive mean bias) and underestimate variability (i.e., their prior variance is lower than the true variance), so eliminating these biases increases the informativeness of the signals received from grades, thereby increasing the value of choosing other options (either switching or dropping out). Third, removing the learning about abilities channel—by setting $\sigma_k^2 = 0$ for $k \in \mathcal{M} \cup \mathcal{S}$ —reduces program switching by roughly 23% relative to the baseline. Beyond this, eliminating the learning about preferences channel—setting $\zeta_{i,m}^\alpha = 0$ for all $i \in \mathcal{I}$ and $m \in \mathcal{M}$ —yields an additional 12% drop in switching and 11% in dropout. Finally, eliminating initial mismatches—by assigning every student to their top true choice—reduces switching by over 94%. Overall, the results indicate that initial mismatches and

learning about abilities are the primary drivers of students' dynamic decisions.

A. Information and Learning Environment

We first evaluate the effects of modifying the information environment to reduce belief biases and front-loading the learning process. We model this intervention as a mandatory year in a Post-Secondary Program (PSP). In some higher education systems (e.g., Canada and France), students are required to complete one- or multi-year preparatory programs before applying to a specific set of degrees; in Chile, the *Programa de Bachillerato/College* offers a similar general first year, after which students transfer into degree programs. This structure enables students to learn about their abilities and preferences prior to specialization without generating crowd-out externalities, albeit at the cost of reducing the terminal period by one year.

The preparatory year resolves two distinct sources of uncertainty that we study jointly: (i) students' major- and subject-specific unknown abilities, and (ii) their preferences. We model learning about abilities by partially debiasing the prior mean and transferring uncertainty from unknown abilities to private abilities, i.e., considering $\hat{\mu}_{i,k} = \rho^A \cdot \tilde{\mu}_{i,k}$, $\hat{\sigma}_{i,k}^2 = \rho^A \cdot \sigma_{i,k}^2$, $\hat{\tilde{\sigma}}_{i,k}^2 = \rho^A \cdot \tilde{\sigma}_{i,k}^2$, and $\hat{\zeta}_{i,k}^2 = \zeta_{i,k}^2 + (1 - \rho^A) \cdot \sigma_{i,k}^2$, for all $i \in \mathcal{I}$ and $k \in \mathcal{M} \cup \mathcal{S}$, with $\rho^A \in [0, 1]$. Similarly, we model learning about preferences by reducing the variance of the learning process about preferences and transferring the resolved heterogeneity to students' initial preferences, i.e., considering $(\hat{\zeta}_{i,m}^\alpha)^2 = \rho^P \cdot (\zeta_{i,m}^\alpha)^2$ and $(\hat{\sigma}_{i,m}^\alpha)^2 = (\sigma_{i,m}^\alpha)^2 + (1 - \rho^P) \cdot (\zeta_{i,m}^\alpha)^2$ for all $i \in \mathcal{I}$ and $m \in \mathcal{M}$, with $\rho^P \in [0, 1]$.²¹ The constants ρ^A and ρ^P allow us to control the level of learning about abilities and preferences, respectively. Then, we can decompose the relative importance of these types of learning by considering different combinations of ρ^A and ρ^P ; for instance, resolving both sources (i.e., setting $\rho^A = 0$ and $\rho^P = 0$) fully recovers the joint intervention described above, while resolving neither (i.e., setting $\rho^A = 1$ and $\rho^P = 1$) isolates the one-year time cost.

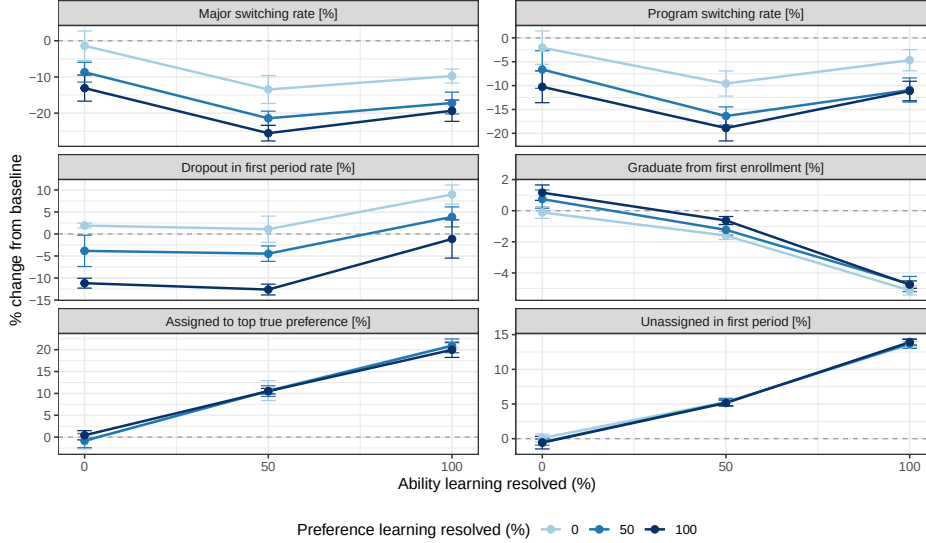
RESULTS.

In Figure 4, we present the estimated effects—measured as percentage changes relative to the baseline—on several outcomes, including switching (program and

²¹Bordon and Fu (2015) analyze a shift from a joint system—where students apply directly to degree programs—to a sequential system, where students first choose a college and later declare a major. Relatedly, Declercq and Verboven (2018) study an open-enrollment system without admission standards (Flanders), in which student-major sorting occurs ex post through first-year performance, and show that introducing moderate ex ante admission standards reduces unsuccessful initial enrollment without decreasing degree completion—front-loading part of the sorting that would otherwise take place through costly reorientation and dropout. In contrast, our analysis modifies the learning environment leaving both the college structure and the assignment process unchanged. Throughout, we hold capacities fixed, consistent with the Chilean rule that vacated seats are neither refilled nor added to future offers.

major), first-period dropout rates, unassignment, graduation rates from the first program of enrollment, and the share of students assigned to their top true preference.²² Each panel plots the effect against the share of ability learning resolved ($1 - \rho^A$), with a separate line for each level of preference learning resolved ($1 - \rho^P$), with both ρ^A and ρ^P taking values in $\{0, 0.5, 1\}$. In Table F2 in Appendix F.F5, we present the detailed results.

Figure 4. Counterfactuals: Information and Learning Environment



We observe that the two sources of learning operate through different channels and target distinct margins. Learning about abilities is central for sorting: it increases the share of students assigned to their top true preference, but it also leads to a higher number of unassigned students in the first period. The reason is that debiasing prior means and eliminating the uncertainty about unknown abilities leads students to submit more “focused” rank-ordered lists, thereby increasing the risk of being unassigned. Moreover, the effect on switching and dropout is non-monotonic, with the largest reductions occurring at intermediate levels of uncertainty resolution: at full resolution, the share of students unassigned in the first period rises by up to 14%, and switching and dropout among enrolled students partially rebound. Learning about preferences, in turn, mainly reduces program switching and first-period dropout, without a comparable increase in

²²We do not report welfare effects for these counterfactuals: because the interventions directly alter students’ beliefs, subjective welfare is evaluated under a different belief system in each counterfactual, so comparisons would conflate changes in outcomes with changes in the beliefs under which welfare is measured. We evaluate welfare below for the market design interventions, which hold the information environment fixed.

unassignment. Both sources contribute to lower major switching.

These results suggest that front-loading learning involves a trade-off that depends on the type of uncertainty resolved: resolving ability uncertainty improves sorting into top true choices at the cost of higher unassignment, whereas resolving preference uncertainty lowers switching and dropout with little change in unassignment.²³

B. Market Design Interventions

We now focus on market design interventions aimed at eliciting the intensity of students' preferences and accounting for their dynamic incentives. We consider two broad types of policy changes. The first targets the assignment mechanism, modifying how preferences are expressed and processed to better capture their intensity—either by restricting the length of applicants' rank-ordered lists (Constrained Deferred Acceptance, CDA) or enabling students to signal one of their preferences to receive a score bonus (Choice-Augmented Deferred Acceptance, CADA (Abdulkadiroğlu, Che and Yasuda, 2015)). The second set of interventions alters the reapplication rules, seeking to address dynamic incentives and improve the system's yield by adjusting how the mechanism treats first-time versus returning applicants, as in the Finnish and Turkish reapplication policies. In addition to the outcomes discussed in the previous section, we also evaluate the welfare implications of these policies. We add two measures of students' welfare: (i) interim and (ii) ex post, which capture welfare before and after students learn about their abilities, respectively.²⁴ Both welfare measures are translated into millions of Chilean pesos as of 2014, and are computed under students' subjective beliefs, which are what drive their choices. Unlike before, we can now meaningfully compare welfare across counterfactuals because these interventions are implemented conditional on the existing information environment—that is, incorporating students' biases. As such, any differences relative to the baseline reflect changes in the mechanism or reapplication rules rather than differences in the information available to students. To understand how these changes affect students' choices and aggregate efficiency, we show several realized outcomes, such as switchings and dropouts, which directly affect vacancy utilization.

ASSIGNMENT MECHANISMS.

As our stylized model shows, dynamic incentives alone elicit the intensity of students' preferences only partially (Lemma 1). Mechanisms that elicit this information at the time of application change the selection of students who obtain

²³Appendix Table F3 reports the same grid without the one-year time cost; the results are very similar, indicating that these effects reflect the resolution of information rather than the mechanical loss of a terminal year.

²⁴Ex post utilities are computed at the end of period two, after students have made all of their choices in the model and updated their beliefs, adding the discounted period-three value function evaluated under the information students hold at that point.

seats in over-demanded programs and can increase the share of students assigned to their top true choice. Since assignment to the top-reported choice causally reduces reapplication and switching—as our regression-discontinuity estimates show (Section III.B)—better-targeted initial assignments translate into fewer vacated seats.

We evaluate the effects of eliciting the intensity of students’ preferences through changes in the assignment mechanism using two alternative approaches (see Appendix F.F1):

- 1) Constrained Deferred Acceptance (CDA): Change the constraint in the length of the ROLs, K . We evaluate $K \in \{1, 2, 3\}$, since most students submit an ROL with length less than or equal to 4.
- 2) Choice-Augmented Deferred Acceptance (CADA): Students can signal one program on their ROL and receive a score bonus (score multiplier $\psi > 1$) applied to their high school GPA component. We implement this mechanism only for first-period applicants; consequently, students who apply in the second period do not receive the bonus.

We focus on CDA and CADA because both mechanisms elicit the intensity of students’ preferences through opportunity costs embedded in the application process. CDA is particularly relevant because restrictions on list length are among the most common design features of centralized admissions systems. By limiting the number of programs that can be included on a rank-order list, CDA forces students to weigh the opportunity cost of each application. CADA, in contrast, allows students to signal only one program, requiring them to decide where the signal is most valuable. We also focus on CADA because it elicits preference intensity without introducing additional strategic considerations into the ranking of programs, confining strategic behavior to the placement of the signal.

Notice that eliciting the intensity of students’ preferences may not necessarily lead to higher yield. On the one hand, if eliciting this information decreases initial mismatches, we would expect to reduce inefficient switching. On the other hand, if the assignment mechanism also elicits the intensity of preferences of students who reapply to the system and these change considerably due to learning, we would see an increase in efficient switching due to a higher value of reapplication. In this sense, we expect that under CADA—which provides a score bonus only in the first period—switching would decrease more than in the case in which the score bonus is applied in both periods, because the policy also gives a comparative advantage to first-period applicants, which increases switching costs through the higher equilibrium cutoff scores produced by the bonus.

REAPPLICATION RULES.

Another policy to reduce the incentives to switch is to provide bonuses to students applying for the first time or to penalize students who reapply and try to

switch programs; these policies have been implemented in Finland and Turkey, respectively. We implement these policies in our counterfactual analysis as follows, where ψ represents a score multiplier applied to the high school GPA component:

- (i) Turkish rule: Applicants currently enrolled in the centralized system receive a score penalty (score multiplier $\psi < 1$).
- (ii) Finnish rule: First-time applicants receive a score bonus (score multiplier $\psi > 1$).

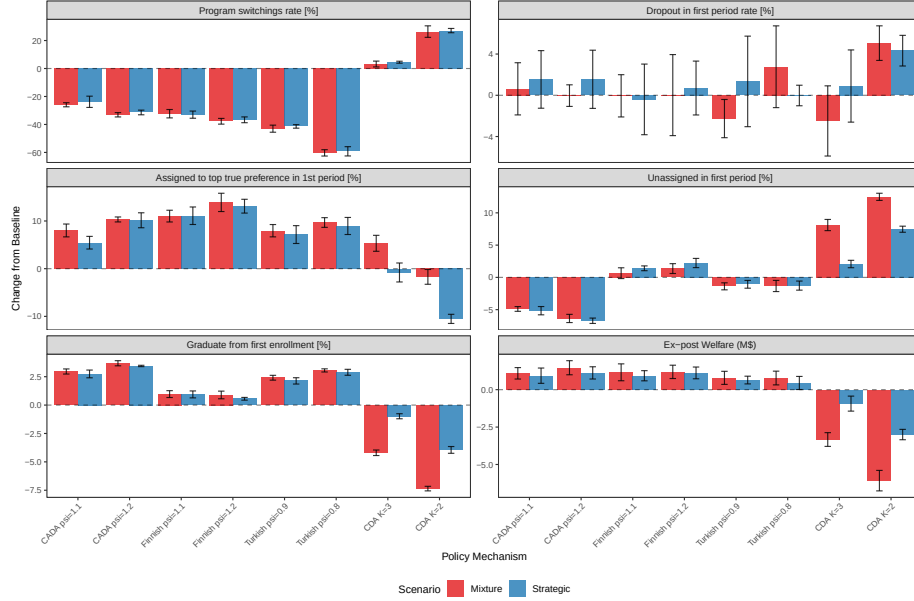
Even though both policies aim to reduce the incentives for switching, they affect students' applications and reapplications in different ways. On the one hand, the Finnish policy directly reduces the incentives to reapply to the system, regardless of the programs students include in their ROLs. As a result, the Finnish policy increases the continuation value of choosing the outside option, and thus increases the fraction of students who wait an extra year to submit their first application. In our simulations, about 4% fewer students apply in the first period under this policy—postponing their entry by a year—a delay that our welfare measures already account for. On the other hand, the Turkish policy reduces the incentives to reapply if students were previously enrolled in a program in the system—i.e., it reduces the incentives to apply to programs if they are very likely to switch from them in the future (e.g., programs for which students have low preference intensity). Hence, the Turkish policy may decrease the fraction of students who enroll in the first period in less preferred programs. Despite these differences, we expect that both policies would decrease the frequency of reapplications and switches. In contrast, the welfare effects of these policies are unclear. Students may benefit from these policies, since both the penalty and the bonus help to address the negative externality that switchers generate in the system. However, since under these policies students face more barriers to switching, the benefits of learning become lower, so students' welfare may decrease.

RESULTS.

In Figure 5, we present a summary of the counterfactual results as percentage changes relative to the baseline. The detailed results are reported in Tables F4 and F5 in Appendix F. Since students' application behavior—i.e., whether they act strategically or as weak truth-tellers—might not be invariant to counterfactuals that heighten strategic incentives, we present bounds on policy effects by considering two benchmark scenarios: (i) all students acting strategically ($\rho = 0$), and (ii) students behaving according to a mixture with the truth-telling probability set to its estimated value ($\rho = \hat{\rho}$), so that students are strategic with probability $1 - \hat{\rho}$. The figure reports both scenarios, while the appendix tables correspond to the fully strategic case.

On the one hand, we find that implementing CDA increases the share of students who remain unassigned, switch programs, and drop out in the first period

Figure 5. Summary of Counterfactual Results



in most scenarios. The stylized model of Section II identifies two opposing forces at play: constraining list length elicits preference intensity, which can improve the selection of students into first-period seats, but it also raises initial mismatches and the risk of remaining unassigned. Our results show that the second force dominates: rather than targeting their most preferred programs, strategic students skip them to improve their chances of admission elsewhere, which significantly decreases the share of students assigned to their top true preference when all students behave strategically and undermines CDA's potential selection benefit. The program switching rate conditional on assignment increases monotonically as K decreases, although the absolute number of switchers peaks at $K = 2$ and declines at $K = 1$, as the sharp rise in unassigned students shrinks the pool of potential switchers. Consequently, CDA leads to significantly lower graduation rates and lower ex post welfare, with ex post welfare declining as K decreases.

On the other hand, CADA and the policies involving changes to the re-application rules can improve student outcomes. From the top panel—showing the percentage change in switching and dropout rates relative to the baseline—we observe that all these policies lead to a sizable reduction in program switching, with the largest effect under the Turkish policy.

From the middle panel—reporting the percentage change in the probability of assignment to the top true preference and the probability of being unassigned in the first period relative to the baseline—we find that CADA and the re-application rule changes significantly increase the share of students assigned to their top true

preference, with the Finnish and CADA policies delivering comparable improvements. In the fully strategic case, we also observe that the Finnish rule causes a small increase in the share of unassigned students in the first period, whereas the Turkish rule and CADA produce a decrease.

From the bottom panel—showing the percentage change in the probability of graduating from the first enrollment and the change in ex post welfare relative to the baseline—we observe that these policies lead to a significant increase in the probability of graduation from the first enrollment, with the largest gains under CADA and the Turkish policy. When considering the mixture of student types, ex post welfare increases under all these policies, with CADA generating the highest welfare improvement, and the Finnish policy being the best at eliciting preference intensity—assigning students to their top-true preference.

Finally, regarding student types, Tables F4 and F5 in Appendix F reveal that males generally benefit more than females across the different policies. However, the effects across income groups depend on the policy’s intensity, underscoring that policy design may entail trade-offs in equity and distributional fairness.

Overall, these results indicate that restricting the length of applicants’ lists can have detrimental effects on both switching behavior and student welfare. In contrast, CADA and the re-application rule policies appear promising for increasing the system’s yield, mitigating the congestion externalities caused by initial mismatches, and enhancing welfare. However, the optimal policy depends on the policymaker’s objective: if the goal is to reduce switching and increase system yield, the Turkish policy performs best, whereas if the aim is to maximize ex post student welfare, CADA delivers superior outcomes.

These results translate into three concrete design lessons. First, restricting the length of application lists is counterproductive: it increases unassignment and initial mismatches and reduces welfare, with welfare losses that are particularly severe for naive truth-tellers. Second, dynamic incentives that price reapplication and switching—score penalties or bonuses, as in Turkey and Finland, or a signal with admission weight, as in CADA—outperform list-length restrictions by discouraging enrollment that is likely to be reversed while preserving students’ ability to correct genuine mismatches. Third, on the information side, front-loading learning as preparatory programs, involves a trade-off that hinges on how much preference versus ability uncertainty they resolve: resolving preference uncertainty reduces switching and dropout without a comparable rise in unassignment, whereas resolving ability uncertainty improves sorting into top true preferences but induces more focused application lists, raising unassignment.

VIII. Conclusions

Introducing dynamics fundamentally changes the incentives generated by centralized college admissions systems. Students enter college with imperfect information and make subsequent decisions as they learn about their abilities, preferences, and opportunities through enrollment. Once reapplication and switching

become possible, application decisions depend not only on the value of the initial assignment but also on future opportunities and the costs of changing programs. As a result, mechanisms such as Deferred Acceptance, which are strategy-proof in static environments, need not remain so in dynamic settings.

Accounting for these incentives requires admissions systems to balance a fundamental trade-off between flexibility and the elicitation of preference intensity. Flexibility allows students to correct initial mismatches and respond to new information, facilitating productive experimentation and reducing ex post mismatch. At the same time, policies that restrict switching or otherwise increase its cost can better elicit preference intensity by forcing students to internalize the opportunity cost of targeting risky alternatives and the congestion externalities created by future reapplications. The central design problem is therefore not whether flexibility is desirable, but rather how much flexibility should be preserved relative to the benefits of eliciting information about the strength of students' preferences.

Striking the right balance between these competing considerations can generate substantial gains in both student welfare and institutional outcomes. Determining the appropriate policy response, however, is far from straightforward, as it depends on the relative importance of the two behavioral channels highlighted in the paper: initial mismatches and learning. Our empirical analysis and estimated model show that both are important determinants of students' educational trajectories. Consequently, the effectiveness of alternative policies depends on the extent to which they affect each channel, with information interventions and market-design policies doing so in fundamentally different ways.

On the information side, front-loading learning through preparatory programs or college-then-major structures can substantially reduce inefficient adjustments, although doing so may require students to delay entry into a degree program by one year. Its effectiveness depends critically on the type of uncertainty that is resolved. Resolving preference uncertainty reduces switching and dropout with little increase in unassignment. In contrast, resolving ability uncertainty improves sorting into students' true preferred programs, but also leads students to submit more concentrated application lists, increasing unassignment. The value of these interventions therefore hinges on the relative importance of preference and ability learning in generating ex post mismatch.

On the market-design side, interventions that elicit preference intensity can reduce inefficiencies, provided they do not unnecessarily restrict student choices. Although list-length restrictions force students to reveal the relative strength of their preferences, our results show that they are ultimately counterproductive, as they increase unassignment, worsen initial mismatches, and reduce welfare, particularly among naive truth-tellers. By contrast, policies that directly incorporate dynamic incentives—such as score penalties or bonuses, or signaling mechanisms—discourage enrollment decisions that are likely to be reversed while preserving students' ability to correct genuine mismatches. As a result, these policies improve retention, increase welfare, and outperform list-length restric-

tions along most dimensions.

Overall, our findings show that the design of college admissions systems cannot be separated from the learning and adjustment processes that follow enrollment. More broadly, our results suggest that market designers should balance the benefits of learning through experience against the congestion and crowd-out externalities generated by mismatch by adopting policies that preserve flexibility while eliciting information about preference intensity. This lesson extends beyond higher education to other matching markets characterized by learning, private information, and repeated participation, including labor, housing, and other allocation environments in which agents continue to learn and revise their choices after the initial match.

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ONLINE APPENDIX

Dynamic College Admissions

APPENDIX TO SECTION II

A1. Proofs.

Throughout this section, we use the following notation: for any $x \in \mathbb{R}$, let $x^+ := \max\{x, 0\}$. Furthermore, when clear from the context, we suppress the type subscript θ and use h, l, ρ and ξ to denote $h_\theta, l_\theta, \rho_\theta$ and ξ_θ , respectively.

Lemma 1 (Application Behavior: DA-U). *Let p_j^t be the period- t admission probability for program j . In period 1, a type θ student ranks h_θ above l_θ whenever both programs are listed. Moreover, the full truthful list $\{h_\theta, l_\theta\}$ is optimal if and only if*

$$(1 - p_{h_\theta}^2)(1 - p_{l_\theta}^2) \geq p_{h_\theta}^2 \min\{\rho_\theta, \tau\}.$$

In period 2, the re-application behavior of a type θ student is as follows: (i) initially assigned to their second choice reapplies to their top choice if and only if $\rho_\theta > \tau$; (ii) initially assigned to their top choice reapplies to their second choice if and only if $\xi_\theta > \rho_\theta + \tau$; (iii) initially unassigned reapplies and ranks both programs truthfully.

Proof. Fix a type θ and suppress the subscript θ . Let h denote the student's top choice and l the second choice. We first compute the continuation values generated by optimal period-2 behavior.

Period 2. If the student holds l , staying yields utility 1. A successful switch to h yields $1 + \rho - \tau$, while an unsuccessful reapplication leaves the student at l . Hence the gain from a successful switch is $\rho - \tau$, so the student reapplies to h if and only if $\rho > \tau$. The value of entering period 2 with l is therefore

$$U_l = 1 + p_h^2(\rho - \tau)^+.$$

If the student holds h and has learned shock ξ , staying yields $1 + \rho - \xi$. A successful switch to l yields $1 - \tau$, while an unsuccessful reapplication leaves the student at h . Thus applying to l is optimal if and only if $1 - \tau > 1 + \rho - \xi$, or equivalently $\xi > \rho + \tau$. Ex ante, before ξ is realized, the value of entering period 2 with h is

$$U_h = \mathbb{E}_\xi [1 + \rho - \xi + p_l^2(\xi - \rho - \tau)^+] = 1 + \rho + p_l^2 \frac{((1 - \rho - \tau)^+)^2}{4}.$$

Finally, an initially unassigned student pays no switching cost and faces a terminal one-shot application problem. The truthful list $\{h, l\}$ yields

$$U_\emptyset = p_h^2(1 + \rho) + (1 - p_h^2)p_l^2.$$

This weakly dominates all other period-2 reports: relative to $\{h\}$ it adds the payoff from obtaining l after rejection from h ; relative to $\{l, h\}$, the gain is $p_h^2 p_l^2 \rho \geq 0$; and the empty list is dominated by any list containing a program with positive admission probability.

Period 1. The period-1 report determines whether the student enters period 2 with h , with l , or unassigned. From the expressions above, $U_h > U_l$: indeed $U_h \geq 1 + \rho$ and $U_l = 1 + p_h^2(\rho - \tau)^+ \leq 1 + \rho$, with strict inequality because $p_h^2 < 1$ when $\rho > 0$ and because $p_l^2 > 0$ and $\tau < 1$ when $\rho = 0$. Also, $U_h > U_\emptyset$: if $\rho > 0$, then $U_\emptyset \leq 1 + p_h^2\rho < 1 + \rho \leq U_h$, while if $\rho = 0$, then $U_\emptyset < 1 < U_h$. The remaining comparison is between receiving the fallback l and remaining unassigned. A direct calculation gives

$$U_l - U_\emptyset = (1 - p_h^2)(1 - p_l^2) - p_h^2 \min\{\rho, \tau\}.$$

To see this, if $\rho \leq \tau$, then $U_l - U_\emptyset = 1 - p_h^2(1 + \rho) - (1 - p_h^2)p_l^2$; if $\rho > \tau$, then $U_l - U_\emptyset = 1 + p_h^2(\rho - \tau) - p_h^2(1 + \rho) - (1 - p_h^2)p_l^2$. These two expressions reduce to the displayed formula.

There are only five payoff-relevant period-1 reports: $\emptyset$, $\{h\}$, $\{l\}$, $\{h, l\}$, and $\{l, h\}$. Their values are

$$\begin{aligned} V_{\{h, l\}} &= p_h^1 U_h + (1 - p_h^1)p_l^1 U_l + (1 - p_h^1)(1 - p_l^1)U_\emptyset, \\ V_{\{l, h\}} &= p_l^1 U_l + (1 - p_l^1)p_h^1 U_h + (1 - p_l^1)(1 - p_h^1)U_\emptyset, \\ V_{\{h\}} &= p_h^1 U_h + (1 - p_h^1)U_\emptyset, \\ V_{\{l\}} &= p_l^1 U_l + (1 - p_l^1)U_\emptyset, \quad V_\emptyset = U_\emptyset. \end{aligned}$$

First,

$$V_{\{h, l\}} - V_{\{l, h\}} = p_h^1 p_l^1 (U_h - U_l) > 0,$$

so whenever both programs are listed, the true order is strictly optimal. Second,

$$V_{\{h, l\}} - V_{\{h\}} = (1 - p_h^1)p_l^1 (U_l - U_\emptyset).$$

Thus the full truthful list is preferred to truncating at the top exactly when $U_l \geq U_\emptyset$, which is equivalent to

$$(1 - p_h^2)(1 - p_l^2) \geq p_h^2 \min\{\rho, \tau\}.$$

Under the strict version of this inequality, $U_l > U_\emptyset$. Since $U_h > U_l > U_\emptyset$, the truthful list also strictly dominates $\{l\}$ and $\emptyset$. If the inequality is reversed, then $U_l < U_\emptyset$, so $V_{\{h\}} > V_{\{h, l\}}$; moreover $\{h\}$ strictly dominates all reports that list l first or omit h , because those reports sometimes replace the strictly better continuation value U_h with either U_l or $U_\emptyset$. This proves the characterization. $\square$

Lemma 2 (Application Behavior: DA-C). *Let p_j^t be the period- t admission probability for program j . In period 1, a type θ student strictly prefers applying to h_θ rather than l_θ if and only if*

$$p_{h_\theta}^1 U_{h_\theta}(\rho_\theta) + (1 - p_{h_\theta}^1) U_\emptyset(\rho_\theta) > p_{l_\theta}^1 U_{l_\theta}(\rho_\theta) + (1 - p_{l_\theta}^1) U_\emptyset(\rho_\theta),$$

where

$$U_{h_\theta}(\rho_\theta) = 1 + \rho_\theta + p_{l_\theta}^2 \frac{((1 - \rho_\theta - \tau)^+)^2}{4}, \quad U_{l_\theta}(\rho_\theta) = 1 + p_{h_\theta}^2 (\rho_\theta - \tau)^+,$$

$$\text{and } U_\emptyset(\rho_\theta) = \max \left\{ p_{h_\theta}^2 (1 + \rho_\theta), p_{l_\theta}^2 \right\}.$$

Proof. Fix a type θ and suppress the subscript θ . Let h denote the student's top choice and l the second choice. Under DA-C, the student can apply to at most one program in each period, so the problem can be solved by backward induction.

Period 2. Suppose first that the student enters period 2 holding h and has learned shock ξ . If she does not reapply, her payoff is $1 + \rho - \xi$. If she applies to l , she obtains $1 - \tau$ with probability p_l^2 and keeps h with probability $1 - p_l^2$. Therefore applying to l raises her payoff by

$$p_l^2 [1 - \tau - (1 + \rho - \xi)] = p_l^2 (\xi - \rho - \tau),$$

which is positive if and only if $\xi > \rho + \tau$. Thus the ex ante value of entering period 2 with h is

$$\begin{aligned} U_h(\rho) &= \mathbb{E}_\xi [1 + \rho - \xi + p_l^2 (\xi - \rho - \tau)^+] \\ &= 1 + \rho + p_l^2 \frac{((1 - \rho - \tau)^+)^2}{4}, \end{aligned}$$

where the last equality uses $\xi \sim U[-1, 1]$.

Suppose next that the student enters period 2 holding l . Staying yields utility 1. Applying to h yields $1 + \rho - \tau$ if successful and leaves the student at l if unsuccessful. Hence the gain from applying is $p_h^2 (\rho - \tau)$, so the student applies to h if and only if $\rho > \tau$, and the value of holding l is

$$U_l(\rho) = 1 + p_h^2 (\rho - \tau)^+.$$

Finally, suppose the student enters period 2 unassigned. Because she can apply to at most one program, applying to h yields expected payoff $p_h^2 (1 + \rho)$, applying to l yields p_l^2 , and not applying yields zero. Since admission probabilities are strictly positive, not applying is dominated, and the unassigned continuation value is

$$U_\emptyset(\rho) = \max \{ p_h^2 (1 + \rho), p_l^2 \}.$$

Period 1. If the student applies to h in period 1, she is admitted with probability p_h^1 and otherwise enters period 2 unassigned. Her value is

$$V_h(\rho) = p_h^1 U_h(\rho) + (1 - p_h^1) U_\emptyset(\rho).$$

If instead she applies to l , her value is

$$V_l(\rho) = p_l^1 U_l(\rho) + (1 - p_l^1) U_\emptyset(\rho).$$

Not applying in period 1 yields $U_\emptyset(\rho)$, which is strictly dominated by applying to h because $U_h(\rho) > U_\emptyset(\rho)$ and $p_h^1 > 0$. To see $U_h(\rho) > U_\emptyset(\rho)$, note that if $\rho > 0$, then $U_\emptyset(\rho) \leq p_h^2(1 + \rho) < 1 + \rho \leq U_h(\rho)$ when the maximum is attained at h , while $U_\emptyset(\rho) = p_l^2 < 1 < 1 + \rho \leq U_h(\rho)$ when the maximum is attained at l ; if $\rho = 0$, then $U_\emptyset(0) < 1 \leq U_h(0)$.

Therefore the only payoff-relevant period-1 comparison is between applying to h and applying to l . The student strictly prefers applying to h if and only if $V_h(\rho) > V_l(\rho)$, which is exactly

$$p_h^1 U_h(\rho) + (1 - p_h^1) U_\emptyset(\rho) > p_l^1 U_l(\rho) + (1 - p_l^1) U_\emptyset(\rho).$$

The reverse strict inequality gives a strict preference for l , and equality gives indifference. Reintroducing the type subscript proves the result. $\square$

Lemma 3 (Application Behavior: DA-NS). *Let p_j^t be the period- t admission probability for program j , and define*

$$\bar{\rho}^{NS} := \frac{(1 - p_{h_\theta}^2)(1 - p_{l_\theta}^2)}{p_{h_\theta}^2}.$$

In period 1, the full truthful list $\{h_\theta, l_\theta\}$ is optimal if $\rho_\theta \leq \bar{\rho}^{NS}$, while truncating to $\{h_\theta\}$ is strictly optimal if $\rho_\theta > \bar{\rho}^{NS}$. In period 2, all initially unassigned students apply reporting their true preferences, while initially assigned students do not re-apply.

Proof. Fix a type θ and suppress the subscript θ . Let h denote the student's top choice and l the second choice. Under DA-NS, students assigned in period 1 cannot apply again in period 2. Thus a student who enters period 2 holding h keeps h and has ex ante payoff

$$\mathbb{E}_\xi[1 + \rho - \xi] = 1 + \rho,$$

while a student who enters period 2 holding l keeps l and has payoff 1.

Period 2. Only initially unassigned students can apply in period 2. The payoffs from the payoff-relevant reports are

$$\begin{aligned} U_{\{h,l\}} &= p_h^2(1 + \rho) + (1 - p_h^2)p_l^2, \\ U_{\{l,h\}} &= p_l^2 + (1 - p_l^2)p_h^2(1 + \rho), \\ U_{\{h\}} &= p_h^2(1 + \rho), \quad U_{\{l\}} = p_l^2, \quad U_\emptyset = 0. \end{aligned}$$

The truthful list weakly dominates these alternatives: $U_{\{h,l\}} - U_{\{l,h\}} = p_h^2 p_l^2 \rho \geq 0$,

$U_{\{h,l\}} - U_{\{h\}} = (1 - p_h^2)p_l^2 > 0$, and $U_{\{h,l\}} - U_{\{l\}} = p_h^2(1 + \rho - p_l^2) > 0$. Hence the continuation value of being unassigned after period 1 is

$$U_\emptyset(\rho) = p_h^2(1 + \rho) + (1 - p_h^2)p_l^2.$$

Period 1. There are five payoff-relevant period-1 reports: $\emptyset$, $\{h\}$, $\{l\}$, $\{h, l\}$, and $\{l, h\}$. Their values are

$$\begin{aligned} V_{\{h,l\}} &= p_h^1(1 + \rho) + (1 - p_h^1)p_l^1 + (1 - p_h^1)(1 - p_l^1)U_\emptyset(\rho), \\ V_{\{l,h\}} &= p_l^1 + (1 - p_l^1)p_h^1(1 + \rho) + (1 - p_l^1)(1 - p_h^1)U_\emptyset(\rho), \\ V_{\{h\}} &= p_h^1(1 + \rho) + (1 - p_h^1)U_\emptyset(\rho), \\ V_{\{l\}} &= p_l^1 + (1 - p_l^1)U_\emptyset(\rho), \quad V_\emptyset = U_\emptyset(\rho). \end{aligned}$$

First,

$$V_{\{h,l\}} - V_{\{l,h\}} = p_h^1 p_l^1 [(1 + \rho) - 1] = p_h^1 p_l^1 \rho \geq 0,$$

so listing h above l is weakly optimal whenever both programs are listed and is strictly optimal when $\rho > 0$. Second,

$$V_{\{h\}} - V_{\{h,l\}} = (1 - p_h^1)p_l^1 [U_\emptyset(\rho) - 1].$$

Thus truncating to $\{h\}$ is strictly preferred to the full truthful list if and only if $U_\emptyset(\rho) > 1$. Since

$$U_\emptyset(\rho) - 1 = p_h^2 \rho - (1 - p_h^2)(1 - p_l^2),$$

this is equivalent to

$$\rho > \frac{(1 - p_h^2)(1 - p_l^2)}{p_h^2}.$$

It remains to verify that no other report can be optimal in this case. If $U_\emptyset(\rho) > 1$, then the displayed inequality implies $\rho > 0$, so $1 + \rho > U_\emptyset(\rho) > 1$. Therefore, after a rejection from h , listing l is worse than remaining unassigned, and any report that omits h or ranks l first sometimes replaces the best payoff $1 + \rho$ with a lower payoff. Hence $\{h\}$ is strictly optimal. If $U_\emptyset(\rho) < 1$, then $1 + \rho \geq 1 > U_\emptyset(\rho)$, so listing l after h is strictly better than stopping after h , and $\{h, l\}$ is optimal. If $U_\emptyset(\rho) = 1$, then adding l after h leaves the value unchanged, so both $\{h\}$ and $\{h, l\}$ are optimal. Reintroducing the type subscript proves the result. $\square$

Proposition 4 (DA-U and DA-C can each dominate). *There exist capacity configurations—i.e., first-period capacities $0 < q_1 < q_2$, second-period scale ϕ , and switching costs $\tau < 1$ —of both of the following types:*

- 1) *Low-congestion configurations for which DA-U yields strictly higher welfare.*
- 2) *High-congestion configurations for which DA-C yields strictly higher welfare.*

Proof. Let p_j^t denote the period- t admission probability to program j under DA-U, and let $\bar{p}_j^t$ denote the period- t admission probabilities to program j under DA-C, respectively. Define

$$\Delta(q_1, q_2, \phi, \tau) := W^{DA-U}(q_1, q_2, \phi, \tau) - W^{DA-C}(q_1, q_2, \phi),$$

where W^{DA-U} and W^{DA-C} denote welfare under DA-U and DA-C, respectively.

We prove each part separately.

Low-congestion. Suppose that $q_1 = 1/4$, $q_2 = 1/2$, $\phi \in [1/3, 1]$, and consider first the boundary case $\tau = 1$. At this switching cost, assigned students never switch, because the largest possible gross gain from switching is one. Hence, we know from Lemma 1 that all top-1 students only apply to program 1 in period 1, while all top-2 students apply to program 2. Thus, program 1 admits one half of top-1 students (since $q_1 = 1/4$ and half of students are top-1) and program 2 admits all top-2 students (since $q_2 = 1/2$). The mass of initially unassigned students is $1/4$, all from the top-1 group. In period 2, these students rank program 1 above program 2. Program 1 admits them with probability ϕ , and the residual demand for program 2 is $(1-\phi)/4$, which is no larger than its capacity $\phi/2$ exactly when $\phi \geq 1/3$. Hence all students rejected by program 1 in period 2 are admitted to program 2. Welfare under DA-U is therefore

$$\begin{aligned} W^U(1/4, 1/2, \phi, 1) &= \frac{1}{2} \cdot \frac{3}{2} + \frac{1}{4} \cdot \frac{3}{2} + \frac{1}{4} \left[\phi \frac{3}{2} + (1-\phi) \right] \\ &= \frac{11}{8} + \frac{\phi}{8}. \end{aligned}$$

Under DA-C, the same first-period behavior is optimal. A top-1 student who applies to program 1 obtains $\frac{1}{2}(1+\rho) + \frac{1}{2}U_\emptyset(\rho)$, where $U_\emptyset(\rho) \geq 1$, while applying to program 2 yields value 1 at the candidate cutoff. Thus, applying to program 1 is strictly optimal for every $\rho > 0$. Top-2 students apply to program 2 and are admitted.

In period 2, the initially unassigned top-1 students can apply to only one program. Let $\bar{p}_1^2$ and $\bar{p}_2^2$ be the DA-C period-2 admission probabilities. In the candidate equilibrium, $\bar{p}_2^2 = 1$, and an unassigned top-1 student applies to program 1 if and only if $\bar{p}_1^2(1+\rho) \geq 1$, or equivalently if and only if $\rho \geq \bar{\rho}^C := (1 - \bar{p}_1^2)/\bar{p}_1^2$. Market clearing at program 1 gives

$$\bar{p}_1^2 \frac{1 - \bar{\rho}^C}{4} = \frac{\phi}{4},$$

and combining this condition with $\bar{\rho}^C = (1 - \bar{p}_1^2)/\bar{p}_1^2$ yields

$$\bar{\rho}^C = \frac{1 - \phi}{1 + \phi}, \quad \bar{p}_1^2 = \frac{1 + \phi}{2}, \quad \bar{p}_2^2 = 1$$

where the latter follows because the mass applying to program 2 is $\bar{\rho}^C/4 \leq \phi/2$ for all $\phi \geq 1/3$. Therefore DA-C welfare is

$$\begin{aligned} W^C(1/4, 1/2, \phi, 1) &= \frac{1}{2} \cdot \frac{3}{2} + \frac{1}{4} \cdot \frac{3}{2} + \frac{1}{4} \left[\bar{\rho}^C + \bar{p}_1^2 \left((1 - \bar{\rho}^C) + \frac{1 - (\bar{\rho}^C)^2}{2} \right) \right] \\ &= \frac{9}{8} + \frac{1}{4} \frac{1 + \phi + \phi^2}{1 + \phi}. \end{aligned}$$

Subtracting the two expressions gives

$$W^U(1/4, 1/2, \phi, 1) - W^C(1/4, 1/2, \phi, 1) = \frac{\phi(1 - \phi)}{8(1 + \phi)} > 0$$

for every $\phi \in [1/3, 1)$. The strict welfare gap at $\tau = 1$, together with continuity of the equilibrium payoffs in τ at the constructed equilibrium, implies that for each fixed $\phi \in [1/3, 1)$ there exists $\underline{\tau}(\phi) < 1$ such that the same strict ranking holds for all $\tau \in (\underline{\tau}(\phi), 1)$.

High-congestion. Let $W_0^m(q_1, q_2)$ denote welfare under mechanism $m \in \{U, C\}$ when $\phi = 0$. Since period-2 capacity is $\phi(q_1 + q_2)$ and utilities are bounded on the maintained parameter space, the total welfare contribution of period-2 assignments and switches is $O(\phi)$ when $\phi \rightarrow 0$. Hence, for each fixed τ ,

$$W^m(q_1, q_2, \phi, \tau) = W_0^m(q_1, q_2) + O(\phi), \quad m \in \{U, C\}.$$

If $W_0^C(q_1, q_2) > W_0^U(q_1, q_2)$, continuity implies that the same strict ranking holds for all sufficiently small positive ϕ . We now show that there are High-congestion regimes where this inequality holds. Specifically, consider $q_1 = \kappa a_1$, $q_2 = \kappa a_2$, with $0 < a_1 < a_2$, and let $\kappa \downarrow 0$. Under DA-U, the period-1 admission probabilities satisfy

$$q_1 = \frac{p_1^1(2 - p_2^1)}{2}, \quad q_2 = \frac{p_2^1(2 - p_1^1)}{2},$$

so $p_j^1 = \kappa a_j + O(\kappa^2)$. As capacities vanish (i.e., $q_1 = \kappa a_1$, $q_2 = \kappa a_2$ with $\kappa \downarrow 0$), the admission probability at each program also goes to zero. In that limit, the event of being admitted to a given program is determined only by having an exceptionally high score at that program, while the score at the other program is asymptotically irrelevant. Since preferences and scores are independent and each preference group has mass $1/2$, among students with a sufficiently high score at program j asymptotically one half rank j first and one half rank it second. Hence, the composition of admits at each program converges to an equal mix of first-choice and second-choice students. The average value of an admitted seat is therefore $(3/2 + 1)/2 = 5/4$, and

$$W_0^U(\kappa a_1, \kappa a_2) = \frac{5}{4} \kappa(a_1 + a_2) + O(\kappa^2).$$

Under DA-C, let $\bar{\rho} \in (0, 1)$ be the limiting cutoff in preference intensity such that top-1 students with $\rho \geq \bar{\rho}$ apply to program 1 and those with $\rho < \bar{\rho}$ apply to program 2. Since $\rho \sim U[0, 1]$, the mass of applicants to program 1 is $(1 - \bar{\rho})/2$, while the mass of applicants to program 2 is $\bar{\rho}/2 + 1/2 = (1 + \bar{\rho})/2$. In the thin-capacity limit, admission probabilities are small, so market clearing gives

$$q_1 = \frac{1 - \bar{\rho}}{2} p_1, \quad q_2 = \frac{1 + \bar{\rho}}{2} p_2.$$

Combining these expressions and replacing $q_j = \kappa a_j$ yields

$$\frac{a_2}{a_1} = \frac{\frac{1+\bar{\rho}}{2} p_2}{\frac{1-\bar{\rho}}{2} p_1} = \frac{(1 + \bar{\rho})^2}{1 - \bar{\rho}}.$$

This equation has a unique solution because the right-hand side is continuous and strictly increasing from 1 to infinity on $(0, 1)$. Program 1 seats are assigned to top-1 students with $\rho \geq \bar{\rho}$, whose average value is $(3 + \bar{\rho})/2$. Program 2 seats are assigned from a pool consisting of all top-2 students, whose average value is $3/2$, and top-1 students with $\rho < \bar{\rho}$, whose value at program 2 is 1. Hence the average value of a program-2 seat is $(3/2 + \bar{\rho})/(1 + \bar{\rho})$, and

$$W_0^C(\kappa a_1, \kappa a_2) = \kappa a_1 \frac{3 + \bar{\rho}}{2} + \kappa a_2 \frac{3/2 + \bar{\rho}}{1 + \bar{\rho}} + O(\kappa^2).$$

Subtracting the DA-U expansion gives

$$W_0^C(\kappa a_1, \kappa a_2) - W_0^U(\kappa a_1, \kappa a_2) = \frac{\kappa}{4} \left[a_1(1 + 2\bar{\rho}) + a_2 \frac{1 - \bar{\rho}}{1 + \bar{\rho}} \right] + O(\kappa^2).$$

The coefficient multiplying κ is strictly positive, so DA-C has strictly higher no-reapplication welfare for all sufficiently small $\kappa > 0$. $\square$

Proposition 5 (DA-U and DA-NS can each dominate). *There exist capacity configurations—i.e., first-period capacities $0 < q_1 < q_2$, second-period scale ϕ , and switching costs $\tau < 1$ —of both of the following types:*

- 1) *Low-congestion configurations for which DA-U yields strictly higher welfare.*
- 2) *High-congestion configurations for which DA-NS yields strictly higher welfare.*

Proof. Let p_j^1 denote the first-period admission probability to program j under truthful reporting in the first-period, and let p_j^2 and $\bar{p}_j^2$ denote the period-2 admission probabilities to program j under DA-U and DA-NS, respectively. Define

$$\Delta(q_1, q_2, \phi, \tau) := W^{DA-U}(q_1, q_2, \phi, \tau) - W^{DA-NS}(q_1, q_2, \phi),$$

where W^{DA-U} and W^{DA-NS} denote welfare under DA-U and DA-NS, respectively. We first show that, whenever truthful reporting is optimal in the first period, the difference in welfare can be expressed as

$$\begin{aligned} \Delta(q_1, q_2, \phi, \tau) = & \frac{(1-\tau)^2}{4} [(1-p_1^1)p_2^1p_1^2 + (1-p_2^1)p_1^1p_2^2] + \frac{(1-\tau)^3}{24} [p_2^1p_1^2 + p_1^1p_2^2] \\ & + \frac{(1-p_1^1)(1-p_2^1)}{2} \left[\frac{3}{2}p_1^2 + (1-p_1^2)p_2^2 + \frac{3}{2}p_2^2 + (1-p_2^2)p_1^2 \right] \\ & - \frac{(1-p_1^1)(1-p_2^1)}{2} \left[\frac{3}{2}\bar{p}_1^2 + (1-\bar{p}_1^2)\bar{p}_2^2 + \frac{3}{2}\bar{p}_2^2 + (1-\bar{p}_2^2)\bar{p}_1^2 \right]. \end{aligned}$$

To see this, note that when the first-period allocation is common, the two mechanisms differ only in period 2. Under DA-U, students initially assigned to their second choice switch to their top choice when $\rho > \tau$. Conditional on a successful switch, the net gain is $\rho - \tau$, so integrating over $\rho \sim U[0, 1]$ gives

$$\int_{\tau}^1 (\rho - \tau) d\rho = \frac{(1-\tau)^2}{2}.$$

Multiplying by the masses of initially mismatched students and by the relevant period-2 admission probabilities gives the first term in $\Delta(q_1, q_2, \phi, \tau)$.

Students initially assigned to their top choice switch after learning when $\xi > \rho + \tau$. Conditional on a successful switch, the net gain is $\xi - \rho - \tau$. Hence

$$\int_0^{1-\tau} \int_{\rho+\tau}^1 (\xi - \rho - \tau) \frac{1}{2} d\xi d\rho = \frac{(1-\tau)^3}{12},$$

and multiplying by the masses assigned to their top choices and by the relevant period-2 admission probabilities gives the second term in $\Delta(q_1, q_2, \phi, \tau)$. The third and fourth terms are the expected assignment values of initially unassigned students under DA-U and DA-NS, respectively. Initially unassigned students pay no switching cost. Combining these mutually exclusive groups gives the displayed decomposition and the sign implications.

We now prove each part separately.

Low-congestion. We show that DA-U can yield strictly higher welfare. Consider a low-congestion configuration in which $q_1 \geq 1/2$, $q_2 \geq 1/2$, and

$$\phi q_j \geq \frac{(1-\tau)^2}{8} \quad \text{for } j = 1, 2.$$

Then every student is assigned to her top choice in period 1 under both mechanisms, so $p_1^1 = p_2^1 = 1$. Under DA-U, the only period-2 applicants are students who were initially assigned to their top choice and then receive a learning shock that makes the other program better net of the switching cost. For each program

j , the mass of such applicants is $(1 - \tau)^2/8$, so the displayed capacity condition implies $p_1^2 = p_2^2 = 1$. Under DA-NS, assigned students cannot reapply; in any case, there are no initially unassigned students because $p_1^1 = p_2^1 = 1$. Substituting these probabilities into the decomposition gives

$$\Delta(q_1, q_2, \phi, \tau) = \frac{(1 - \tau)^3}{12} > 0.$$

Thus DA-U strictly dominates DA-NS in this Low-congestion regime.

High-congestion. We show that DA-NS can yield strictly higher welfare when the second-period market is thin. Fix q_1, q_2 such that $p_1^1, p_2^1 \in (0, 1)$, and fix $\tau < 1$. For sufficiently small ϕ , period-2 admission probabilities are close to zero, so the no-truncation conditions in Lemmas 1 and 3 hold and the decomposition above applies.

The argument is a comparison of the marginal value of a scarce period-2 seat. Under DA-NS, only initially unassigned students apply. To first order in ϕ , a seat in program j is assigned either to a student who ranks j first, with expected value $3/2$, or to a student who ranks j second after being rejected by her top choice, with value 1. Because the two preference groups have equal mass, the first-order value of a DA-NS seat is $5/4$. Hence DA-NS creates period-2 welfare

$$\frac{5}{4} \phi(q_1 + q_2) + O(\phi^2).$$

Under DA-U, the same initially unassigned students compete with assigned students who want to switch. Fix program j , and let j' be the other program. To first order in ϕ , the applicant mass for program j under DA-U is

$$D_j(\tau) = (1 - p_1^1)(1 - p_2^1) + \frac{1}{2}(1 - p_j^1)p_{j'}^1(1 - \tau) + \frac{1}{8}p_{j'}^1(1 - \tau)^2.$$

The first term is the mass of initially unassigned applicants. Their average welfare gain from a seat in program j is $5/4$. The second term is the mass of students initially assigned to their second choice who want to switch to j ; conditional on applying, their average net gain is $(1 - \tau)/2$. The third term is the mass of students initially assigned to their top choice j' who want to switch after learning; conditional on applying, their average net gain is $(1 - \tau)/3$. Therefore the first-order average value of a DA-U seat in program j is

$$\frac{\frac{5}{4}(1 - p_1^1)(1 - p_2^1) + \frac{1}{4}(1 - p_j^1)p_{j'}^1(1 - \tau)^2 + \frac{1}{24}p_{j'}^1(1 - \tau)^3}{D_j(\tau)}.$$

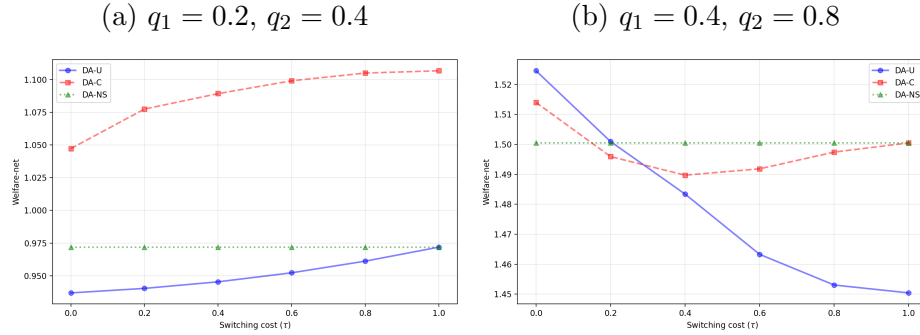
Since $\tau < 1$, the two switcher gains $(1 - \tau)/2$ and $(1 - \tau)/3$ are both strictly below $5/4$, and the switcher applicant mass is strictly positive. Thus the first-order average value of a DA-U seat is strictly below $5/4$ for each j . It follows

that $\Delta(q_1, q_2, \phi, \tau) < 0$ for all sufficiently small positive ϕ . Hence DA-NS strictly dominates DA-U when the second-period market is sufficiently thin. $\square$

A2. Simulations.

We present simulation results to illustrate the welfare and switching behavior under different mechanisms and congestion levels as a function of the switching cost τ . The simulations are based on the model described in Section II, with parameters chosen to reflect various market conditions. Specifically, throughout all these simulations, we consider $N = 1,000,000$ students, $\phi = 0.25$, and we vary the first-period capacities q_1 and q_2 to represent different congestion levels. Then, for each simulation, we first solve for the equilibrium cutoffs to obtain both the first and second period allocations and compute the outcomes under different mechanisms and switching costs. The results are summarized in Figures A1 and A2, which present the net welfare and the masses of switchers (for each channel). The figures illustrate how each mechanism can be preferable depending on the environment.

Figure A1. Net-Welfare as a function of the switching cost τ for different mechanisms and capacity levels.



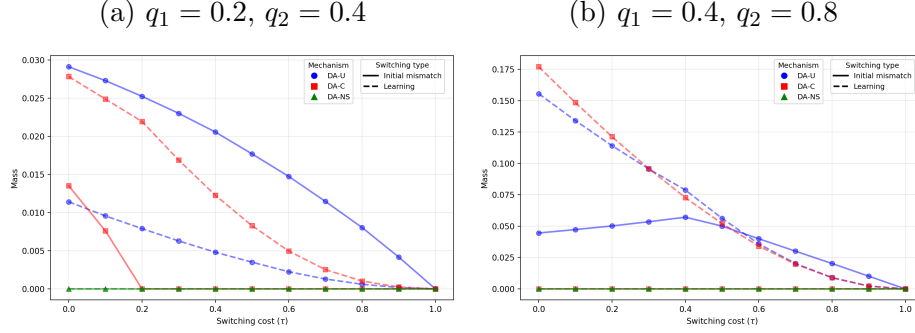
APPENDIX TO SECTION III

B1. The Chilean Mechanism

The Chilean mechanism is a variant of the student-proposing Deferred Acceptance algorithm²⁵ in which students who tie for the last seat of a program are

²⁵Before 2014, the algorithm used was the university-proposing version. The assignment differences between both implementations of the algorithm are negligible Rios et al. (2021).

Figure A2. Switchers as a function of the switching cost τ for different mechanisms and capacity levels.



not rejected if vacancies are exceeded. More formally, the allocation rule can be described as follows:

Step 1. Each student proposes to their first choice according to their submitted ROL. Each program rejects any unacceptable student, and if the number of proposals exceeds its vacancies (q), rejects all students whose scores are strictly less than the q -th most preferred student.

Step $k \geq 2$. Any student rejected in step $k - 1$ proposes to the next program in their submitted ROL. Each program rejects any unacceptable student, and if the number of proposals exceeds its vacancies (q), rejects all students whose score is strictly less than the q -th most preferred student.

The algorithm terminates either when there are no new proposals or when all rejected students have exhausted their preference lists. The final allocation is obtained by assigning each student to the most preferred program on their ROL that did not reject them. As a side outcome of this, the algorithm generates a set of cutoffs $\{\bar{s}_j\}_{j \in \mathcal{J}}$, where $\bar{s}_j$ is the minimum application score among students matched to program $j \in \mathcal{J}$. Hence, student i with ROL R_i and set of scores $\{s_{i,j}\}_{j \in \mathcal{J}}$ gets allocated to program j if and only if (i) $s_{i,j} \geq \bar{s}_j$ and (ii) $s_{i,j'} < \bar{s}_{j'}$ for all $j' \in R_i$ such that $j' \succ_i j$, i.e., student i 's score is less than the cutoff in all programs they prefer over j . The fact that the allocation can be described in terms of cutoffs is known as the *cutoff structure* of the algorithm, and it is a key feature of DA that we exploit in our empirical analysis and estimation.

B2. Surveys

All post-application surveys discussed in Section III.A were sent to all students who submitted an application to the centralized system (about 150,000 each year); the 2023 pre-application survey was instead sent to registered students. In 2019

and 2020, MINEDUC sent the survey right at the end of the application time window; in 2023, we sent the pre-application survey a few days before the application window opened and the post-application survey right after it closed (similar to those in previous years).

To identify model parameters for the 2014 admissions cohort using preference and belief moments from these surveys, we assume that the distributions of preferences and beliefs are stationary across cohorts, conditional on key observables such as gender, income, test scores, and broad major. This stationarity assumption allows us to map belief and preference data from later cohorts onto the 2014 sample.

About 39,000 students completed the 2019 survey and 38,000 the 2020 survey—response rates of 25% and 26% of centralized-system applicants, respectively. Table B1 compares respondents with the population of applicants in each year: differences are modest—respondents are somewhat more likely to be female and have entrance-exam scores about 0.1 standard deviations higher—and the moments we construct from the surveys condition on these observables (see [Dutz et al., 2026](#), for a general treatment of nonresponse bias).

Because our estimation combines these surveys with administrative data for the 2014 cohort, we also document that the admissions environment changed little over this period. Although the centralized system expanded—the number of participating programs grew by roughly 40%—the expansion added programs and institutions without materially affecting the set relevant to our sample: over 90% of the programs in our estimation sample were still offered, under the same code, in each survey year. Table B2 shows that, on this common set, both the distribution of first-listed applications across programs and the programs’ selectivity ranking remain highly correlated with their 2014 counterparts, declining only gradually relative to the adjacent-year benchmark. Application behavior is similarly stable: the median list length is four to five programs in every cohort between 2014 and 2022, and the share of applicants who re-apply the following year moves between 17% and 20% with no trend (17.7% for the 2014 cohort, 18.7% for 2019). In 2023, the admission test, its score scale, and the maximum list length changed with the transition to the PAES system, so we end these comparisons in 2022.

Finally, the objects elicited in the surveys are themselves stable across waves: the distribution of students’ most-preferred (top-true) programs across fields of study is nearly identical in 2019 and 2020 (correlation 0.998).

B3. Regression discontinuities

We use a regression discontinuity design that exploits the algorithm’s cutoff structure to estimate the causal effect of being assigned to a higher preference on students’ outcomes. In particular, we estimate the following specification:

$$(B1) \quad y_{ip} = f_p(d_{ip}) + \delta_p \cdot Z_{ip} + \epsilon_{ip},$$

Table B1—Survey Respondents and the Applicant Population

	2019				2020			
	Survey		Applicants		Survey		Applicants	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Female	0.62	(0.49)	0.56	(0.50)	0.61	(0.49)	0.57	(0.49)
Income below median	0.57	(0.50)	0.55	(0.50)	0.53	(0.50)	0.53	(0.50)
Average score (Math–Verbal)	563.28	(86.83)	553.95	(83.04)	561.48	(88.73)	552.05	(84.99)
High school GPA score	603.86	(100.54)	589.34	(99.24)	614.70	(102.49)	600.12	(101.15)
Observations	38,980		154,366		38,093		146,438	

Note: “Survey” columns consider students who completed the corresponding survey and applied to the centralized system; “Applicants” columns consider all students who submitted an application in that year. Income below median is based on reported family income brackets. Scores are on the 150–850 scale.

Table B2—Stability of the Admissions Environment, 2014–2022

Comparison	Programs offered		Top-listed shares	Cutoff scores
	N	% of sample	Rank corr.	Rank corr.
2014 vs 2015 (adjacent years)	422	96.1	0.96	0.98
2014 vs 2019	407	92.7	0.87	0.92
2014 vs 2020	403	91.8	0.82	0.92
2014 vs 2022	397	90.4	0.80	0.88

Note: Each row compares 2014 with the indicated year on the set of programs from our estimation sample offered in both years. “Programs offered” reports the number and share of the 439 programs in our sample also offered in the comparison year. “Top-listed shares” refers to the distribution of first-listed applications across these common programs, and “Cutoff scores” to the minimum weighted score among admitted students in each program. Both correlations are Spearman rank correlations.

where y_{ip} is the outcome of interest for student i applying to preference p ; f_p is a smooth function of the distance d_{ip} between student i 's application score and the cutoff of their preference p , estimated as a local-linear function with separate slopes on each side of the cutoff; Z_{ip} is an indicator equal to 1 if student i 's score is greater than or equal to the cutoff of their p -th preference and 0 otherwise; and ϵ_{ip} is an error term. The figures display binned means of y_{ip} as a function of the distance to the cutoff.

Notice that many of the outcomes we consider—e.g., switches and dropouts, among others—rely on students enrolling in the centralized system. If there are significant differences in the enrollment rates between students right above and below the cutoff, then the two samples would not be directly comparable, leading to a selection problem (Dong, 2017). To show that this is not the case, in Figure B1a we show the binned means of the probability of enrolling in the centralized system as a function of distance to the cutoff. In addition, the line represents the predicted values obtained from estimating the regression discontinuity model described in (B1), considering as dependent variable the probability of enrolling in the centralized system. As Figure B1a and Column (1) in Table B4 show, there are no statistically significant differences at the 5% level in the enrollment probabilities between students above and below the cutoff, so we conclude that the selection problem is not a concern.

Table B3 reports standard diagnostics for this design. Because application scores are fixed before students submit their lists and cutoffs are realized only after the assignment is run, students cannot precisely manipulate their position relative to the cutoff; we therefore assess the validity of the design through covariate balance at the cutoff, following Abdulkadiroğlu et al. (2022). Predetermined characteristics—gender, income, high-school GPA, and the NEM score—are balanced at the cutoff (Panel A), and the estimated effects on reapplication and switching are stable across bandwidths between 5 and 30 points (Panel B). The estimates in Table B4 use a bandwidth of 15 points, which is conservative relative to the MSE-optimal bandwidths of roughly 50 and 40 points, respectively.

Figure B1 also displays binned means for different outcomes as a function of the distance between the cutoffs in their top preference and the students' scores, and Table B4 reports the corresponding estimation results. In all these analyses, we consider students who applied to at least two programs in the centralized system and focus on the top preference of each student for simplicity. Similar results follow from considering other preferences. As shown in Figure B1b and Column (2) in Table B4, exceeding the cutoff increases the probability of enrollment in the top preference by 51.6 percentage points. Notice that this is not 100% for two reasons: (i) students whose score exceeds the cutoff may decide not to enroll, and (ii) students whose score was below the cutoff may end up enrolling after being pulled from the wait-list. Figures B1c and B1d are discussed in Section III.B, and show that being above the cutoff significantly reduces the probability of reapplying and switching programs within the system. These results are confirmed in

Table B3—Regression Discontinuity Diagnostics

<i>Panel A: Covariate balance at the cutoff (bandwidth ± 15)</i>					
	Female	Income below median	High school GPA	NEM score	
Z_{ip}	0.013 (0.025)	−0.009 (0.024)	−2.565 (2.217)	−5.268 (4.557)	
Observations	5,993	5,993	5,993	5,993	
<i>Panel B: Bandwidth sensitivity</i>					
Bandwidth ($\pm$ points)	5	10	15	20	30
Reapplication	−0.071** (0.032)	−0.067*** (0.024)	−0.084*** (0.020)	−0.090*** (0.017)	−0.090*** (0.014)
Observations	2,108	4,128	5,993	7,797	10,965
Switching	−0.067* (0.035)	−0.065*** (0.024)	−0.073*** (0.020)	−0.056*** (0.018)	−0.049*** (0.015)
Observations	1,671	3,229	4,652	6,003	8,293

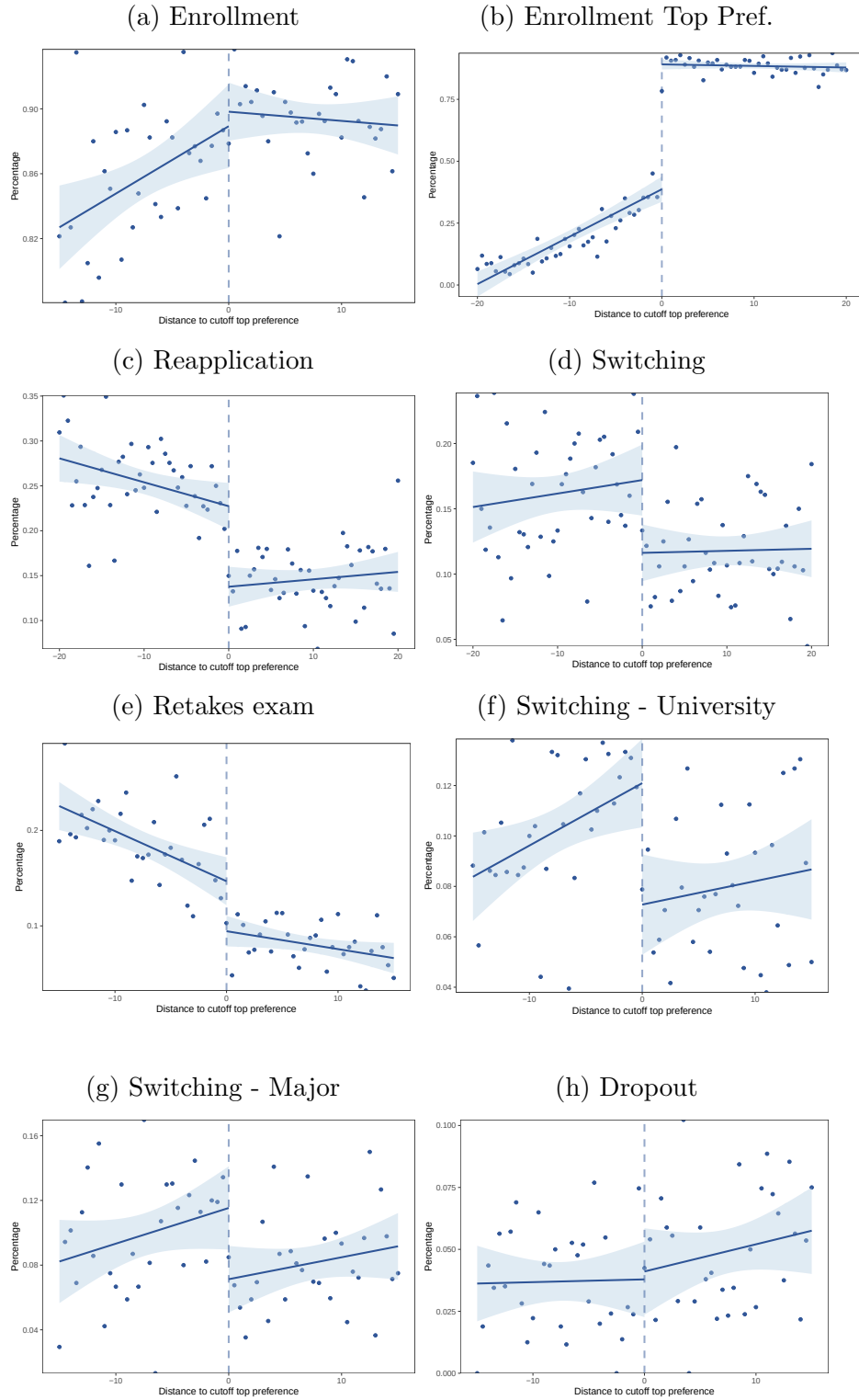
Note: Local-linear estimates with separate slopes on each side of the cutoff and heteroskedasticity-robust standard errors in parentheses. Panel A reports the estimated discontinuity in predetermined characteristics at the top-preference cutoff (bandwidth ± 15 points). Panel B re-estimates the reapplication and switching discontinuities of Table B4 across bandwidths. Significance levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Columns (3) and (4) in Table B4. Figure B1e shows a similar discontinuity in retaking the entrance exam: students just below the cutoff are about 5.7 percentage points more likely to retake it, a relative increase of roughly 60% over those just above. Figure B1g and Column (5) in Table B4 exhibit a similar pattern, as it shows that the probability of major-switching also decreases among students above the cutoff. Similarly, Column (6) shows that university switching is also significantly reduced. Finally, in Figure B1h, we show that there is no effect of exceeding the cutoff on dropout rates.

REGRESSION DISCONTINUITIES WITH TRUE PREFERENCES

Our previous analysis focuses on the effect of being above or below the cutoff of the top reported preference on different outcomes. Using the 2019 cohort and our nationwide survey, we can perform a similar analysis to estimate the causal effect of being above or below the cutoff of students' top-true preference on their outcomes. In Figures B2a and B2b, being above the cutoff significantly reduces the probability of reapplying to the system and being assigned to a different program the next year, consistent with those reported in Figure B1.

Figure B1. Effect of Cutoff Crossing



Note: Binned means as a function of the distance between the student's application score and the cutoff of their top-reported program; lines are local-linear fits within ± 15 points. Samples as in Table B4.

Table B4—Causal Effect of Crossing the Cutoff in Top Reported Preference

	Enroll (1)	Enroll Top Pref. (2)	Reapp. (3)	Switch (4)	Switch Major (5)	Switch University (6)	Dropout (7)
Z_{ip}	0.030* (0.018)	0.516*** (0.021)	-0.084*** (0.020)	-0.073*** (0.020)	-0.044*** (0.017)	-0.047*** (0.017)	0.007 (0.011)
Constant	0.868*** (0.014)	0.371*** (0.018)	0.228*** (0.015)	0.183*** (0.016)	0.114*** (0.013)	0.121*** (0.014)	0.035*** (0.008)
Mean	0.866	0.567	0.197	0.139	0.090	0.090	0.043
Observations	5,372	5,372	5,993	4,652	4,652	4,652	4,652

Note: Heteroskedasticity-robust standard errors in parentheses. Local-linear estimates with separate slopes on each side of the cutoff, using a bandwidth of 15 points around the top-preference cutoff.

Enrollment outcomes (Columns 1–2) are estimated on assigned applicants who listed at least two programs, re-application (Column 3) on all applicants who listed at least two programs, and switching and dropout (Columns 4–7) on those assigned and enrolled. See Table B3 for sensitivity to the bandwidth. Significance levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure B2. Effect of Cutoff Crossing for Top-True Preference

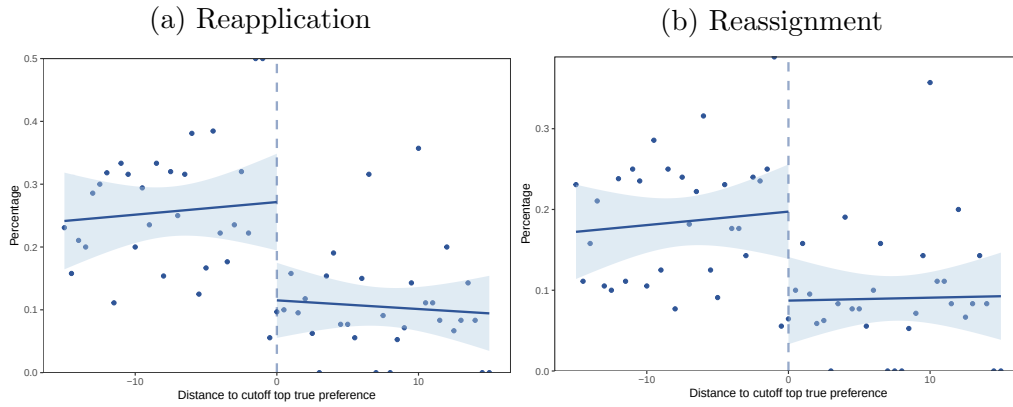


Figure B3. Students flow across states

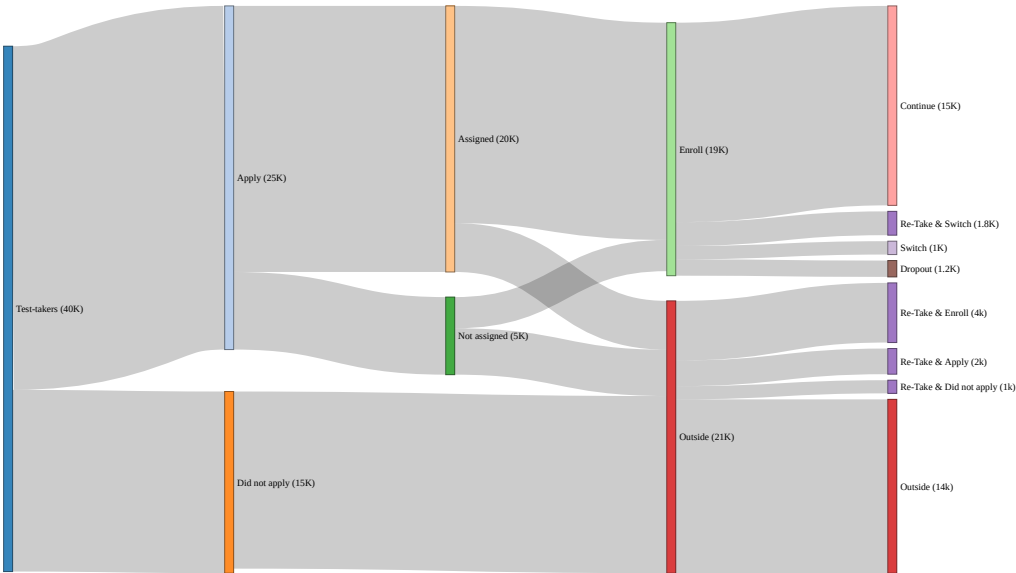
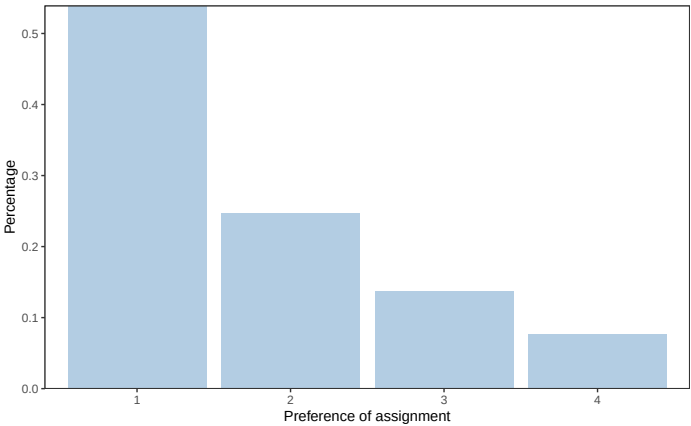


Figure B4. Distribution Year of First Switching

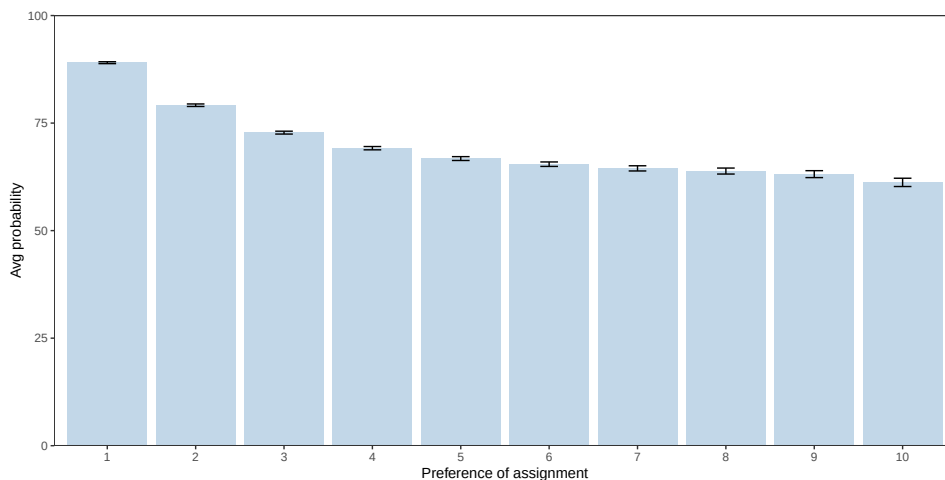


B4. Additional Evidence

PERCEIVED PERSISTENCE AND PREFERENCE OF ASSIGNMENT

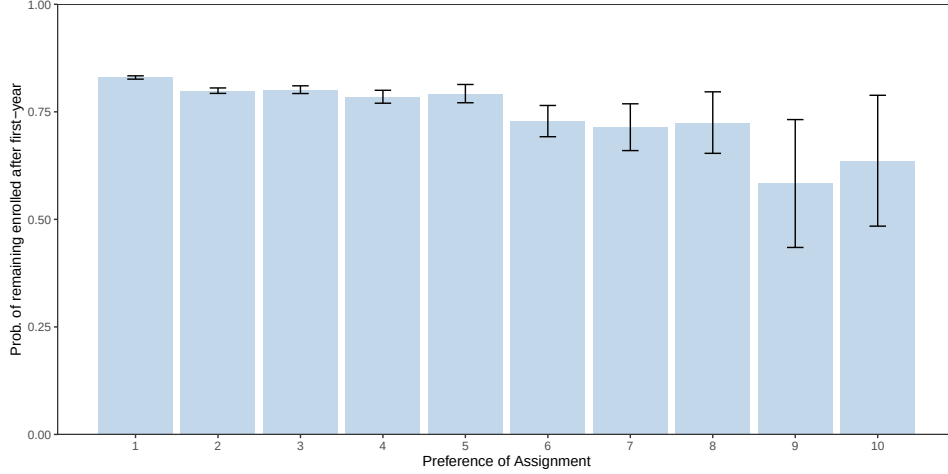
The regression discontinuity results show a causal effect of preference of assignment on students' persistence with respect to their initial assignment. To show that this is partially explained by the mismatch channel, we use the 2020 survey on students' preferences and beliefs. Figure B5 shows that the average “perceived” probability of remaining enrolled in the same program after the first year is significantly lower for lower-ranked preferences. On average, students believe that there is an 85% probability of staying enrolled in their top reported preference, whereas it is close to 65% for programs ranked below the fourth choice. Furthermore, comparing Figure B5 with Figure B6, which provides actual probabilities of remaining enrolled, shows evidence of forward-looking behavior and suggests that—on aggregate—students' subjective beliefs are close to being correct.

Figure B5. Average “perceived” probability of remaining enrolled in the same program, by preference of enrollment



The previous evidence does not guarantee that there are match effects that are correlated with college persistence. For instance, a similar pattern would arise if all students had identical preferences over programs—determined solely by program-level characteristics that are highly correlated with persistence—so that lower-ranked programs systematically have lower retention rates regardless of individual student-program matches. To rule out this alternative explanation and provide evidence of match effects, we exploit the panel structure of students' ROLs—we elicited perceived persistence probability for every program listed in

Figure B6. Actual probability of remaining enrolled in the same program, by preference of enrollment



the ROL—and estimate the following model:

$$\tilde{P}_{i,R_i(l)}^r = \psi_i^r + \psi_l^r + \psi_{R_i(l)}^r + \psi_X^r X_{i,R_i(l)}^r + \epsilon_{i,l}^r,$$

where R_i is student i 's ROL and $R_i(l)$ is the program listed in the l -th position; $\tilde{P}_{i,R_i(l)}^r$ is their perceived probability of remaining enrolled in program $R_i(l)$, with $l \in \{1, \dots, |R_i|\}$; ψ_i^r is students i 's fixed effect; ψ_l^r is position l 's fixed effect; ψ_j^r is program j 's fixed effect; $X_{i,j}^r$ are time-invariant observable characteristics of student i and program j (third-degree polynomial of the application score of student i in program j); and $\epsilon_{i,l}^r$ is an i.i.d shock.

In Table B5, we report the estimation results. We observe that the preference of enrollment has a significant and strong effect on the perceived probability of remaining enrolled, so we conclude that there are match effects in the setting, which exhibit a strong correlation with students' college persistence. Overall, these results show that (i) there is a significant effect of the preference of assignment on the switching behavior of students, (ii) a significant fraction of students correctly forecast this, and (iii) that match effects (at least) partially explain these patterns.

APPENDIX FOR SECTION IV

Proposition 6. Let $\{\tilde{\mu}_{i,j',t}, \tilde{\sigma}_{i,j',t}\}_{j' \in \mathcal{J}}$ be the sets of prior means and variances of student i 's unobserved ability in programs $j' \in \mathcal{J}$ at time t , respectively. In addition, let $j \in \mathcal{J} \cup \{\emptyset\}$ be the program of enrollment of student i in period t , and $a_{i,j,t}$ be the observed signal (if $j \in \mathcal{J}$). Then, the posterior mean of student

Table B5—Two-way Fixed Effects Regression Results

<i>Dependent variable: Prob. of Persistence</i>		
	Estimate	Std. Error
Preference 2	-9.891	8.3652e-03
Preference 3	-16.844	8.7019e-03
Preference 4	-21.355	9.3909e-03
Preference 5	-24.831	1.0631e-02
Preference 6	-27.148	1.1805e-02
Preference 7	-29.164	1.3202e-02
Preference 8	-30.329	1.5004e-02
Preference 9	-31.995	1.6876e-02
Preference 10	-34.757	1.9483e-02
Constant	89.181	1.0186e-02
Observations	159,894	
R ²	0.095	
Adjusted R ²	0.095	

Notes: Standard errors in parentheses. The table reports estimates from a two-way fixed effects regression of students' perceived probability of remaining enrolled in the same program on preference rank indicators, controlling for student fixed effects, program fixed effects, and a third-degree polynomial of the application score. The dependent variable is the perceived probability of remaining enrolled (0-100 scale) elicited from the 2020 survey.

i in program $j' \in \mathcal{J}$ is given by

$$\tilde{\mu}_{i,j',t+1} = \begin{cases} \tilde{\mu}_{i,j',t} + \frac{(a_{i,j,t} - \tilde{\mu}_{i,j,t})}{\tilde{\sigma}_{i,j,t}^2 + \sigma_G^2} \cdot \left[\tilde{\sigma}_{i,m(j),t}^2 + \sum_{s \in \mathcal{S}} \omega_{j's} \omega_{js} \tilde{\sigma}_{i,s,t}^2 \right] & \text{if } m(j') = m(j), j \in \mathcal{J} \\ \tilde{\mu}_{i,j',t} + \frac{(a_{i,j,t} - \tilde{\mu}_{i,j,t})}{\tilde{\sigma}_{i,j,t}^2 + \sigma_G^2} \cdot \left[\sum_{s \in \mathcal{S}} \omega_{j's} \omega_{js} \tilde{\sigma}_{i,s,t}^2 \right] & \text{if } m(j') \neq m(j), j \in \mathcal{J} \\ \tilde{\mu}_{i,j',t} & \text{if } j \notin \mathcal{J} \end{cases}$$

Proof of Proposition 6. Let W, X, Y, Z be independent random variables normally distributed with means $\mu_W, \mu_X, \mu_Y, \mu_Z$ and variances $\sigma_W^2, \sigma_X^2, \sigma_Y^2, \sigma_Z^2$, and let $A = \alpha_W W + \alpha_X X + \alpha_Y Y + \alpha_Z Z$. Using properties of the normal distribution, we know that A is also normally distributed with mean $\mu_A := \alpha_W \mu_W + \alpha_X \mu_X + \alpha_Y \mu_Y + \alpha_Z \mu_Z$ and variance $\sigma_A^2 := \alpha_W^2 \sigma_W^2 + \alpha_X^2 \sigma_X^2 + \alpha_Y^2 \sigma_Y^2 + \alpha_Z^2 \sigma_Z^2$. Moreover, we know that

$$\mathbb{E}[X \mid A = a] = \mu_X + \frac{\text{cov}(X, A)}{\sigma_X \cdot \sigma_A} \cdot \frac{\sigma_X}{\sigma_A} \cdot (a - \mu_A).$$

By definition, we know that

$$\begin{aligned}
\text{cov}(X, A) &= \mathbb{E}[X \cdot A] - \mathbb{E}[X] \cdot \mathbb{E}[A] \\
&= \mathbb{E}[X \cdot (\alpha_W W + \alpha_X X + \alpha_Y Y + \alpha_Z Z)] \\
&\quad - \mathbb{E}[X] \cdot \mathbb{E}[\alpha_W W + \alpha_X X + \alpha_Y Y + \alpha_Z Z] \\
&= \alpha_X \mathbb{E}[X^2] + \alpha_W \mathbb{E}[XW] + \alpha_Y \mathbb{E}[XY] + \alpha_Z \mathbb{E}[XZ] \\
&\quad - \alpha_X \mathbb{E}[X]^2 - \alpha_W \mathbb{E}[X] \mathbb{E}[W] - \alpha_Y \mathbb{E}[X] \mathbb{E}[Y] - \alpha_Z \mathbb{E}[X] \mathbb{E}[Z] \\
&= \alpha_X \mathbb{E}[X^2] - \alpha_X \mathbb{E}[X]^2 \\
&= \alpha_X \sigma_X^2
\end{aligned}$$

where the first equality is by definition; the second follows from the linearity of the expectation; the third follows from independence; and the fourth follows from the definition of the variance. Hence,

$$(C1) \quad \mathbb{E}[X \mid A = a] = \mu_X + \frac{\alpha_X \sigma_X^2}{\sigma_X \cdot \sigma_A} \cdot \frac{\sigma_X}{\sigma_A} \cdot (a - \mu_A) = \mu_X + \alpha_X \cdot \frac{\sigma_X^2}{\sigma_A^2} \cdot (a - \mu_A).$$

Next, let $\tilde{\mu}'_{\theta j'}(a) := \mathbb{E}[A_{\theta j'}^u \mid a_{ijt} = a]$, where we may have that $j \neq j'$ (if the student switches). Then, if $\theta(i) = \theta$,

$$\begin{aligned}
\tilde{\mu}'_{\theta j'}(a) &= \mathbb{E} \left[\tilde{A}_{\theta m_{j'}}^u + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \cdot \tilde{A}_{\theta k}^u + \varepsilon_{ij't+1} \mid a_{ijt} = a \right] \\
&= \mathbb{E} \left[\tilde{A}_{\theta m_{j'}}^u + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \cdot \tilde{A}_{\theta k}^u \mid a_{ijt} = a \right] \\
&= \mathbb{E} \left[\tilde{A}_{\theta m_{j'}}^u \mid a_{ijt} = a \right] + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \cdot \mathbb{E} \left[\tilde{A}_{\theta k}^u \mid a_{ijt} = a \right]
\end{aligned}$$

Note that, since the student knows their observable characteristics and their private abilities, the signal is such that

$$\begin{aligned}
a_{ijt} &= G_{ijt} - \gamma_{m_j} - \gamma_2 \cdot A_{ij}^o - A_{ij}^p \\
&= \tilde{A}_{\theta m_j}^u + \sum_{k \in \{s_m, s_v\}} \omega_{jk} \cdot \tilde{A}_{\theta k}^u + \varepsilon_{ijt} \\
&\sim N \left(\tilde{\mu}_{\theta m_j} + \sum_{k \in \{s_m, s_v\}} \omega_{jk} \cdot \tilde{\mu}_{\theta k}, \quad \tilde{\sigma}_{\theta m_j}^2 + \sum_{k \in \{s_m, s_v\}} \omega_{jk}^2 \cdot \tilde{\sigma}_{\theta k}^2 + \sigma_g^2 \right) \\
&\sim N \left(\mu_{\theta j}, \quad \tilde{\sigma}_{\theta j}^2 + \sigma_g^2 \right)
\end{aligned}$$

where $\tilde{\mu}_{\theta j} := \tilde{\mu}_{\theta m_j} + \sum_{k \in \{s_m, s_v\}} \omega_{jk} \cdot \tilde{\mu}_{\theta k}$ and $\tilde{\sigma}_{\theta j}^2 := \tilde{\sigma}_{\theta m_j}^2 + \sum_{k \in \{s_m, s_v\}} \omega_{jk}^2 \cdot \tilde{\sigma}_{\theta k}^2$.

Note that we have four independent random variables $\tilde{A}_{\theta m_j}^u, \tilde{A}_{\theta s_m}^u, \tilde{A}_{\theta s_v}^u$ and ε_{ijt} , and we have another random variable a_{ijt} given by the weighted sum of these four ones considering as weights $\{1, \omega_{js_m}, \omega_{js_v}, 1\}$. Hence, we can use (C1) to compute $\tilde{\mu}'_{\theta j'}$. Specifically, noticing that $\tilde{A}_{\theta m_{j'}}^u \perp \tilde{A}_{\theta m_j}^u, \tilde{A}_{\theta s_m}^u, \tilde{A}_{\theta s_v}^u, \varepsilon_{ijt}$ if $m_{j'} \neq m_j$ and $\tilde{A}_{\theta m_{j'}}^u \perp \tilde{A}_{\theta s_m}^u, \tilde{A}_{\theta s_v}^u, \varepsilon_{ijt}$ otherwise, we know that

$$\mathbb{E} \left[\tilde{A}_{\theta m_{j'}}^u \mid a_{ijt} = a \right] = \begin{cases} \tilde{\mu}_{\theta m_{j'}} & \text{if } m_j \neq m_{j'} \\ \tilde{\mu}_{\theta m_{j'}} + \frac{\tilde{\sigma}_{\theta m_{j'}}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \cdot (a - \tilde{\mu}_{\theta j}) & \text{if } m_j = m_{j'} \end{cases}$$

Similarly, for $k \in \{s_m, s_v\}$,

$$\mathbb{E} \left[\tilde{A}_{\theta k}^u \mid a_{ijt} = a \right] = \tilde{\mu}_{\theta k} + \omega_{jk} \cdot \frac{\tilde{\sigma}_{\theta k}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \cdot (a - \tilde{\mu}_{\theta j})$$

Combining these two equations we obtain that

$$\begin{aligned} \tilde{\mu}'_{\theta j'}(a) &= \mathbb{E} \left[\tilde{A}_{\theta m_{j'}}^u \mid a_{ijt} = a \right] + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \cdot \mathbb{E} \left[\tilde{A}_{\theta k}^u \mid a_{ijt} = a \right] \\ &= \begin{cases} \tilde{\mu}_{\theta m_{j'}} + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \left(\tilde{\mu}_{\theta k} + \omega_{jk} \cdot \frac{\tilde{\sigma}_{\theta k}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \cdot (a - \tilde{\mu}_{\theta j}) \right) & \text{if } m_j \neq m_{j'} \\ \tilde{\mu}_{\theta m_{j'}} + \frac{\tilde{\sigma}_{\theta m_{j'}}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \cdot (a - \tilde{\mu}_{\theta j}) + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \left(\tilde{\mu}_{\theta k} + \omega_{jk} \cdot \frac{\tilde{\sigma}_{\theta k}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \cdot (a - \tilde{\mu}_{\theta j}) \right) & \text{if } m_j = m_{j'} \end{cases} \\ &= \begin{cases} \tilde{\mu}_{\theta j'} + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \omega_{jk} \cdot \frac{\tilde{\sigma}_{\theta k}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \cdot (a - \tilde{\mu}_{\theta j}) & \text{if } m_j \neq m_{j'} \\ \tilde{\mu}_{\theta j'} + \left[\frac{\tilde{\sigma}_{\theta m_{j'}}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} + \sum_{k \in \{s_m, s_v\}} \omega_{j'k} \omega_{jk} \cdot \frac{\tilde{\sigma}_{\theta k}^2}{\tilde{\sigma}_{\theta j}^2 + \sigma_g^2} \right] \cdot (a - \tilde{\mu}_{\theta j}) & \text{if } m_j = m_{j'} \end{cases} \end{aligned}$$

C1. Implementation Details

Timing. For estimation, we consider a three-period model. Periods 1 and 2 correspond to the first and second years of college, while period 3 begins in the third year and aggregates all subsequent years until graduation, including discounted labor market payoffs. In period 1, students apply to colleges, receive their assignment, decide whether to retake the PSU, obtain first-year grades, and update their beliefs. In period 2, they may reapply and, given their current enrollment, new assignment, and what they learned earlier, decide whether to remain, switch programs, or drop out. In period 3, students face dropout and graduation probabilities and enter the labor market.

Heterogeneity. The learning process described above incorporates heterogeneity at the student level, allowing the learning parameters—related to private signals ($\varsigma_{i,k}$ for $k \in \mathcal{M} \cup \mathcal{S}$), unknown abilities ($\sigma_{i,k}$, $\tilde{\mu}_{i,k}$, $\tilde{\sigma}_{i,k}$ for $k \in \mathcal{M} \cup \mathcal{S}$), and preferences ($\mu_{i,m}^\alpha$, $\sigma_{i,m}^\alpha$, and $\zeta_{i,m}^\alpha$ for $m \in \mathcal{M}$)—to vary across students and majors/subjects. To ease estimation, we group majors into four *broad major*—

Science (Science, Farming, and Technology), Social Sciences (Social Sciences, Business and Management, Art and Architecture, and Law), Education and Humanities, and Health. Major-specific structural parameters vary at the broad major level, while random variables (e.g., major-specific abilities) vary at the *major* level. We also classify programs as *math intensive* (*verbal intensive*) if their math weight exceeds the verbal one (otherwise) and use the average math weight within each type, i.e., $\bar{\omega}_{s(j)} = \frac{\sum_{j' \in \mathcal{J}} \omega_{j', s_m} \cdot \mathbb{1}_{\{s(j')=s(j)\}}}{\sum_{j' \in \mathcal{J}} \mathbb{1}_{\{s(j')=s(j)\}}}$, where $s(j)$ indicates the type of program j . Then, the unobserved ability of student i in program j is $A_{ij}^u = A_{i, m(j)}^u + \bar{\omega}_{s(j)} A_{i, s_m}^u + (1 - \bar{\omega}_{s(j)}) A_{i, s_v}^u$.

Furthermore, we capture heterogeneity at the student-level using a small set of observable characteristics that are likely to influence the learning process and biases in students' beliefs. Specifically, we model each of the mean parameters mentioned above (i.e., $\tilde{\mu}_{i,k}$ for $k \in \mathcal{M} \cup \mathcal{S}$ and $\mu_{i,m}^\alpha$ for $m \in \mathcal{M}$) as linear functions of gender, income level, and their average ability, measured as the normalized average between the Math and Verbal entrance exams. We apply a similar approach to the logarithm of the variance parameters ($\varsigma_{i,k}$, $\sigma_{i,k}$ for $k \in \mathcal{M} \cup \mathcal{S}$ and $\sigma_{i,m}^\alpha$, $\varsigma_{i,m}^\alpha$ for $m \in \mathcal{M}$). For instance, for the true variance of unknown abilities, we assume $\log(\sigma_{i,k}^2) = \gamma_k^\sigma + \gamma_F^\sigma \cdot \text{female}_i + \gamma_{LI}^\sigma \cdot \text{low-income}_i + \gamma_A^\sigma \cdot \bar{A}_i$, where female_i , low-income_i are dummy variables for whether student i is female or their income below the median, respectively, and $\bar{A}_i = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} A_{i,s}^o$ is i 's average observable ability.²⁶

Functional Forms. We assume that the grade, flow utility, and log-wage functions (in (1), (2) and (4), respectively) are linear in their arguments, i.e.,

$$(C2a) \quad G(X_{i,j}^G, A_{i,j}^p, A_{i,j}^u, \epsilon_{i,j,t}^G) = \beta_X^G X_{i,j}^G + A_{i,j}^p + A_{i,j}^u + \epsilon_{i,j,t}^G,$$

$$(C2b) \quad U(X_{i,j}^U, \alpha_{i, m(j), t}, \epsilon_{i,j,t}^U) = \beta_X^U X_{i,j}^U + \alpha_{i, m(j), t} + \epsilon_{i,j,t}^U,$$

$$(C2c) \quad \log W(X_{i,j}^W, G_{i,j,t'}, \Lambda_{j,t-t'}) = \beta_X^W X_{i,j}^W + \beta_{G,i}^W G_{i,j,t'} + \Lambda_{m(j), t-t'} + \epsilon_{i,j,t}^W,$$

where $\Lambda_{m(j), t-t'} = \beta_{m(j),1}^\Lambda (t - t') + \beta_{m(j),2}^\Lambda (t - t')^2$. Furthermore, following our survey evidence, we assume that the effect of grades on wages, $\beta_{G,i}^W$, are heterogeneous across students and can be modeled as a random coefficient that is log-normally distributed with mean $\log(\mu_{W \sim G})$ and variance $\sigma_{W \sim G}^2$. Finally, note that for estimation, we use beliefs on expected wages and grades as opposed to their actual values, so we estimate (C2c) considering $\tilde{W}_{i,j,t}$, $\tilde{G}_{i,j,t}$ and adding a measurement error term. This approach, made possible by our rich survey data, allows us to relax the rational expectations assumption and enable us to more accurately capture the (potentially biased) information that students actually used when making their decisions (Wiswall and Zafar, 2015a,b). Furthermore, it al-

²⁶We use the average instead of the program-specific observed ability (multiplying each $A_{i,s}^o$ by the corresponding weight $\omega_{j,s}$) to avoid program-specific heterogeneity in defining students' types and simplify the estimation.

lows us to understand the role of biases (in both mean and variances) in affecting the dynamics of college admissions and how different interventions can mitigate these. The full set of controls entering each function is described in the *Controls* paragraph of Section IV.C, and Appendix C.C4 provides additional details.

C2. Additional Decisions

In addition to the initial application decision discussed in Section IV.C, students also make the following decisions:

ENROLLMENT.

Consider a student i who is enrolled in program j at the beginning of period $t = 2$ and decides to reapply, getting assigned to some program $j' \in \mathcal{J} \cup \{\emptyset\}$ as a result. Then, the expected utility of student i is given by

$$(C3) \quad \Psi_{i,j,t}(j') := (v_{i,j',t} - C_{i,t}^S) \cdot P_{i,t}^E + \max \{v_{i,j,t}, v_{i,\emptyset,t}\} \cdot (1 - P_{i,t}^E),$$

where $P_{i,t}^E$ is the enrollment probability of student i in period t ; $C_{i,t}^S$ is the cost of starting a new program; and $v_{i,j',t}$ is the expected utility that student i receives from being enrolled in program j' from period t onwards, which is given by

$$(C4) \quad v_{i,j,t} = \sum_{t'=t}^T \left[\delta^{t'-t+1} \left(\mathbb{E}[V_{i,j,t'}] \cdot \frac{P_{i,j,t'}^g}{P_{i,j,t'}^g + P_{i,j,t'}^d} + \mathbb{E}[V_{i,\emptyset,t'}] \cdot \frac{P_{i,j,t'}^d}{P_{i,j,t'}^g + P_{i,j,t'}^d} \right) + \sum_{\tau=t}^{t'} \delta^{\tau-t} \cdot \mathbb{E}[U_{i,j,\tau}] \right] \cdot P_{i,j,t'}^{g \vee d},$$

where $P_{i,j,t'}^g$ and $P_{i,j,t'}^d$ are the probabilities of graduating and dropping out from program j in period t' conditional on being enrolled, respectively, and $P_{i,j,t'}^{g \vee d} := (P_{i,j,t'}^g + P_{i,j,t'}^d) \cdot \prod_{\tau=1}^{t'-1} (1 - P_{i,j,\tau}^g - P_{i,j,\tau}^d)$ is the unconditional probability of student i graduating or dropping out from program j in period t' . The first term in (C4) captures the expected workforce utility of graduating from program j in period t' , while the second term captures the expected utility of dropping out in period t' . The third term captures the present value of the flow utility that student i receives from attending program j from period t to t' . Finally, if $j' = \emptyset$ (i.e., the student does not get assigned to a new program), we consider $\Psi_{i,j,t}(\emptyset) = \max \{v_{i,j,t+1}, v_{i,\emptyset,t+1}\}$, and with a slight abuse of notation, we define

$$\Psi_{i,j,t}(R) := \sum_{l=1}^{|R|} p_{i,R(l),t} \cdot \bar{p}_{i,R(l),t} \cdot \Psi_{i,j,t}(R(l)) + \prod_{l'=1}^{|R|} (1 - p_{i,R(l'),t}) \cdot \Psi_{i,j,t}(\emptyset),$$

where $p_{i,j,t}$ is the probability of getting assigned to program j in period t ; $\bar{p}_{i,R(l),t} = \prod_{l'=1}^{l-1} (1 - p_{i,R(l'),t})$ is the probability of not being assigned to any of the top $l-1$

programs listed in R , with $\bar{p}_{i,R(1),t} = 1$.²⁷ With a slight abuse of notation, we denote by $p_{j,t}$ the function that returns the CDF of the admission probability to program j in period t , i.e., $p_{j,t}(s)$ is the probability of admission given a score s .

RE-APPLICATION.

In period $t = 2$, each student i who is strategic and decides to re-apply solves the following optimal portfolio problem:

$$R_{i,j,t}^* \in \operatorname{argmax}_{R \subseteq \mathcal{R}_{i,j,t}} \left\{ \Psi_{i,j,t}(R) - C_{i,t}^A(R) \right\}$$

where

$$\mathcal{R}_{i,j,t} := \left\{ R \in \mathcal{R}(\mathcal{J} \setminus \{j\}) : |R| \leq K, v_{i,j',t} - C_{i,t}^S > v_{i,j,t}, p_{i,j',t} > 0, \forall j' \in R \right\}$$

is the set of ROLs that include at most K programs that have positive admission probability and that are strictly preferred to program j from period t onward, accounting for the cost $C_{i,t}^S$ that student i incurs upon starting a new program; and $C_{i,t}^A(R) = c|R|$ is the cost of applying to ROL R . With a slight abuse of notation, we say that $R_{i,j,t}^* = \{\emptyset\}$ if student i does not re-apply in period t . Note that students include programs in their ROL only if it is strictly profitable, so they will skip programs for which their chances are zero. This assumption is for simplicity, as it prevents from having multiplicity of optimal ROLs involving zero-chance programs.

EXAM RETAKING.

We assume that students endogenously decide whether to retake the entrance exam in period $t = 1$, weighing the expected benefits—e.g., increasing their chances of admission into a more preferred program—against the associated costs. This decision shapes the application behavior discussed above, which is conditional on a set of scores determining admission probabilities $\{p_{i,j,t}\}_{j \in \mathcal{J}}$. Formally, student i decides whether to retake the exam in period $t = 1$ by solving:

$$(C5) \quad \max_{x \in \{0,1\}} \left\{ \delta \mathbb{E} \left[\Psi_{i,j,t+1}(R_{i,j,t+1}^*) - C_{i,t+1}^A(R_{i,j,t+1}^*) \mid x \right] - x \cdot C_{i,t}^T \right\}.$$

The first term is the expected utility of re-applying in period $t + 1$, while the second is the cost of taking the exam. Importantly, the expectation is taken over the distribution of the signal about the student's unknown ability (learning about abilities), the preferences shocks and random coefficients (learning about preferences), as well as the distribution of test scores that the student receives, which is affected by the decision of taking the exam. Note that the latter affects

²⁷For simplicity, we assume that enrollment, graduation and dropout occur according to a semi-exogenous statistical process that depends on observable characteristics, as discussed in Appendix E.E4. Furthermore, we assume that students neglect potential correlations across cutoff distributions and that strategic students have correct beliefs about the marginal admission probabilities. Finally, note that (C4) correctly captures the indirect utility of enrolling in program j , as there are no decisions after period 2.

the admission probabilities considered in determining $R_{i,j,t+1}^*$ (with $R_{i,j,t+1}^* = \{\emptyset\}$ representing the option of not re-applying), evaluating $\Psi_{i,j,t+1}$, and also in the observed ability that student i considers for later outcomes. Finally, with a slight abuse of notation, let $x_{i,j}^*$ be the optimal solution of (C5) and $\Psi_{i,j}^T$ the corresponding value of the objective.

INITIAL APPLICATION.

The objects above determine the continuation value of the initial application—the term $\Psi_{i,R(l)}^*$ in Equation (5). Conditional on being assigned to program $j = R(l)$ in period $t = 1$, the student enrolls with probability $P_{i,1}^E$, in which case she receives the expected flow utility of her first year and the value of the exam-retaking problem (C5)—which embeds the subsequent re-application, switching, and dropout decisions, discounted to the first period, and whose expectation integrates over the grade signal and the evolution of preferences realized during the first year. With probability $1 - P_{i,1}^E$, she instead starts from the outside option and faces the analogous retaking and re-application problem of an unenrolled student. Hence, the value of being assigned to program j in the initial application is

$$\Psi_{i,j}^{\star,1} := P_{i,1}^E \cdot (\mathbb{E}[U_{i,j,1}] + \Psi_{i,j}^T) + (1 - P_{i,1}^E) \cdot (\mathbb{E}[U_{i,\emptyset,1}] + \Psi_{i,\emptyset}^T),$$

where $\Psi_{i,j}^T$ is the value of the retaking problem defined above for a student who begins period $t = 2$ enrolled in program j , and $\Psi_{i,\emptyset}^T$ is its analogue for a student who begins from the outside option. The value of not being assigned in the initial application is the second term alone, and the expected value of an initial ROL follows by weighting these assignment values with the admission probabilities, as in Equation (5).

C3. Continuation values.

The model discussed above involves computing the expectation of the continuation value that students receive after solving a series of complex decision problems—e.g., deciding to take the exam or solving an optimal portfolio problem—for each possible state of the world in each period t —summarized in $\omega_{i,t} := \{\vec{s}_{i,t}, a_{i,j,t}, \vec{\epsilon}_{i,j,t}, \vec{\alpha}_{i,t}\} \in \Omega$, which includes the scores obtained $\vec{s}_{i,t}$ conditional on taking the exam; the signal $a_{i,j,t}$ received; the random shocks $\vec{\epsilon}_{i,t}$, which comprise the grade shock $\epsilon_{i,j,t}^G$ and the preference shocks $\{\epsilon_{i,j',t}^U\}_{j' \in \mathcal{J} \cup \{\emptyset\}}$; and random coefficients $\vec{\alpha}_{i,t} = \{\alpha_{i,m,t}\}_{m \in \mathcal{M}}$. Such computation imposes several challenges that we address in the following way.

First, to avoid taking expectation over all preference shocks $\vec{\epsilon}_{i,t}^U = \{\epsilon_{i,j',t}^U\}_{j' \in \mathcal{J} \cup \{\emptyset\}}$, we adapt the results in [Fack, Grenet and He \(2019\)](#) to our dynamic setting. Specifically, [Fack, Grenet and He \(2019\)](#) show that, under mild assumptions, the

outcome of constrained DA satisfies pairwise stability relative to students' true preferences and, consequently, the optimal portfolio for student i in period t is equivalent to choosing the most preferred program $r_{i,j,t}^*(\omega)$ among those that are ex post feasible, i.e., those for which their application score $s_{i,j,t}(\omega)$ is above the program's cutoff $\bar{s}_{j,t}(\omega)$. Combining this result with (C3), we know that

$$\begin{aligned} \mathbb{E}_\omega [\Psi_{i,j,t}(R_{i,j,t}^*(\omega), \omega) \mid x] &= (\mathbb{E}_{\epsilon^U} [\bar{v}_{i,r^*(\omega),t+1} + \epsilon_{i,r^*(\omega),t}] - C_{i,t}^S) \cdot P_{i,t+1}^E \\ &\quad + \mathbb{E}_{\epsilon^U} [\max \{\bar{v}_{i,j,t+1} + \epsilon_{i,j,t}, \bar{v}_{i,\emptyset,t+1} + \epsilon_{i,\emptyset,t}\}] \cdot (1 - P_{i,t+1}^E) \end{aligned}$$

where $\bar{v}_{i,j',t+1} = \mathbb{E}_{\omega \setminus \epsilon^U} [v_{i,j',t+1} \mid x]$ is the expectation of the continuation values taken over all sources of randomness except of the preference shocks, and

$$r^*(\omega) = r_{i,j,t}^*(\omega) \in \underset{\substack{j' \in \mathcal{J} \setminus \{j\} \\ s_{i,j',t}(\omega) \geq \bar{s}_{j',t}(\omega)}}{\operatorname{argmax}} \left\{ \Psi_{i,j,t}(j', \omega) \right\}.$$

Leveraging the fact that the preference shocks ϵ are i.i.d. type-I extreme value, the expression in (C.C3) can be approximated as

$$\begin{aligned} \mathbb{E}_{\omega \setminus \epsilon^U} [\Psi_{i,j,t}(R_{i,j,t}^*(\omega), \omega) \mid x] &\approx (\kappa \log(\exp(\bar{v}_{i,r^*(\omega),t+1}/\kappa)) - C_{i,t}^S + \kappa\gamma) \cdot P_{i,t+1}^E \\ &\quad + (\kappa \log(\exp(\bar{v}_{i,j,t+1}/\kappa) + \exp(\bar{v}_{i,\emptyset,t+1}/\kappa)) + \kappa\gamma) \cdot (1 - P_{i,t+1}^E) \end{aligned}$$

where γ is Euler's constant and κ is the scale of the preference shocks. This approximation allows us to compute the expected utility of student i from program j in period t without having to integrate over all preference shocks, which would be computationally infeasible.

C4. Observable Characteristics

C5. Evidence Supporting Implementation Choices

APPENDIX FOR SECTION V

D1. Identifying Equations

GRADES.

The variance of the first-year grades is given by

$$\begin{aligned} \mathbb{V}(G_{i,j,1} \mid \theta(i) = \theta) &= \underbrace{\varsigma_{\theta,m(j)}^2 + \sum_{s \in \mathcal{S}} \omega_{j,s}^2 \cdot \sigma_{\theta,s}^2}_{:= \mathbb{V}(A_{ij}^p \mid \theta(i) = \theta)} + \underbrace{\sigma_{\theta,m(j)}^2 + \sum_{s \in \mathcal{S}} \omega_{j,s}^2 \cdot \sigma_{\theta,s}^2}_{:= \mathbb{V}(A_{ij}^u \mid \theta(i) = \theta)} + \underbrace{\sigma_G^2}_{:= \mathbb{V}(\epsilon_{ij,1}^G)} \end{aligned}$$

We use variation in the variances and covariance of the noisy signals that students who stayed enrolled in the same program received across periods; all moments

Table C1—Time-invariant observable characteristics included in the model

Equation	Coefficient	Description
Grades (β_X^G)	$\beta_{m(j)}^G$	Broad major-fixed effects
	β_A^G	Observed-ability ($A_{i,j}^o$)
Flow-utility (β_X^U)	β_i^U	Program-fixed effects
	β_A^U	Observed-ability ($A_{i,j}^o$)
	β_D^U	Distance between student i ; and program j 's municipality ($D_{i,j}$)
	β_Q^U	Distance to program mean quality ($\frac{A_{ij}^o - \bar{A}_j^o}{\sigma_j}$)
	β_C^U	Net cost (i.e., tuition - scholarships) that student i pays for program j ($C_{i,j}$)
Wage (β_X^W)	$\beta_{m(j)}^W$	Broad major-fixed effects
	β_W^W	Average wage posted at the moment of enrollment
	β_F^W	Female fixed effect
	β_{LI}^W	Low-income fixed effect
	β_G^W	Expected grades at graduation

Note: Broad majors include Science, Social Sciences, Education and Humanities and Health.

below implicitly condition on the student's type, their observable characteristics, and their private ability A_{ij}^p , which the signal nets out. Specifically, denote by $j_{i,1}$ and $j_{i,2}$ the programs of enrollment of student i in the first and second periods, respectively, and write “stay” for the event $j = j_{i,1} = j_{i,2}$. Staying is decided at the end of the first period, so it depends on the first-period signal $a_{i,j,1}$ (through updated beliefs) and on shocks that are independent of the unknown ability and of the second-period noise. Because $\epsilon_{i,j,2}^G$ is therefore unaffected by selection into staying,

$$(D1) \quad \mathbb{V}(a_{i,j,2} \mid \text{stay}) = \mathbb{V}(A_{ij}^u \mid \text{stay}) + \sigma_G^2.$$

The first-period signal admits no analogous decomposition: selection on $a_{i,j,1}$ distorts the conditional variance of $\epsilon_{i,j,1}^G$ and induces a covariance between A_{ij}^u and $\epsilon_{i,j,1}^G$ among stayers. We instead use the unconditional linear projection of the unknown ability on the first-period signal. By joint normality,

$$A_{ij}^u - \mathbb{E}[A_{ij}^u] = \lambda_a \cdot (a_{i,j,1} - \mathbb{E}[a_{i,j,1}]) + u_{ij}, \quad \lambda_a := \frac{\mathbb{V}(A_{ij}^u)}{\mathbb{V}(A_{ij}^u) + \sigma_G^2},$$

where u_{ij} is independent of $a_{i,j,1}$ with $\mathbb{V}(u_{ij}) = \lambda_a \cdot \sigma_G^2$. Since staying depends on the first-period signal only, together with shocks independent of u_{ij} and $\epsilon_{i,j,2}^G$, the distribution of u_{ij} is unaffected by selection. Taking conditional variances and

Table C2—Grades and Wages: Actual vs. Beliefs

	Grades				Wages
	Mean		Log-Variance		Mean
	Actual	Belief	Actual	Belief	Belief
Social Sciences	−0.012** (0.006)	0.011 (0.007)	0.081*** (0.016)	0.015 (0.039)	0.024 (0.016)
Education and Humanities	0.009 (0.009)	0.011 (0.009)	−0.187*** (0.026)	−0.004 (0.054)	−0.085*** (0.028)
Health	−0.025*** (0.007)	0.022*** (0.006)	−0.404*** (0.021)	0.006 (0.036)	−0.006 (0.016)
Female	0.061*** (0.005)	0.009* (0.005)	−0.251*** (0.013)	−0.071*** (0.026)	0.009 (0.012)
Low-Income	0.000 (0.005)	0.009** (0.005)	0.155*** (0.013)	−0.006 (0.026)	−0.034*** (0.011)
Observed Ability	0.083*** (0.004)	0.019*** (0.003)	−0.050*** (0.001)	−0.301*** (0.016)	0.078*** (0.007)
Omega Verbal	0.051 (0.033)	−0.000 (0.041)			−0.144 (0.097)
Omega Verbal Squared			0.652*** (0.067)	−0.151 (0.154)	
Avg. Log-Income					0.624*** (0.021)
Belief on Grades					0.038*** (0.007)
Constant	0.378*** (0.023)	0.618*** (0.029)	−0.345*** (0.035)	−1.777*** (0.084)	0.795*** (0.104)
Mean	0.523	0.64	−0.649	−1.885	2.668
Observations	15,710	10,423	11,024	7,198	13,351

Note: Standard errors in parentheses. The “Actual” columns report coefficients from regressions of realized outcomes on student characteristics estimated on the sample of students who enrolled in their top-reported preference. The “Belief” columns report coefficients from regressions of students’ stated beliefs (from survey data) on the same characteristics estimated on the sample of students who reported beliefs over their top-reported preference. Science is the omitted broad major category.

Significance levels: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

covariances of the projection then gives

$$(D2) \quad \mathbb{V}(A_{ij}^u \mid \text{stay}) = \lambda_a^2 \cdot \mathbb{V}(a_{i,j,1} \mid \text{stay}) + \lambda_a \cdot \sigma_G^2$$

and, since $a_{i,j,2} = A_{ij}^u + \epsilon_{i,j,2}^G$ with $\epsilon_{i,j,2}^G$ independent of $a_{i,j,1}$ and of the staying event, the covariance condition

$$(D3) \quad \text{Cov}(a_{i,j,1}, a_{i,j,2} \mid \text{stay}) = \lambda_a \cdot \mathbb{V}(a_{i,j,1} \mid \text{stay}).$$

Equation (D3) identifies the signal-to-noise ratio λ_a as the slope of a regression of second-period on first-period signals among stayers. Given λ_a , Equations (D1) and (D2) identify the grade noise variance, $\sigma_G^2 = (\mathbb{V}(a_{i,j,2} \mid \text{stay}) - \lambda_a^2 \cdot \mathbb{V}(a_{i,j,1} \mid \text{stay})) / (1 + \lambda_a)$, and hence the variance of the true unknown abilities, both conditional on staying (from Equation (D1)) and unconditional, $\mathbb{V}(A_{ij}^u) = \frac{\lambda_a}{1 - \lambda_a} \cdot \sigma_G^2$. In practice, the estimation exploits this variation through simulated moments, which integrate over the distribution of private abilities (known to students but not observed by the econometrician). The empirical counterparts of these moments, the variances of estimated grade-equation residuals among stayers and the coefficient on first-period grades in the second-period grade equation, are included in our moment vector (see Appendix E.E3).

WAGES.

Since $\log(\tilde{W}_{i,j,t}) = \beta_{G,i}^W \cdot \tilde{G}_{i,j,t'} + \beta_X^W X_{i,j}^W + \Lambda_{m(j),t-t'} + \epsilon_{i,j,t}^{W \sim M}$ —where the latter term is the measurement error of wages—, and we have modeled $\beta_{G,i}^W = \mu_{WG} \times \exp(\epsilon_i^{W \sim G})$, where $\epsilon_i^{W \sim G} \sim N(0, \sigma_{W \sim G}^2)$; the residual of this regression is given by

$$e_{i,j,t} = (\beta_{G,i}^W - \bar{\beta}_{W \sim G}) \tilde{G}_{i,j,t'} + \varepsilon_{i,j,t}^{W \sim M},$$

where the pooled slope is given by

$$\bar{\beta}_{W \sim G} \equiv \mathbb{E}[\beta_{G,i}^W] = \mu_{W \sim G} \exp\left(\frac{1}{2} \sigma_{W \sim G}^2\right).$$

Define $v_\beta \equiv \text{Var}(\beta_{G,i}^W) = \mu_{W \sim G}^2 (e^{2\sigma_{W \sim G}^2} - e^{\sigma_{W \sim G}^2})$. Assuming $\varepsilon_{i,j,t}^{W \sim M}$ is mean-zero, independent of $(\beta_{G,i}^W, \tilde{G}_{i,j,t'})$, and uncorrelated across programs, we obtain the system of moments:

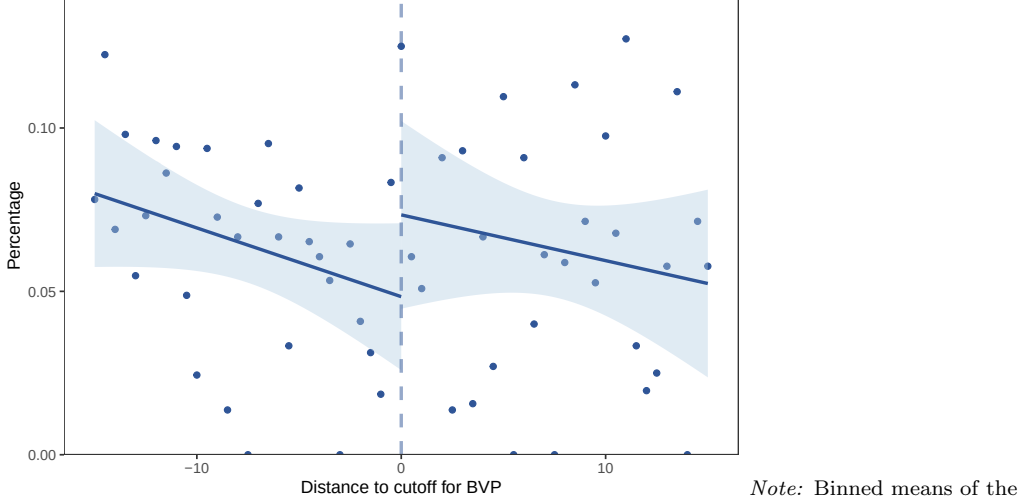
$$(D4) \quad \mathbb{E}[e_{i,j,t}^2] = v_\beta \cdot \mathbb{E}[\tilde{G}_{i,j,t'}^2] + \sigma_{W \sim M}^2,$$

$$(D5) \quad \mathbb{E}[e_{i,j,t} e_{i,j',t'}] = v_\beta \cdot \mathbb{E}[\tilde{G}_{i,j,t'} \tilde{G}_{i,j',t'}], \quad (j \neq j').$$

Thus, after identifying the pooled slope by OLS, we have three moments and three unknowns.

D2. Additional Evidence

Figure D1. Application to Education around Cutoff for BVP



share of applicants applying to an Education program as a function of the distance between their average Math–Verbal score and the 600-point BVP eligibility threshold; lines are local-linear fits with separate slopes on each side, with 95% confidence bands. Students in the BEA admission track are excluded.

APPENDIX FOR SECTION VI

E1. Estimator

We now introduce the estimator, closely following [Bruins et al. \(2018\)](#). Let $y_i := (y_{it}, \dots, y_{iQ})$ be a collection of Q outcomes for student i , and let $\mathbf{y} := \{y_i\}_{i=1}^N$ denote the aggregate outcomes of all students $i \in \{1, \dots, N\}$. Similarly, let x_i and $\mathbf{x}$ be individual and aggregate students' and programs' characteristics, and η_i and η be individual and aggregate random shocks. Let $\hat{\beta}_n$ be the vector of parameter estimates of the auxiliary model; that is,

$$\hat{\beta} := \operatorname{argmax}_{\beta} \mathcal{L}(\mathbf{y}, \mathbf{x}; \beta) = \operatorname{argmax}_{\beta} \frac{1}{N} \sum_{i=1}^N l(y_i, x_i; \beta),$$

where $l(\cdot; \beta)$ is the log-likelihood function given the vector β . Let $\eta^b := \{\eta_i^b\}_{i=1}^N$ denote a set of simulated draws for the random shocks of the structural model for simulations $s = 1, \dots, S$, where each set of draws is simulated independent of

each other. Let $\theta \in \Theta$ be a vector of parameters from the structural model, with $d_\theta \leq d_\beta$. Given the observable characteristics $\mathbf{x}$ and a parameter vector $\theta \in \Theta$, we can use the structural model to simulate data $\mathbf{y}^b(\theta) := \{y_i^b(\theta)\}_{i=1}^N$ and estimate the auxiliary model on each simulated sample:

$$\hat{\beta}^b(\theta) := \underset{\beta}{\operatorname{argmax}} \mathcal{L}(\mathbf{y}^b(\theta), \mathbf{x}; \beta).$$

Let $\bar{\beta}(\theta)$ be the average of these estimates, i.e., $\bar{\beta}(\theta) := \frac{1}{S} \sum_{b=1}^S \hat{\beta}^b(\theta)$. Then, the II estimator minimizes the following function:

$$Q(\theta) := \left(\bar{\beta}(\theta) - \hat{\beta} \right)^T W \left(\bar{\beta}(\theta) - \hat{\beta} \right)$$

where W is a positive-definite weighting matrix.

For a given value of the parameters θ , and given the first-stage estimates (students' beliefs and enrollment, dropout, graduation, and employment probabilities), computing the objective function $Q(\theta)$ involves solving the model via Backward Induction and then forward simulating outcomes.²⁸ Solving the model is computationally expensive and especially computing the continuation value terms, since they depend on realization of the random coefficients $\{\alpha_i\}_{i=1}^N$ (which are known to students), which restricts the number of draws of the random coefficients we can use to evaluate the objective function. To reduce the noise due to a small number of draws, we consider a larger number of draws for those shocks that do not affect the backward induction. The current set of estimation results uses 50 draws for shocks that do not affect the backward-induction procedure (preference shocks, enrollment shocks, etc.) and 5 draws for random coefficient shocks per student.²⁹ In Algorithm 1, we describe in detail how we perform the estimation and discuss some related technical considerations.

²⁸In which we have suppressed the dependency on the first-stage estimators for readability.

²⁹We have run robustness checks estimating the model with up to 10 realizations of random coefficient shocks per student, and estimation results are relatively similar.

Algorithm 1 Computing $Q(\theta)$

Input. Value of the structural parameters θ and first-stage estimates $\hat{p}$, $\hat{P}^e$, $\hat{P}^d$, $\hat{P}^g$, and $\hat{P}^w$.

Output. Value of the objective function $Q(\theta)$.

Step 1. For each student i , program j , and simulation b

Step 1.a. Draw a vector of random coefficients α_i^{mrc} ,

Step 1.b. Solve the model by Backward Induction,

Step 1.c. For each simulation in N_s and for each date, draw a vector of preference shocks $\varepsilon_i^{m_s, m_{rc}}$, enrollment shocks $\varepsilon_i^{e, m_s, m_{rc}}$, wage shocks $\epsilon_i^{m_s, m_{rc}}$, vector of random cutoff scores $P^{m_s, m_{rc}}$ from the empirical distribution of cutoffs, vector of PSU score shocks $\nu_i^{m_s, m_{rc}}$, vector of unknown abilities $A_i^{u, m_s, m_{rc}}$, and grade shocks $\varepsilon_i^{g, m_s, m_{rc}}$

Step 1.d. Forward simulate the model and obtain a set of outcomes $y_i^{m_s, m_{rc}}$,

Step 2. For each simulation, estimate the *auxiliary* model parameters, $\hat{\beta}^{m_s, m_{rc}}(\theta)$, on the simulated sample

Step 3. Compute $\bar{\beta}(\theta) = \frac{1}{N_{rc} \times N_s} \sum_{m_{rc}} \sum_{m_s} \hat{\beta}^{m_s, m_{rc}}(\theta)$

Step 4. Return $Q(\theta) := \left(\bar{\beta}(\theta) - \hat{\beta} \right)^T W \left(\bar{\beta}(\theta) - \hat{\beta} \right)$

E2. Estimation Moments Summary

Tables E1 and E2 provide a comprehensive summary of all moment conditions used in the estimation procedure, grouped by outcome type following the organization of our goodness-of-fit validation exercises.

E3. Auxiliary models

We use as an *auxiliary* model a combination of regression models—including data analogs of the grade equations, wage equations, linear probabilities models of graduation, linear probability models of switching and dropout, and RDD models—and a vector of moment conditions. The parameters of the model are identified jointly by the moment conditions generated by the auxiliary models. We describe now in detail the regressions and moment conditions we use in the estimation and the sets of parameters that explain most of each moment's variation.

Grades. The auxiliary models targeting the grade equations are:

$$\begin{aligned}
 G_{i,j,1} &= \beta_{m(j),1}^G + \beta_{A,1}^G A_{i,j}^o + \beta_{TP,1}^G \mathbb{1}_{\{j=R_i^1(1)\}} + \beta_{M,1}^G M_{i,m(j),1} + \epsilon_{i,j,1}^G \\
 G_{i,j,2} &= \beta_{m(j),2}^G + \beta_{A,2}^G A_{i,j}^o + \beta_{TP,2}^G \mathbb{1}_{\{j=R_i^2(1)\}} + \beta_{M,2}^G M_{i,m(j),2} + \beta_{CP,2}^G \mathbb{1}_{\{j_i^1=j_i^2\}} + \epsilon_{i,j,2}^G \\
 G_{i,j,2} &= \beta_{m(j),3}^G + \beta_{A,3}^G A_{i,j}^o + \beta_{TP,3}^G \mathbb{1}_{\{j=R_i^2(1)\}} + \beta_{M,3}^G M_{i,m(j),3} + \beta_{CP,3}^G \mathbb{1}_{\{j_i^1=j_i^2\}} \\
 &\quad + \beta_{a,3}^G a_{i,j_i^1,1} + \left(\beta_{CM,3}^G \mathbb{1}_{\{m(j_i^1)=m(j_i^2)\}} + \beta_{CT,3}^G \mathbb{1}_{\{s(j_i^1)=s(j_i^2)\}} \right) a_{i,j_i^1,1} + \epsilon_{i,j,3}^G
 \end{aligned}$$

where R_i^t is the ROLs of student i in periods $t \in \{1, 2\}$ (with $R_i^2 = R_i^1$ if student

Table E1—Estimation moments summary (Panel A)

Group	Moment description
<i>Application and Preference Moments</i>	Top true preference reported first in R_1 Top true preference reported first in R_2 Share of students who apply in period 1 Share of students who apply in period 2 Share changing broad major preference Share changing broad major from original order
<i>Additional Application and Assignment Characteristics</i>	Share unassigned in period 1 Share assigned to first reported preference Share assigned to second reported preference Share assigned to third reported preference Share assigned to top true preference Share of reapplicants assigned to top true preference Share with top true preference changed Student observed ability (top reported preference) Program observed ability (top reported preference) Tuition at top reported preference Distance to top reported preference Relative position at top reported preference Portfolio risk in R_1 Share of broad majors within R_1 (enrolled) Correlation between preference changes and grades (major) Correlation between preference changes and grades (broad major) Correlation between preference changes and grades (math type)
<i>Switching and Outcome Moments</i>	Share switching programs Share switching majors Share switching major within math (period 1) Share switching math within major (period 1) Share switching broad majors Share switching universities Share switching math types Share switching up (period 1) Share switching down (period 1) Share switching out (feasible) Share switching out (unfeasible) Share retaking the PSU Share dropouts Share of reapplicants Share of reapplicants assigned top reported preference Share enrolling for the first time in period 2 Share dropout at the end of period 1 Share first-year students in period 2 Share second-year students in period 2
<i>Market Structure Moments</i>	Major shares in R_1 (10 majors) Major shares in R_2 (10 majors) Major inclusion indicators in R_1 Major inclusion indicators in R_2 Broad major shares in R_1 Broad major shares in R_2 Market shares by major (period 1) Market shares by major (period 2) Market shares by broad major (period 1) Market shares by broad major (period 2)

Notes: Moments are grouped following the comprehensive goodness-of-fit tables used in the estimation and validation exercises.

Table E2—Estimation moments summary (Panel B)

Group	Moment description
<i>Portfolio and Reapplication Structure</i>	Norm difference of college-type shares for reapplicants Norm difference of major shares for reapplicants Norm difference of math-type shares for reapplicants Norm difference of ω shares for reapplicants Applications by major and college type (gender groups, R_1) Applications by major and college type (gender groups, R_2)
<i>Grade and Academic Performance</i>	Grade equation coefficients (year 1) Grade equation coefficients (year 2) Grade time-series coefficients (no switchers) Grade time-series coefficients (switchers) Grade SSR (year 1) Grade SSR (year 2) Residual variance of grades for stayers (years 1-2)
<i>Signal and Belief Moments</i>	Signal correlations with graduation and dropout Signal correlations with switching decisions Difference between subjective and realized grades (broad majors & demographics) Log variance of ability beliefs (realized) Log variance of ability beliefs (subjective) Learning regression on change in broad major norm
<i>Wage Regression and Expectations</i>	Wage regression coefficients (broad majors, ability, demographics) Wage regression residual diagnostics Expected log wage regression (top true vs random) Belief-based wage dispersion (dependent mass within quintile) Expected log wages: top true vs random assignment
<i>PSU Score Evolution</i>	Mean and variance of log score ratios (Language, Math) Mean and variance of log score ratios (History, Science takers) Mean and variance of log score ratios (History, Science non-takers)
<i>RDD Auxiliary Moments</i>	RDD switching intercepts RDD switching discontinuities RDD reapplication intercepts RDD reapplication discontinuities RDD switch up/out intercepts RDD switch up/out discontinuities RDD BVP intercepts RDD BVP discontinuities

Notes: Moments are grouped following the comprehensive goodness-of-fit tables used in the estimation and validation exercises.

i does not reapply); j_i^t is the program of enrollment of student i in period t ; $M_{i,m(j),t} = \frac{\sum_{l=1}^{|R_i^t|} \mathbb{1}_{\{m(R_i^t(l))=m(j)\}}}{|R_i^t|}$ is the fraction of programs in student i 's ROL in period t that share the same major as program j ($m(j)$); $a_{i,j_i^1,1}$ is the first-year grade signal of student i in program j_i^1 ; and $s(j)$ captures the type of program j (i.e., whether or not it is math intensive). We also estimate autoregressions of second-period grades on first-period grades separately for students who switch programs and those who do not, to capture grade persistence dynamics. Additionally, we use the sum of squared residuals (SSR) and residual variances from the grade equations as auxiliary moments.

Wages. The auxiliary models targeting the initial wage equation is:

$$\begin{aligned} \log(\tilde{W}_{i,j}) = & \beta_{m(j),1}^W + \beta_{A,1}^W A_{i,j}^o + \beta_{TP,1}^W \mathbb{1}_{\{j=R_i^1(1)\}} \\ & + \beta_{G,1}^W \tilde{G}_{i,j} + \beta_{\bar{W},1}^W \bar{W}_{m(j)} + \beta_{F,1}^W fem_i + \beta_{LI,1}^W li_i + \epsilon_{i,j}^W \end{aligned}$$

where $\tilde{W}_{i,j}$ and $\tilde{G}_{i,j}$ are the subjective wage and grade reported by student i for program j (both elicited from the survey); $\bar{W}_{m(j)}$ is the average wage of graduates from major $m(j)$ after four years of graduation (same horizon as that considered in the survey); fem_i is a dummy variable that equals one if student i is a female student; and li_i is a dummy variable that equals one if student i 's family income is below the median of the income distribution.

To target the parameters related to the evolution of wages with tenure, we use the following auxiliary model:

$$\bar{W}_{j,\tau} = \beta_{m(j),1}^\Lambda + \beta_{m(j),2}^\Lambda \tau + \beta_{m(j),3}^\Lambda \tau^2 + \epsilon_{j,\tau}^\Lambda,$$

where $\tau > 4$ is the number of years after graduation; and $\bar{W}_{j,\tau}$ is the average wage. We use the latter instead of beliefs on wages because we only elicited beliefs at the fourth year after graduation.

Nonpecuniary utilities. The auxiliary model targeting the non-pecuniary payoffs is:

$$grad_{i,j} = \beta_{m(j)}^U + \beta_A^U A_{i,j}^o + \beta_{TP}^U \mathbb{1}_{\{j=R_i^1(1)\}} + \beta_M^U M_{i,m(j)} + \beta_G^U G_{i,j,1} + \epsilon_{i,j}^U,$$

where $grad_{i,j} = \mathbb{1}_{\{j_i^g=j_i^1\}}$, with $j_i^g \in \mathcal{J} \cup \{\emptyset\}$ being the program of graduation of student i and j_i^1 being the program of enrollment of student i in their first year; the regression is estimated over realized first-year enrollments, i.e., $j = j_i^1$.

Learning. The auxiliary models targeting students' learning process are:

$$y_{i,j}^o = \beta_{m(j),1}^o + \beta_{A,1}^o A_{i,j}^o + \beta_{TP,1}^o \mathbb{1}_{\{j=R_i^1(1)\}} + \beta_{M,1}^o M_{i,m(j),1} + \beta_{a,1}^o a_{i,j_i^1,1} + \epsilon_{i,j,1}^o$$

where $y_{i,j}^o$ equals 1 if outcome o holds at the end of the first year, where o is

switching program, dropout, among others; the regression is estimated over realized first-year enrollments ($j = j_i^1$); and $a_{i,j_i^1,1}$ is the first-year grade signal of student i in program j_i^1 .

To separately identify learning about abilities from learning about preferences, we estimate regressions of preference changes on first-year grade signals. These regressions help distinguish belief updating based on grade information (ability learning) from shifts in tastes (preference learning):

$$\mathbb{1}\{\Delta\text{Pref}_i > \tau\} = \beta_{m(j_i^1)}^{\Delta p} + \beta_{A,1}^{\Delta p} A_{i,j_i^1}^o + \beta_{F,1}^{\Delta p} fem_i + \beta_{LI,1}^{\Delta p} li_i + \beta_a^{\Delta p} a_{i,j_i^1,1} + \varepsilon_i^{\Delta p},$$

where $\mathbb{1}\{\Delta\text{Pref}_i > \tau\}$ equals one if the change in the norm of student i 's reported preferences between the initial application and reapplication exceeds threshold τ , and zero otherwise; $a_{i,j_i^1,1}$ is the first-year grade signal of student i in the initial program of enrollment j_i^1 ; $A_{i,j_i^1}^o$ is student i 's observed ability in that program; fem_i and li_i are indicators for female and low-income students. The coefficient $\beta_a^{\Delta p}$ captures how preference changes correlate with grade signals, while the major fixed effects $\beta_{m(j_i^1)}^{\Delta p}$ capture baseline preference drift that is independent of grade information.

We use the following auxiliary models to separately target the parameters related to subjective and actual variance of the unknown ability:

$$\log(\tilde{\sigma}_{i,j}^2) = \beta_{m(i)}^{\tilde{\sigma}} + \beta_{A,1}^{\tilde{\sigma}} A_{i,j}^o + \beta_{F,1}^{\tilde{\sigma}} fem_i + \beta_{LI,1}^{\tilde{\sigma}} li_i + \omega_{j,s_v}^2 \beta_{s_v}^{\tilde{\sigma}} + \epsilon_{i,j}^{\tilde{\sigma}},$$

$$\log(\sigma_{i,j}^2) = \beta_{m(i)}^{\sigma} + \beta_{A,1}^{\sigma} A_{i,j}^o + \beta_{F,1}^{\sigma} fem_i + \beta_{LI,1}^{\sigma} li_i + \omega_{j,s_v}^2 \beta_{s_v}^{\sigma} + \epsilon_{i,j}^{\sigma},$$

where $\log(\tilde{\sigma}_{i,j}^2)$ is the log of the subjective variance of student i 's prior beliefs about unknown abilities in program j , $\log(\sigma_{i,j}^2)$ is the log of the actual variance, and ω_{j,s_v}^2 is the squared weight that program j places in the verbal test.

Bias. The auxiliary models targeting the mean bias are:

$$\begin{aligned} G_{i,j,t} &= \left(\beta_{m(i),1}^{\tilde{\mu}} + \beta_{A,1}^{\tilde{\mu}} A_{i,j}^o + \beta_{F,1}^{\tilde{\mu}} fem_i + \beta_{LI,1}^{\tilde{\mu}} li_i \right) \\ &\quad + \omega_{j,s_m} \left(\beta_{s_m,1}^{\tilde{\mu}} + \beta_{A,2}^{\tilde{\mu}} A_{i,j}^o + \beta_{F,2}^{\tilde{\mu}} fem_i + \beta_{LI,2}^{\tilde{\mu}} li_i \right) + \epsilon_{i,j,1}^{\tilde{\mu}}, \\ \tilde{G}_{i,j,t} &= \left(\beta_{m(i),2}^{\tilde{\mu}} + \beta_{A,3}^{\tilde{\mu}} A_{i,j}^o + \beta_{F,3}^{\tilde{\mu}} fem_i + \beta_{LI,3}^{\tilde{\mu}} li_i \right) \\ &\quad + \omega_{j,s_m} \left(\beta_{s_m,2}^{\tilde{\mu}} + \beta_{A,4}^{\tilde{\mu}} A_{i,j}^o + \beta_{F,4}^{\tilde{\mu}} fem_i + \beta_{LI,4}^{\tilde{\mu}} li_i \right) + \epsilon_{i,j,2}^{\tilde{\mu}}. \end{aligned}$$

Unobserved preferences. The auxiliary models targeting unobserved and persistent preferences are:

$$y_{i,j}^o = \beta_{1,2}^o (s_{i,j,1} - \bar{s}_{j,1}) + \beta_{2,2}^o \mathbb{1}_{\{(s_{i,j,1} - \bar{s}_{j,1}) > 0\}} + \beta_{3,2}^o (s_{i,j,1} - \bar{s}_{j,1}) \mathbb{1}_{\{(s_{i,j,1} - \bar{s}_{j,1}) > 0\}} + \epsilon_{i,j,2}^o$$

where $s_{i,j,1}$ is the application score of student i in program j in the first period; and $\bar{s}_{j,1}$ is the cutoff of program j in the first period. Note that we estimate these equations (for each outcome) only considering students whose application score is around the cutoff of their top reported preference.

E4. Exogenous Components

ADMISSION PROBABILITIES

To compute the admission probabilities, we use a bootstrap procedure similar to that in [Agarwal and Somaini \(2018\)](#). Specifically, we perform 1,000 bootstrap simulations, each consisting of the following steps:

- 1) Sample with replacement a number of applicants equal to the total number of students that applied in the admissions process.
- 2) Run the assignment mechanism used in the Chilean system (see [Rios et al. \(2021\)](#)).
- 3) Compute the cutoff of each program for both the regular and BEA admission tracks.

As a result of this procedure, we obtain two matrices (for the regular and BEA processes) with 1,000 cutoffs for each program. The next step is to estimate the distribution of the cutoff of each program in each admission track. To accomplish this, we estimate the parameters of a truncated normal distribution for each program and admission track via maximum likelihood. Then, using the estimated distributions, we evaluate the CDF on the application score of the student to obtain an estimate of the admission probability, taking into account whether the student participates only in the regular process or also in the BEA track.

As discussed in Section [IV.C](#), these estimated admission probabilities are considered by strategic students when making their application decisions. In contrast, we assume that truth-tellers consider an admission probability equal to one for all programs and continuation values obtained assuming pairwise stability. Although there is evidence that students have biased beliefs and make applications mistakes ([Larroucau et al., 2025](#)), our counterfactuals do not aim to correct these and, instead, focus on the dynamic aspects of the mechanism, re-application rules, and the information environment over future outcomes. For this reason, we abstract from this for simplicity and assume correct beliefs.

ENROLLMENT

As discussed in Section [C.C2](#), we assume for simplicity that the decision to enroll in a newly assigned program is exogenously given, while the decision to remain in the current program or take the outside option conditional on not enrolling in the newly assigned program is endogenous. Specifically, our model assumes that

students enroll in their newly assigned program with some probability $P_{i,t}^e$ that depends on a vector of time-invariant observable characteristics X_i^e as follows:

$$P_{i,t}^e = \frac{\exp(X_i^e \psi^e)}{1 + \exp(X_i^e \psi^e)}.$$

For estimation, we include in X_i^e a constant, student's gender, a dummy variable that identifies if the student's family income is below the median of the income distribution, and student's High-school GPA.

Note that we only incorporate students' observable characteristics in the estimation of enrollment probabilities. This is because these probabilities multiply the continuation values as shown in Equation C3, so making them program specific would not allow us to remove those from the expectation and, more importantly, it would break the type I extreme value distribution of shocks and, consequently, the inclusive value formulas would not hold, preventing us from using the approach discussed in Section C.C3.

DROPOUT AND GRADUATION

We assume that graduation and dropout after period 2 are exogenously determined. Specifically, we assume that conditional on being enrolled in program j for the last τ periods, student i 's probability of graduating or dropping out depends on vectors of time-invariant observable characteristics $X_{i,j}^{g\vee d}$, which may include characteristics of student i , program j , and the interaction between their characteristics. Furthermore, we estimate these probabilities using a multinomial logit model, as follows:

$$P_{i,j,\tau}^g = \frac{\exp(X_{i,j}^{g\vee d} \psi_\tau^g)}{1 + \sum_{a \in \{g,d\}} \exp(X_{i,j}^{g\vee d} \psi_\tau^a)}, \quad \text{and} \quad P_{i,j,\tau}^d = \frac{\exp(X_{i,j}^{g\vee d} \psi_\tau^d)}{1 + \sum_{a \in \{g,d\}} \exp(X_{i,j}^{g\vee d} \psi_\tau^a)}$$

where $P_{i,j,\tau}^g$ and $P_{i,j,\tau}^d$ represent the probability of student i graduating or dropping out from program j in after τ periods of enrollment, respectively; and ψ_τ^g, ψ_τ^d are vectors of parameters that need to be estimated. For estimation, we include in $X_{i,j}^{g\vee d}$ a constant, student's gender, a dummy variable that identifies whether the student's family income is below the median of the income distribution, and student's high school GPA. Note that we do not include students' GPA during college for two reasons: (i) we only have data for their first two years, so it may not be representative of their grades upon graduation / dropout; and (ii) to keep consistency in the observable characteristics considered with those in the enrollment probabilities.

Conditional on dropping out—or never enrolling—students enter an absorbing outside state. Because we do not observe labor-market outcomes for students who leave the centralized system, we model the value of this state as a function

of observable characteristics: from the period of exit onward, students receive a constant flow utility $\alpha_i^{\text{dropout}}$ —linear in gender, income, and observed ability—discounted until the terminal period. The associated parameters are reported in Table E3.

EVOLUTION OF SCORES.

Since students can retake the test before re-applying, we need to model the evolution of (i) their scores, and (ii) their beliefs about the weights that programs will use in the future. For simplicity, we assume that students have correct beliefs about these weights,³⁰ while we assume that student i 's scores in period $t + 1$ (conditional on re-taking the test), $\vec{s}_{i,t+1}$, are exogenously given and evolve depending on their initial scores according to the following process:

$$s_{i,k,t+1} = (\psi_k^s + \nu_{i,t+1}) \cdot s_{i,k,t} + \mathbb{1}_{\{s_{i,k,t}=0\}} (\psi_{k,0}^s + \nu_{i,t+1}) \cdot \bar{s}_{i,t},$$

where $s_{i,k,t}$ is the score of student i in subject k in period t ; $\bar{s}_{i,t}$ is the average Math-Verbal score of student i in period t ; $\nu_{i,t+1} \sim N(0, \sigma_s^2)$ is a random shock common across subjects; and $\{\psi_k^s, \psi_{k,0}^s\}_k$ and σ_s^2 are estimated jointly with the structural model by matching the distribution of score changes among retakers. The process conditions only on students' initial scores; in particular, the selectivity of the program a retaker attends in the interim adds no explanatory power for score gains conditional on these.

E5. MIA

Chade and Smith (2006) show that, when admission probabilities are independent and the cost of submitting ROL R only depends on its cardinality—i.e., $C_i^A(R) = c(|R|)$ for some function c —the unconstrained problem is Downward Recursive, and the optimal solution is given by the MIA algorithm presented next.

MIA: Marginal Improvement Algorithm

- Initialize $S_0 = \emptyset$.
- Select $j_n = \arg \max_{j \in M \setminus S_{n-1}} \{U(S_{n-1} \cup j)\}$.
- If $U(S_{n-1} \cup j_n) - U(S_{n-1}) < c(S_{n-1} \cup j_n) - c(S_{n-1})$, then STOP.
- Set $S_n = S_{n-1} \cup j_n$.

Olszewski and Vohra (2016) show that MIA also returns the optimal ROL when the number of applications is constrained and when $c(S)$ is supermodular. Note that, if some programs have zero-admission chance, the strict inequality in MIA's

³⁰This assumption is likely to hold in practice, since admission weights are relatively stable over time.

stopping criteria becomes a weak inequality, which could lead to multiplicity of best responses (He (2012)). We discuss this potential identification threat in Larroucau and Rios (2018).

E6. *Weighting Matrix and Standard Errors*

WEIGHTING MATRIX

The weighting matrix W is a diagonal matrix that combines two components. First, moments are grouped by outcome type, and each group receives a fixed share of the total weight; the largest shares go to the groups that identify the learning and initial-mismatch channels (directional switching and its correlations with grade signals, the regression-discontinuity coefficients, and assignment to the top-true preference) and to those pinning down the market shares of majors. Second, within each group, weights are proportional to the inverse of each moment's bootstrap variance, so noisier moments contribute less. We do not use the optimal weighting matrix because of the numerical complexities involved in computing the derivatives of the objective function $Q(\theta)$. Therefore, our estimator will be consistent but not efficient.

STANDARD ERRORS

We compute standard errors for 145 out of 165 structural parameters using a simulated method of moments (SMM) framework with finite-simulation correction. The estimation proceeds in three steps. First, we construct the Jacobian matrix G by computing numerical derivatives of all moment conditions with respect to free parameters at the estimated optimum $\hat{\theta}$, using parallel finite-difference approximations with common random numbers to reduce simulation noise. Second, we build the asymptotic variance of moments Σ by combining the data moment covariance matrix Ω (a diagonal matrix with the bootstrap variances of the data moments) with a finite-simulation correction factor. Third, we compute standard errors from the sandwich formula $\widehat{\text{Var}}(\hat{\theta}) = N_{\text{data}}^{-1}(G'WG)^{-1}G'W\Sigma WG(G'WG)^{-1}$, which properly accounts for both sampling variation in the data and simulation error. This formula does not account for the effect of first-stage estimation error.

E7. *Results*

APPENDIX FOR SECTION VII

F1. *Additional Details*

CADA.

We choose to implement CADA only in the first period to avoid solving for the continuation values under this mechanism, which would add a high computational burden to the model. To implement this mechanism, we need to specify

Table E3—Estimation Results - Main Structural Parameters

Parameter	Value	Std. Error
<i>Application behavior and Dropout</i>		
Share of strategic ROLs ($1 - \rho$)	0.88	0.04
Cost of retaking PSU (C^T)	21.84	< 0.01
Dropout flow-utility for females ($\alpha_{female}^{dropout}$)	38.64	0.10
Dropout flow-utility for males ($\alpha_{male}^{dropout}$)	-12.73	0.05
Dropout flow-utility for low-income ($\alpha_{low-income}^{dropout}$)	34.64	0.20
Dropout observed ability effect ($\alpha_{obs}^{dropout}$)	9.40	< 0.01
First-time enrollment cost (C^S)	206.55	< 0.01
<i>Flow-utility</i>		
Tuition (β_3^U)	-3.28	0.16
Relative position (β_2^U)	1.35	0.17
Distance (β_1^U)	-1.00	(-)
Student observed ability (β_4^U)	13.46	0.08
Female effect by major (Δ_m female)	(13.85, 15.71, 10.45, 9.68)	(0.11, 0.12, < 0.01, 0.07)
Low-income effect by major (Δ_m low-income)	(-19.02, -31.31, 0.69, 31.90)	(0.01, 0.01, < 0.01, 0.14)
Avg. observed ability effect by major (Δ_m avg. obs. ability)	(10.42, 20.98, -1.68, 19.48)	(0.05, 0.10, < 0.01, 0.09)
Variance major random coefficient (σ_α^2)	1134.98	< 0.01
Scale gumbel parameter (κ)	17.78	0.21
<i>Grade equations</i>		
Constant by major ($\beta_{m,j}^G$)	(4.03, 4.54, 4.41, 4.00)	(0.25, 0.22, 0.25, 0.24)
Student observed ability (β_1^G)	0.90	0.14
Variance grade shock (σ_g^2)	0.11	0.02
<i>Evolution of scores</i>		
Std. of ν (σ_s)	0.08	0.23
Mean prop. change ($\{\psi_k^s\}_k$)	(1.04, 1.02, 1.03, 1.03)	(0.13, 0.18, 0.26, 0.20)
Mean prop. change from zero ($\{\psi_{k,0}^s\}_k$)	(1.05, 1.02)	(0.05, 0.04)

Notes: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator. Parameters from external regressions are indicated by (-).

Table E4—Estimation Results - Work Utility and Wage Parameters

Parameter	Value	Std. Error
<i>Non-pecuniary work utility</i>		
Random coefficient by major (β_1^V)	6.43	0.12
Observed ability (β_2^V)	311.54	< 0.01
Unknown and private ability by major (β_3^V)	(553.65, 571.55, 741.39, 691.94)	(0.03, < 0.01, < 0.01, 0.02)
Major fixed effects ($\beta_{m_j}^V$)	(-3.40, 44.45, 25.84, -9.23)	(< 0.01, 0.01, < 0.01, < 0.01)
<i>Wage parameters</i>		
Constant by major ($\beta_{m_j}^W$)	(-0.08, 0.08, -0.17, -0.10)	(0.20, 0.20, 0.19, 0.20)
Grades (β_G^W , mean)	0.08	0.07
Gender effects (β_1^W)	-0.01	0.21
Wage low income effect (β_2^W)	-0.02	0.24
Avg. posted wage (β_3^W)	0.56	0.13
Variance of wage grades effect ($\sigma_{W \sim G}^2$)	0.43	0.07
Measurement error log wages variance (σ_W^2)	0.27	0.19
Variance wage shock (σ_w^2)	0.09	0.02
<i>Wage growth</i>		
Linear term by major ($\beta_{m_j,1}^A$)	(0.23, 0.16, 0.15, 0.27)	(-)
Quadratic term by major ($\beta_{m_j,2}^A$)	(-0.01, -0.01, -0.01, -0.02)	(-)

Notes: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator. Wage growth parameters are fixed from auxiliary regressions (indicated by (-)).

Table E5—Estimation Results - Subjective Prior Means Bias

Major Unknown Ability			Subject Unknown Ability		
Parameter	Value	Std. Error	Parameter	Value	Std. Error
<i>By major and subject type</i>					
Science ($\tilde{\mu}_{m,0}$)	0.63	(-)	Verbal type ($\tilde{\mu}_{s,v}$)	-0.05	(-)
Social Sciences ($\tilde{\mu}_{m,1}$)	0.66	(-)			
Education and Humanities ($\tilde{\mu}_{m,2}$)	0.36	(-)			
Health ($\tilde{\mu}_{m,3}$)	0.84	(-)			
<i>Covariates</i>					
Low income ($\tilde{\mu}_{m,low}$)	0.08	(-)	Low income ($\tilde{\mu}_{s,low}$)	0.00	(-)
Female ($\tilde{\mu}_{m,f}$)	-0.19	(-)	Female ($\tilde{\mu}_{s,f}$)	0.00	(-)
Avg. obs. ability ($\tilde{\mu}_{m,\bar{a}}$)	-0.21	(-)	Avg. obs. ability ($\tilde{\mu}_{s,\bar{a}}$)	0.00	(-)

Notes: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator. Parameters from external regressions are indicated by (-).

Table E6—Estimation Results - Log-Variances: True vs Subjective Bias

True Log-Variances			Subjective Bias Log-Variances		
Parameter	Value	Std. Error	Parameter	Value	Std. Error
<i>Major unknown ability (by major)</i>					
Science ($\log \sigma_{m,0}^2$)	-0.27	0.19	Science ($\log \tilde{\sigma}_{m,0}^2$)	-1.28	0.21
Social Sciences ($\log \sigma_{m,1}^2$)	-0.20	0.22	Social Sciences ($\log \tilde{\sigma}_{m,1}^2$)	-2.03	0.13
Education and Humanities ($\log \sigma_{m,2}^2$)	-0.58	0.26	Education and Humanities ($\log \tilde{\sigma}_{m,2}^2$)	-0.74	0.16
Health ($\log \sigma_{m,3}^2$)	-1.44	0.18	Health ($\log \tilde{\sigma}_{m,3}^2$)	-1.99	0.09
<i>Major unknown ability (covariates)</i>					
Low income ($\log \sigma_{m,low}^2$)	0.12	0.09	Low income ($\log \tilde{\sigma}_{m,low}^2$)	-0.19	0.21
Female ($\log \sigma_{m,f}^2$)	-0.37	0.10	Female ($\log \tilde{\sigma}_{m,f}^2$)	0.20	0.25
Avg. obs. ability ($\log \sigma_{m,\bar{a}}^2$)	-0.80	0.16	Avg. obs. ability ($\log \tilde{\sigma}_{m,\bar{a}}^2$)	-1.90	0.08
<i>Subject unknown ability</i>					
Verbal type ($\log \sigma_{s,v}^2$)	-3.86	0.04	Verbal type ($\log \tilde{\sigma}_{s,v}^2$)	-3.61	0.03
Low income ($\log \sigma_{s,low}^2$)	0.00	(-)	Low income ($\log \tilde{\sigma}_{s,low}^2$)	0.00	(-)
Female ($\log \sigma_{s,f}^2$)	0.00	(-)	Female ($\log \tilde{\sigma}_{s,f}^2$)	0.00	(-)
Avg. obs. ability ($\log \sigma_{s,\bar{a}}^2$)	0.00	(-)	Avg. obs. ability ($\log \tilde{\sigma}_{s,\bar{a}}^2$)	0.00	(-)
<i>Learning about preferences - Major (by major)</i>					
Science ($\log(\zeta_{m,0}^\alpha)^2$)	0.57	0.09			
Social Sciences ($\log(\zeta_{m,1}^\alpha)^2$)	1.14	0.01			
Education and Humanities ($\log(\zeta_{m,2}^\alpha)^2$)	0.48	< 0.01			
Health ($\log(\zeta_{m,3}^\alpha)^2$)	2.69	0.02			
<i>Learning about preferences - Major (covariates)</i>					
Low income ($\log(\zeta_{m,low}^\alpha)^2$)	2.47	0.02			
Female ($\log(\zeta_{m,f}^\alpha)^2$)	-0.04	0.02			
Avg. obs. ability ($\log(\zeta_{m,\bar{a}}^\alpha)^2$)	1.20	0.03			

Notes: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator. Parameters from external regressions are indicated by (-).

Table E7—Estimation Results - Private Abilities

Log-Variiances			Means		
Parameter	Value	Std. Error	Parameter	Value	Std. Error
<i>Private ability - Subjects</i>					
Intercept ($\log \zeta_{s,0}^2$)	-0.75	0.11	Verbal type ($\mu_{s,v}^p$)	-1.16	0.03
Low income ($\log \zeta_{s,low}^2$)	0.00	(-)	Low income ($\mu_{s,low}^p$)	0.35	0.03
Female ($\log \zeta_{s,f}^2$)	0.00	(-)	Female ($\mu_{s,f}^p$)	1.79	0.01
Avg. observed ability ($\log \zeta_{s,\bar{a}}^2$)	0.00	(-)	Avg. observed ability ($\mu_{s,\bar{a}}^p$)	-0.02	0.11
<i>Private ability - Major (by major)</i>					
Science ($\log \zeta_{m,0}^2$)	-2.04	0.08	Science ($\mu_{m,0}^p$)	0.00	(-)
Social Sciences ($\log \zeta_{m,1}^2$)	-2.02	0.08	Social Sciences ($\mu_{m,1}^p$)	0.00	(-)
Education and Humanities ($\log \zeta_{m,2}^2$)	-0.97	0.19	Education and Humanities ($\mu_{m,2}^p$)	0.00	(-)
Health ($\log \zeta_{m,3}^2$)	0.05	0.14	Health ($\mu_{m,3}^p$)	0.00	(-)
<i>Private ability - Major (covariates)</i>					
Low income ($\log \zeta_{m,low}^2$)	-0.27	0.06	Low income ($\mu_{m,low}^p$)	-0.36	0.00
Female ($\log \zeta_{m,f}^2$)	-0.72	0.01	Female ($\mu_{m,f}^p$)	-0.20	0.01
Avg. observed ability ($\log \zeta_{m,\bar{a}}^2$)	-0.72	0.15	Avg. observed ability ($\mu_{m,\bar{a}}^p$)	-0.35	0.01

Notes: Standard errors (S.E.) are computed using the asymptotic variance-covariance of the estimator. Parameters from external regressions are indicated by (-).

how to find the optimal ROL for each student, given their preferences and beliefs. Algorithm 3 in Appendix F describes a procedure to accomplish this. See Abdulkadiroğlu, Che and Yasuda (2015) for details.

F2. Market Structure and Capacities

To construct the counterfactual market, we scale from our estimation sample of 4,000 students to the full market of approximately 40,000 students. We first back out program capacities that are consistent with the empirical distribution of cutoffs observed in the data, given our sample composition. After recomputing these capacities for the entire market, we solve for the baseline equilibrium by running Algorithm 2 instead of using the empirical distribution of cutoffs for assignments. This yields an equilibrium cutoff distribution that serves as the baseline for all counterfactual experiments. We fix this vector of capacities throughout all counterfactual analyses, allowing us to isolate the effects of policy changes on student behavior and outcomes while holding market structure constant.

F3. Finding equilibrium beliefs

Index each counterfactual experiment and the baseline model by τ ; then the rational expectations equilibrium cutoff distributions, $\hat{p}(\tau)$, can be computed with the following algorithm:

Algorithm 2 Computing $\hat{p}(\tau)$

Input. Structural parameter estimates $\hat{\theta}$, first-stage estimates $\hat{p}$, $\hat{P}^e$, $\hat{P}^d$, $\hat{P}^g$, and $\hat{P}^w$, and tolerance level ϵ_{tol} .

Output. Rational expectations equilibrium cutoff distributions $\hat{p}(\tau)$

Step 1. For each program j

Step 1.a. Solve the model and simulate outcomes given the rules implied by counterfactual τ and the estimated objects $(\hat{\theta}, \hat{p}, \hat{P}^e, \hat{P}^d, \hat{P}^g, \hat{P}^w)$

Step 1.b. Obtain a set of simulated ROLs and scores $(R_1^0, R_2^0, s_1^0, s_2^0)$

Step 1.c. For each program j , estimate the mean and standard deviation of the cutoff distributions $\delta_j^0 \equiv (\hat{\mu}_j^0, \hat{\sigma}_j^0)$

Step 2. $\delta_{diff} = 2\epsilon_{tol}$, $k = 1$, $\rho = 0.9$

Step 3. While $\delta_{diff} > \epsilon_{tol}$

Step 3.a. For each student i , solve the model via Backward Induction given τ , $\hat{\theta}$, $\hat{P}^e$, $\hat{P}^d$, $\hat{P}^g$, $\hat{P}^w$, and $\hat{p}^{k-1}$

Step 3.b. Forward simulate first period ROL R_{i1}^k given those objects

Step 3.c. For each program j , estimate the mean and standard deviation $\delta_j^k \equiv (\hat{\mu}_j^k, \hat{\sigma}_j^k)$

Step 3.d. Given R_1^k , R_2^{k-1} , s_1^k , and s_2^{k-1} , run the Chilean matching mechanism and obtain μ^k

Step 3.e. Given μ^k and τ , simulate second-period ROLs R_{i2}^k

Step 3.f. Bootstrap to estimate $\tilde{p}^k$ and update $\hat{p}^k = \rho^k \tilde{p}^{k-1} + (1 - \rho^k) \tilde{p}^k$

Step 3.g. Compute $\delta_{diff} = \|\delta^k - \delta^{k-1}\|$

$\hat{p}(\tau) = \hat{p}^{k-1}$; $k++$

Algorithm 3 Constrained Deferred Acceptance with signal and bonus ψ

Input. Indirect utilities v , application scores s , cutoff distributions P , and application score bonus ψ

Output. Optimal ROL $R(v, s, P, \psi_\tau)$

Step 1. For each program j

Step 1.a. Compute admission probabilities given cutoff distributions P and application scores $\tilde{s}(j) = \{s_1, \dots, s_{j-1}, s_j(\psi_\tau), s_{j+1}, \dots, s_J\}$, where $s_j(\psi_\tau)$ is program j 's application score recomputed after multiplying the student's high-school GPA components (NEM and rank scores) by ψ_τ

Step 1.b. Compute and store optimal ROL $R(v, \tilde{p}(j))$ using MIA

Step 2. Compute optimal signal

$$j^* = \underset{j}{\operatorname{argmax}} \{EU(R(v, \tilde{p}(j)))\},$$

where $EU(R)$ is the expected value of ROL R under the corresponding admission probabilities

Step 3. Return the optimal ROL $R(v, \tilde{p}(j^*))$ associated with the optimal signal j^*

F4. Hypothetical scenario

In 2022, we conducted a survey to elicit preferences and beliefs, which is similar to the 2019-2021' versions. In this version, we included a question to elicit information about how students would change their application lists if they could apply to a single program—i.e., applying under CDA with $K = 1$. Figure F1 shows the distribution of the chosen program in the hypothetical scenario relative to their submitted ROL. Labels from 1-10 identify the share of students whose hypothetical program coincides with their k-th reported preference; *TT*, *BT*, and *Other* identify the share of students who report in the hypothetical scenario a program outside their submitted ROL; finally, *NR* identifies the share of students who did not respond to the survey question. We observe that under CDA with $K = 1$, a significant fraction of students would choose a program that is not their top-reported program on their current list (close to 40%). This suggests that a significant fraction of students would react strategically and take into account their admission probabilities when facing a binding constraint on the length of the list, consistent with our modeling assumptions.

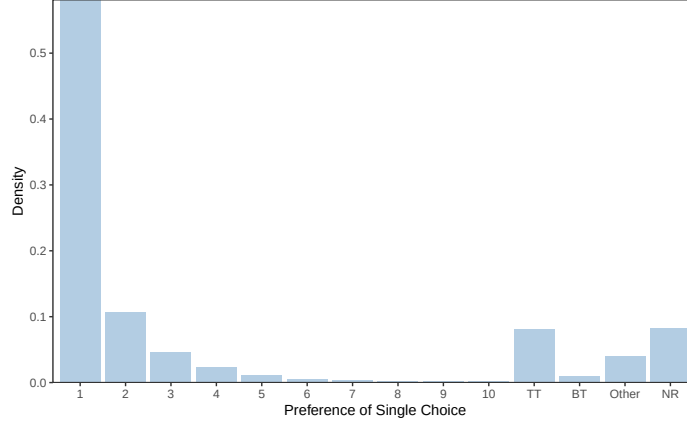
F5. Additional Results

APPENDIX FOR SECTION VI.C

Table F1—Understanding Behavioral Channels - Percentage Changes (Exogenous Version)

Outcome	Baseline (exogenous)	No bias mean	No bias	No learning abilities	No learning prefs	No mismatch
Re-applicants [%]	18.80 (0.041)	-5.20 (0.258)	19.35 (0.196)	-7.99 (0.240)	-10.78 (0.200)	-54.86 (0.203)
Program switchings [%]	4.21 (0.055)	-15.62 (0.505)	80.80 (1.452)	-22.80 (0.904)	-34.51 (1.118)	-93.97 (0.225)
Program switching rate [%]	6.69 (0.093)	-8.35 (0.614)	89.02 (1.518)	-20.25 (0.987)	-31.42 (1.214)	-95.35 (0.170)
Major switchings [%]	3.44 (0.043)	-20.35 (0.427)	93.42 (1.798)	-26.41 (0.765)	-39.68 (0.985)	-93.29 (0.326)
Major switching rate [%]	5.47 (0.072)	-13.48 (0.532)	102.22 (1.877)	-23.98 (0.823)	-36.83 (1.056)	-94.83 (0.248)
Retakes PSU [%]	7.65 (0.020)	-8.51 (0.394)	-8.03 (0.205)	5.34 (0.585)	4.09 (0.590)	-69.77 (0.485)
Dropouts [%]	7.45 (0.119)	3.82 (0.893)	23.29 (0.806)	-0.99 (0.767)	-12.31 (0.813)	0.20 (1.905)
Dropout rate [%]	11.83 (0.178)	12.78 (0.966)	28.90 (0.829)	2.29 (0.687)	-8.15 (0.748)	-22.71 (1.393)
Dropout in first period rate [%]	5.27 (0.131)	23.99 (2.459)	55.97 (2.251)	-2.71 (2.095)	-27.70 (2.050)	-7.01 (2.988)
Applicants in first period [%]	76.58 (0.089)	-7.74 (0.033)	-4.86 (0.025)	-3.82 (0.094)	-4.94 (0.104)	19.89 (0.125)
Unassigned in first period [%]	29.79 (0.116)	18.90 (0.232)	10.27 (0.046)	7.61 (0.339)	10.72 (0.380)	-72.53 (0.215)
Mean assigned preference rank	1.59 (0.009)	-1.74 (0.114)	-1.83 (0.079)	-5.60 (0.151)	-5.58 (0.166)	-37.11 (0.341)
Assigned to top true preference in period 1 [%]	11.60 (0.137)	19.67 (0.550)	16.79 (0.464)	62.02 (1.360)	64.74 (1.382)	761.77 (10.175)
Graduate from 1st enrollment [%]	45.67 (0.046)	-8.50 (0.113)	-13.73 (0.093)	-1.59 (0.092)	-0.67 (0.086)	45.79 (0.151)

Note: Baseline column shows levels; other columns show percentage changes relative to the baseline. Standard errors computed using the delta method for percentage changes, accounting for correlation between baseline and counterfactual simulations (common seeds across 5 simulations). Switching and dropout rates are computed relative to first-period enrollees. This table uses the exogenous version of the counterfactuals (empirical distribution of cut-offs rather than solving for equilibrium cut-offs under the policy change). Counterfactual columns build on one another: “No bias” adds the variance debiasing to the mean debiasing; “No learning prefs” eliminates learning about preferences in addition to learning about abilities; and “No mismatch” further assigns every student to their top true choice.

Figure F1. Distribution of applications under CDA with $K = 1$ 

Note: Bars report shares of respondents to the 2022 survey.

Table F2—PSP: % Change from Baseline (Information and Learning Grid)

		Major sw.	Program sw.	Dropout	Graduate	Top true	Unassigned
Baseline (levels)		3.29	4.26	3.70	48.45	11.09	28.56
Ability 0%	Pref. 0%	-1.4	-2.0	1.9	-0.1	-0.9	0.1
		(2.09)	(1.78)	(0.28)	(0.19)	(0.92)	(0.29)
	Pref. 50%	-8.7	-6.6	-3.8	0.7	-0.8	-0.5
		(1.40)	(2.01)	(1.81)	(0.30)	(0.82)	(0.22)
	Pref. 100%	-13.1	-10.3	-11.2	1.2	0.4	-0.6
		(1.84)	(1.70)	(0.57)	(0.25)	(0.54)	(0.46)
Ability 50%	Pref. 0%	-13.5	-9.6	1.1	-1.6	10.6	5.3
		(1.97)	(1.34)	(1.52)	(0.13)	(1.16)	(0.31)
	Pref. 50%	-21.4	-16.4	-4.5	-1.2	10.5	5.2
		(1.00)	(0.97)	(0.89)	(0.17)	(0.62)	(0.27)
	Pref. 100%	-25.6	-18.9	-12.6	-0.6	10.5	5.2
		(1.10)	(1.38)	(0.62)	(0.13)	(0.32)	(0.21)
Ability 100%	Pref. 0%	-9.7	-4.7	9.0	-5.1	21.0	13.6
		(0.99)	(1.13)	(1.11)	(0.15)	(0.26)	(0.21)
	Pref. 50%	-17.2	-10.9	3.9	-4.7	20.9	13.7
		(1.54)	(1.29)	(1.16)	(0.25)	(0.80)	(0.34)
	Pref. 100%	-19.3	-11.1	-1.1	-4.7	20.0	13.9
		(1.50)	(1.04)	(2.21)	(0.12)	(0.89)	(0.21)

Note: The top row reports baseline levels (the no-information, no-cost corner). Remaining rows report percentage changes from baseline for the PSP policy (with the one-year time cost), for each combination of ability and preference learning *resolved* ($1 - \rho^A$ and $1 - \rho^P$, respectively). Outcomes are: major and program switching rates, first-period dropout rate, graduation from first enrollment, assignment to the top true preference, and unassignment in the first period; switching and dropout rates are computed relative to first-period enrollees. Standard errors (in parentheses) are computed using the delta method over 5 seeds with common draws.

Table F3—PSP: % Change from Baseline (Information and Learning Grid, No Time Cost)

		Major sw.	Program sw.	Dropout	Graduate	Top true	Unassigned
Baseline (levels)		3.29	4.26	3.70	48.45	11.09	28.56
Ability 0%	Pref. 0%	0.0	0.0	0.0	0.0	0.0	0.0
	Pref. 50%	-6.1 (0.95)	-5.2 (0.77)	-3.3 (1.25)	0.4 (0.12)	-0.5 (0.89)	-0.3 (0.20)
	Pref. 100%	-10.2 (1.58)	-9.0 (1.21)	-10.6 (0.66)	0.9 (0.14)	-1.2 (0.44)	-0.5 (0.30)
Ability 50%	Pref. 0%	-13.2 (1.16)	-10.4 (0.93)	-1.3 (0.74)	-1.5 (0.25)	9.7 (0.53)	5.0 (0.15)
	Pref. 50%	-18.3 (1.76)	-13.5 (1.33)	-3.8 (0.84)	-1.1 (0.23)	12.2 (0.81)	5.1 (0.26)
	Pref. 100%	-24.9 (0.99)	-19.3 (0.78)	-10.1 (1.33)	-0.7 (0.03)	11.1 (0.83)	4.9 (0.27)
Ability 100%	Pref. 0%	-7.1 (1.97)	-3.3 (1.14)	7.5 (1.85)	-5.4 (0.17)	21.5 (0.48)	14.3 (0.22)
	Pref. 50%	-15.0 (1.01)	-8.6 (0.81)	7.6 (0.60)	-5.4 (0.24)	21.8 (1.26)	14.1 (0.43)
	Pref. 100%	-17.0 (1.50)	-10.1 (1.56)	-0.3 (1.64)	-4.6 (0.05)	20.4 (0.53)	13.3 (0.32)

Note: The top row reports baseline levels (the no-information, no-cost corner). Remaining rows report percentage changes from baseline for the PSP policy *without* the one-year time cost, for each combination of ability and preference learning *resolved* ($1 - \rho^A$ and $1 - \rho^P$, respectively); the (Ability 0%, Pref. 0%) cell coincides with the baseline, so its entries are zero by construction. Outcomes are: major and program switching rates, first-period dropout rate, graduation from first enrollment, assignment to the top true preference, and unassignment in the first period; switching and dropout rates are computed relative to first-period enrollees. Standard errors (in parentheses) are computed using the delta method over 5 seeds with common draws.

Table F4—Counterfactual Policies: CDA and CADA Mechanisms (% Change from Baseline)

Outcome	Baseline	CDA Rules			CADA Rules		
		$K = 3$	$K = 2$	$K = 1$	$\psi = 1.1$	$\psi = 1.2$	$\psi = 1.3$
Re-applicants [%]	15.06 (0.096)	3.23 (0.494)	11.67 (0.633)	56.12 (0.390)	-10.37 (0.417)	-13.85 (0.571)	-15.70 (0.510)
Program switchings [%]	2.68 (0.020)	3.65 (0.572)	23.44 (1.576)	11.50 (0.864)	-22.31 (1.199)	-29.72 (1.265)	-34.15 (1.741)
Program switching rate [%]	4.19 (0.037)	4.50 (0.583)	27.12 (1.590)	28.68 (0.907)	-23.80 (1.151)	-31.42 (1.260)	-36.04 (1.766)
Major switchings [%]	2.09 (0.018)	2.11 (0.719)	17.18 (1.845)	2.57 (1.016)	-20.38 (1.695)	-27.03 (1.683)	-31.66 (2.343)
Major switching rate [%]	3.27 (0.033)	2.95 (0.736)	20.66 (1.866)	18.38 (1.045)	-21.91 (1.633)	-28.79 (1.679)	-33.63 (2.375)
Retakes PSU [%]	5.56 (0.062)	2.12 (1.383)	9.70 (1.058)	47.14 (0.224)	-16.62 (1.018)	-20.43 (0.657)	-22.66 (1.279)
Dropouts [%]	6.48 (0.075)	-1.95 (1.298)	-3.30 (1.523)	-9.94 (1.545)	-1.07 (2.016)	-1.69 (1.864)	-2.28 (0.868)
Dropout rate [%]	10.12 (0.100)	-1.13 (1.207)	-0.41 (1.390)	3.96 (1.449)	-2.96 (1.849)	-4.05 (1.771)	-5.08 (0.871)
Dropout in first period rate [%]	3.72 (0.053)	0.90 (1.142)	4.33 (1.195)	14.62 (2.815)	1.54 (2.356)	1.55 (2.717)	2.39 (0.621)
Applicants in first period [%]	75.21 (0.094)	-0.01 (0.108)	-0.20 (0.292)	0.25 (0.151)	1.00 (0.116)	1.71 (0.153)	2.25 (0.134)
Unassigned in first period [%]	28.60 (0.094)	2.07 (0.163)	7.46 (0.324)	34.55 (0.070)	-5.16 (0.377)	-6.72 (0.303)	-7.77 (0.131)
Assigned to top true preference in first period [%]	11.03 (0.105)	-0.79 (0.286)	-10.53 (1.212)	-26.16 (0.866)	5.45 (1.184)	10.15 (0.513)	13.80 (0.552)
Graduate - first enrollment [%]	48.43 (0.065)	-0.98 (0.156)	-3.95 (0.165)	-13.99 (0.140)	2.74 (0.079)	3.45 (0.096)	4.27 (0.035)
Difference in Interim Welfare Relative to Baseline							
Overall	—	-0.63 (0.204)	-1.80 (0.190)	-4.51 (0.107)	0.90 (0.204)	1.30 (0.087)	1.97 (0.124)
Difference in Ex-Post Welfare Relative to Baseline							
Overall	—	-0.93 (0.150)	-3.00 (0.161)	-11.03 (0.112)	0.94 (0.104)	1.12 (0.135)	1.74 (0.112)
Males	—	-0.90 (0.254)	-3.14 (0.138)	-11.66 (0.258)	1.33 (0.238)	1.93 (0.134)	2.15 (0.187)
Females	—	-0.96 (0.258)	-2.86 (0.300)	-10.38 (0.306)	0.54 (0.285)	0.30 (0.363)	1.33 (0.283)
Low income	—	-0.60 (0.332)	-1.77 (0.214)	-6.55 (0.299)	0.99 (0.166)	1.19 (0.213)	1.48 (0.278)
High income	—	-1.28 (0.504)	-4.29 (0.341)	-15.74 (0.316)	0.88 (0.281)	1.05 (0.301)	2.01 (0.455)

Note: Baseline column shows levels; other columns show percentage changes relative to the baseline for outcomes, and absolute differences for welfare measures. Standard errors computed using the delta method for percentage changes and error propagation for welfare differences, accounting for correlation between baseline and counterfactual simulations (common seeds across 5 simulations). Switching and dropout rates are computed relative to first-period enrollees. Welfare measures are in millions of Chilean pesos (2014) and are computed under subjective beliefs. All columns correspond to the fully strategic scenario ($\rho = 0$); Figure 5 reports both scenarios.

Table F5—Counterfactual Policies: Turkish and Finnish Mechanisms (% Change from Baseline)

Outcome	Baseline	Turkish Rules			Finnish Rules		
		$\psi = 0.9$	$\psi = 0.8$	$\psi = 0.7$	$\psi = 1.1$	$\psi = 1.2$	$\psi = 1.3$
Re-applicants [%]	15.06 (0.096)	-21.62 (0.494)	-31.98 (0.633)	-35.46 (0.390)	-23.15 (0.417)	-25.24 (0.571)	-26.24 (0.510)
Program switchings [%]	2.68 (0.020)	-41.20 (0.572)	-59.03 (1.576)	-71.30 (0.864)	-33.35 (1.199)	-37.25 (1.265)	-40.92 (1.741)
Program switching rate [%]	4.19 (0.037)	-41.39 (0.583)	-59.19 (1.590)	-71.45 (0.907)	-32.99 (1.151)	-36.67 (1.260)	-40.11 (1.766)
Major switchings [%]	2.09 (0.018)	-38.09 (0.719)	-56.23 (1.845)	-68.61 (1.016)	-31.08 (1.695)	-33.82 (1.683)	-39.28 (2.343)
Major switching rate [%]	3.27 (0.033)	-38.29 (0.736)	-56.41 (1.866)	-68.78 (1.045)	-30.70 (1.633)	-33.22 (1.679)	-38.46 (2.375)
Retakes PSU [%]	5.56 (0.062)	-12.53 (1.383)	-19.44 (1.058)	-22.20 (0.224)	-5.58 (1.018)	-2.56 (0.657)	1.30 (1.279)
Dropouts [%]	6.48 (0.075)	-1.81 (1.298)	-1.68 (1.523)	-2.69 (1.545)	-0.68 (2.016)	0.07 (1.864)	-0.52 (0.868)
Dropout rate [%]	10.12 (0.100)	-2.12 (1.207)	-2.08 (1.390)	-3.20 (1.449)	-0.12 (1.849)	1.01 (1.771)	0.85 (0.871)
Dropout in first period rate [%]	3.72 (0.053)	1.35 (1.142)	-0.02 (1.195)	0.93 (2.815)	-0.40 (2.356)	0.70 (2.717)	2.50 (0.621)
Applicants in first period [%]	75.21 (0.094)	0.06 (0.108)	-0.26 (0.292)	-0.20 (0.151)	-4.13 (0.116)	-4.33 (0.153)	-4.50 (0.134)
Unassigned in first period [%]	28.60 (0.094)	-1.07 (0.163)	-1.27 (0.324)	-1.32 (0.070)	1.41 (0.377)	2.23 (0.303)	3.17 (0.131)
Assigned to top true preference in first period [%]	11.03 (0.105)	7.15 (0.286)	8.95 (1.212)	9.87 (0.866)	11.10 (1.184)	13.10 (0.513)	13.36 (0.552)
Graduate - first enrollment [%]	48.43 (0.065)	2.13 (0.156)	2.89 (0.165)	3.61 (0.140)	0.94 (0.079)	0.55 (0.096)	0.29 (0.035)
Difference in Interim Welfare Relative to Baseline							
Overall	–	0.52 (0.204)	0.47 (0.190)	0.60 (0.107)	1.27 (0.204)	1.89 (0.087)	2.22 (0.124)
Difference in Ex-Post Welfare Relative to Baseline							
Overall	–	0.64 (0.150)	0.45 (0.161)	0.74 (0.112)	0.93 (0.104)	1.12 (0.135)	1.01 (0.112)
Males	–	0.77 (0.254)	0.87 (0.138)	1.01 (0.258)	1.41 (0.238)	1.43 (0.134)	1.41 (0.187)
Females	–	0.51 (0.258)	0.01 (0.300)	0.46 (0.306)	0.44 (0.285)	0.81 (0.363)	0.62 (0.283)
Low income	–	0.52 (0.332)	0.02 (0.214)	0.78 (0.299)	0.73 (0.166)	0.99 (0.213)	0.77 (0.278)
High income	–	0.77 (0.504)	0.90 (0.341)	0.69 (0.316)	1.15 (0.281)	1.26 (0.301)	1.27 (0.455)

Note: Baseline column shows levels; other columns show percentage changes relative to the baseline for outcomes, and absolute differences for welfare measures. Standard errors computed using the delta method for percentage changes and error propagation for welfare differences, accounting for correlation between baseline and counterfactual simulations (common seeds across 5 simulations). Switching and dropout rates are computed relative to first-period enrollees. Welfare measures are in millions of Chilean pesos (2014) and are computed under subjective beliefs. All columns correspond to the fully strategic scenario ($\rho = 0$); Figure 5 reports both scenarios.

Table G1—Goodness of Fit: Application and Preference Moments

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
Top true pref. is reported first (Period 1)	0.24	0.44	0.28	0.40	0.27	0.38	0.26	0.52
Top true pref. is reported first (Period 2)	0.44	0.45	0.43	0.45	0.42	0.40	0.51	0.51
Share that applies (Period 1)	0.77	0.66	0.74	0.68	0.58	0.56	0.89	0.77
Share that applies (Period 2)	0.21	0.22	0.21	0.25	0.20	0.20	0.18	0.24
Share changed broad major preference	0.09	0.06	0.13	0.06	0.10	0.07	0.10	0.05
Share changed broad major from original order	0.10	0.04	0.13	0.03	0.09	0.09	0.07	0.00

Notes: This table presents moments related to application behavior and reported preferences. Moments are shown across four demographic groups: Overall population, Female students, Low-income students, and High-ability students.

Table G2—Goodness of Fit: Switching and Outcome Moments

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
Share switching math type	0.01	0.02	0.01	0.02	0.02	0.01	0.02	0.02
Share switching major within math (Period 1)	0.02	0.03	0.02	0.03	0.02	0.02	0.02	0.04
Share switching math within major (Period 1)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Share switching up (Period 1)	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01
Share switching down (Period 1)	0.00	0.01	0.00	0.01	0.00	0.01	0.00	0.02
Share switching out feasible (Period 1)	0.03	0.03	0.02	0.03	0.03	0.02	0.03	0.04
Share switching out unfeasible (Period 1)	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.01
Share switching major	0.04	0.05	0.03	0.04	0.04	0.03	0.03	0.06
Share switching broad major	0.03	0.03	0.02	0.03	0.03	0.02	0.03	0.04
Share switching university	0.03	0.04	0.02	0.04	0.03	0.03	0.04	0.06
Share retakes	0.08	0.21	0.08	0.23	0.08	0.20	0.07	0.21
Share dropouts	0.07	0.06	0.06	0.06	0.10	0.07	0.07	0.07
Share program switching	0.04	0.08	0.03	0.07	0.04	0.05	0.05	0.10
Share reapplicants assigned top reported preference	0.43	0.25	0.47	0.23	0.41	0.26	0.52	0.26
Share reapplicants	0.19	0.15	0.17	0.16	0.16	0.13	0.17	0.18
Share major switching	0.04	0.05	0.03	0.04	0.04	0.03	0.03	0.06
Share enrolling for the first time in period 2	0.09	0.17	0.11	0.17	0.08	0.14	0.07	0.19
Share dropout end of period 1	0.03	0.03	0.03	0.03	0.06	0.03	0.04	0.03
Share first year students (Period 2)	0.13	0.22	0.14	0.23	0.12	0.18	0.12	0.26
Share second year students (Period 2)	0.56	0.39	0.54	0.39	0.34	0.30	0.70	0.49

Notes: This table presents moments related to switching behavior and enrollment dynamics across the four demographic groups.

Table G3—Goodness of Fit: Grade Moments

Coefficient	Grade Yr 1		Grade Yr 2		Coefficient	Time Series No Switchers		Time Series Switchers	
	Model	Data	Model	Data		Model	Data	Model	Data
Observed ability	0.423	0.456	0.408	0.408	Intercept	0.664	0.854	3.731	3.560
Top preference	-0.000	0.076	0.003	0.045	Year 1 grade	0.865	0.808	0.282	0.279
Science	3.720	3.691	3.806	3.820					
Social Science	4.434	4.058	4.520	4.234					
Education and Humanities	4.582	4.272	4.692	4.453					
Health	4.283	4.261	4.386	4.605					
ROL share same major	0.269	0.170	0.209	0.130					
Signal period 1			0.820	0.702					
Switch major $\times$ Signal 1			-0.725	-0.556					
Switch math type $\times$ Signal 1			0.015	-0.127					

Notes: This table presents coefficients from grade equations (year 1 and year 2 for students enrolled both periods) and time series for grades equations (for no switchers and switchers). Each coefficient is labeled in the first and middle columns.

Table G4—Goodness of Fit: Regression Discontinuity Design (RDD) Moments

Coefficient	RDD Switching		RDD Switch Up/Out		RDD Reapplications		RDD BVP	
	Model	Data	Model	Data	Model	Data	Model	Data
Intercept	0.120	0.167	0.067	0.048	0.276	0.234	0.059	0.046
Causal effect	-0.088	-0.051	-0.063	-0.034	-0.144	-0.087	0.042	0.025

Notes: This table presents coefficients from four regression discontinuity designs (RDD). RDD Switching examines the effect of admission to your top-reported preference on switching. RDD Switch Up/Out examines switching up or out to unfeasible programs. RDD Reapplications examines reapplication decisions. RDD BVP examines the effect of scoring above the 600 threshold on applying to BVP-eligible programs (Beca Vocación de Profesor, a teacher scholarship). Each RDD shows the intercept and the causal effect (treatment effect of being above the cutoff).

Table G5—Goodness of Fit: Major Market Shares - Majors in ROL

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
<i>Average Share - Majors in ROL 1</i>								
Business & Management	0.06	0.13	0.03	0.11	0.02	0.11	0.09	0.13
Agriculture	0.03	0.02	0.05	0.02	0.04	0.02	0.03	0.02
Arts & Architecture	0.11	0.07	0.12	0.08	0.07	0.05	0.10	0.07
Basic Sciences	0.06	0.05	0.04	0.04	0.05	0.04	0.06	0.05
Social Sciences	0.10	0.12	0.16	0.14	0.06	0.12	0.13	0.12
Law	0.03	0.05	0.05	0.05	0.02	0.04	0.03	0.05
Education	0.09	0.07	0.10	0.09	0.11	0.09	0.06	0.07
Humanities	0.01	0.02	0.02	0.03	0.02	0.02	0.01	0.02
Health	0.23	0.21	0.23	0.30	0.37	0.23	0.21	0.21
Technology	0.28	0.24	0.21	0.12	0.25	0.27	0.29	0.25
<i>Average Share - Majors in ROL 2</i>								
Business & Management	0.03	0.11	0.01	0.08	0.01	0.09	0.05	0.11
Agriculture	0.02	0.03	0.04	0.03	0.03	0.03	0.01	0.03
Arts & Architecture	0.06	0.06	0.08	0.06	0.04	0.05	0.05	0.06
Basic Sciences	0.03	0.04	0.03	0.04	0.02	0.04	0.04	0.05
Social Sciences	0.06	0.14	0.09	0.16	0.04	0.14	0.08	0.14
Law	0.02	0.04	0.04	0.04	0.02	0.04	0.02	0.04
Education	0.14	0.09	0.19	0.11	0.16	0.12	0.06	0.08
Humanities	0.02	0.02	0.04	0.03	0.03	0.02	0.01	0.02
Health	0.38	0.26	0.29	0.36	0.49	0.28	0.35	0.26
Technology	0.24	0.19	0.19	0.09	0.16	0.20	0.34	0.20
<i>Average Dummy - Majors in ROL 1</i>								
Business & Management	0.09	0.22	0.05	0.19	0.03	0.19	0.13	0.22
Agriculture	0.05	0.04	0.07	0.05	0.05	0.04	0.03	0.04
Arts & Architecture	0.17	0.12	0.19	0.14	0.10	0.10	0.15	0.12
Basic Sciences	0.13	0.15	0.10	0.14	0.11	0.13	0.12	0.16
Social Sciences	0.17	0.22	0.23	0.25	0.10	0.20	0.19	0.22
Law	0.06	0.09	0.11	0.09	0.04	0.07	0.06	0.09
Education	0.11	0.13	0.13	0.16	0.14	0.16	0.07	0.12
Humanities	0.03	0.06	0.04	0.08	0.04	0.05	0.01	0.06
Health	0.27	0.27	0.26	0.38	0.42	0.30	0.25	0.27
Technology	0.37	0.35	0.28	0.21	0.33	0.38	0.36	0.36
<i>Average Dummy - Majors in ROL 2</i>								
Business & Management	0.05	0.18	0.03	0.14	0.02	0.16	0.07	0.18
Agriculture	0.04	0.06	0.06	0.06	0.04	0.06	0.02	0.06
Arts & Architecture	0.10	0.11	0.14	0.11	0.07	0.09	0.08	0.11
Basic Sciences	0.09	0.12	0.07	0.11	0.06	0.13	0.09	0.13
Social Sciences	0.10	0.23	0.15	0.25	0.06	0.23	0.12	0.23
Law	0.06	0.07	0.10	0.06	0.05	0.06	0.05	0.07
Education	0.17	0.16	0.23	0.18	0.20	0.20	0.07	0.15
Humanities	0.06	0.06	0.10	0.07	0.07	0.07	0.01	0.06
Health	0.43	0.33	0.33	0.43	0.54	0.35	0.39	0.33
Technology	0.31	0.29	0.25	0.17	0.21	0.30	0.41	0.29

Notes: This table shows major shares and dummies in rank-ordered lists (ROL) in periods 1 and 2, across four demographic groups. “Average Share” is the average within-student fraction of a ROL in a given major. “Average Dummy” is the proportion of students who list at least one program from that major.

Table G6—Goodness of Fit: Broad Major and Math Type Shares in ROL

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
<i>Average Share - Broad Majors in ROL 1</i>								
Science	0.37	0.37	0.30	0.39	0.34	0.32	0.37	0.37
Social Science	0.30	0.31	0.35	0.19	0.16	0.33	0.35	0.32
Education and Humanities	0.10	0.09	0.12	0.12	0.13	0.11	0.07	0.09
Health	0.23	0.21	0.23	0.30	0.37	0.23	0.21	0.21
<i>Average Share - Broad Majors in ROL 2</i>								
Science	0.30	0.35	0.26	0.34	0.21	0.31	0.39	0.35
Social Science	0.16	0.26	0.22	0.16	0.10	0.27	0.19	0.27
Education and Humanities	0.16	0.12	0.22	0.14	0.20	0.14	0.07	0.11
Health	0.38	0.26	0.29	0.36	0.49	0.28	0.35	0.26
<i>Average Dummy - Broad Majors in ROL 1</i>								
Science	0.43	0.50	0.35	0.51	0.40	0.45	0.42	0.50
Social Science	0.33	0.45	0.38	0.32	0.19	0.46	0.39	0.46
Education and Humanities	0.12	0.17	0.14	0.21	0.16	0.19	0.08	0.17
Health	0.27	0.27	0.26	0.38	0.42	0.30	0.25	0.27
<i>Average Dummy - Broad Majors in ROL 2</i>								
Science	0.35	0.47	0.30	0.45	0.25	0.43	0.45	0.47
Social Science	0.19	0.39	0.25	0.28	0.12	0.40	0.22	0.40
Education and Humanities	0.18	0.20	0.25	0.23	0.21	0.24	0.07	0.19
Health	0.43	0.33	0.33	0.43	0.54	0.35	0.39	0.33
<i>Average Share - Math Types in ROL 1</i>								
Low math intensity	0.40	0.39	0.69	0.47	0.47	0.41	0.37	0.38
High math intensity	0.60	0.61	0.31	0.53	0.53	0.59	0.63	0.62
<i>Average Share - Math Types in ROL 2</i>								
Low math intensity	0.38	0.45	0.69	0.53	0.48	0.49	0.28	0.43
High math intensity	0.62	0.55	0.31	0.47	0.52	0.51	0.72	0.57
<i>Average Dummy - Math Types in ROL 1</i>								
Low math intensity	0.46	0.36	0.75	0.45	0.54	0.37	0.42	0.36
High math intensity	0.66	0.64	0.39	0.55	0.59	0.63	0.67	0.64
<i>Average Dummy - Math Types in ROL 2</i>								
Low math intensity	0.45	0.53	0.75	0.61	0.55	0.59	0.31	0.50
High math intensity	0.68	0.47	0.39	0.39	0.58	0.41	0.75	0.50

Notes: This table shows broad major and math type shares and dummies in rank-ordered lists (ROL) in periods 1 and 2, across four demographic groups.

Table G7—Goodness of Fit: Market Shares in Periods 1 and 2

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
<i>Major Market Shares - Period 1</i>								
No College	0.36	0.26	0.39	0.28	0.55	0.30	0.21	0.22
Business & Management	0.04	0.12	0.02	0.11	0.01	0.11	0.08	0.11
Agriculture	0.02	0.02	0.03	0.02	0.02	0.01	0.02	0.02
Arts & Architecture	0.08	0.06	0.08	0.07	0.03	0.04	0.08	0.06
Basic Sciences	0.04	0.04	0.03	0.04	0.03	0.04	0.05	0.06
Social Sciences	0.07	0.07	0.10	0.09	0.03	0.06	0.11	0.08
Law	0.02	0.04	0.03	0.04	0.01	0.02	0.02	0.04
Education	0.06	0.05	0.06	0.06	0.05	0.06	0.05	0.05
Humanities	0.01	0.01	0.01	0.02	0.01	0.01	0.01	0.02
Health	0.13	0.12	0.13	0.18	0.15	0.12	0.15	0.13
Technology	0.18	0.20	0.12	0.09	0.11	0.22	0.22	0.21
<i>Major Market Shares - Period 2</i>								
No College	0.31	0.12	0.32	0.11	0.54	0.15	0.18	0.09
Business & Management	0.04	0.15	0.02	0.14	0.01	0.14	0.08	0.14
Agriculture	0.02	0.02	0.04	0.02	0.02	0.02	0.02	0.02
Arts & Architecture	0.08	0.07	0.09	0.08	0.03	0.05	0.08	0.07
Basic Sciences	0.04	0.04	0.03	0.04	0.02	0.04	0.05	0.05
Social Sciences	0.07	0.09	0.11	0.12	0.03	0.08	0.11	0.10
Law	0.02	0.04	0.03	0.04	0.01	0.03	0.03	0.05
Education	0.07	0.07	0.08	0.09	0.06	0.08	0.05	0.07
Humanities	0.01	0.02	0.02	0.02	0.02	0.02	0.00	0.02
Health	0.16	0.16	0.16	0.24	0.19	0.17	0.17	0.17
Technology	0.18	0.23	0.11	0.10	0.08	0.24	0.23	0.23
<i>Broad Major Market Shares - Period 1</i>								
No College	0.36	0.26	0.39	0.28	0.55	0.30	0.21	0.22
Science	0.24	0.26	0.18	0.15	0.16	0.27	0.30	0.28
Social Science	0.20	0.28	0.22	0.31	0.08	0.24	0.29	0.30
Education and Humanities	0.07	0.06	0.08	0.08	0.07	0.07	0.06	0.07
Health	0.13	0.12	0.13	0.18	0.15	0.12	0.15	0.13
<i>Broad Major Market Shares - Period 2</i>								
No College	0.31	0.12	0.32	0.11	0.54	0.15	0.18	0.09
Science	0.24	0.29	0.18	0.16	0.12	0.29	0.30	0.30
Social Science	0.21	0.35	0.24	0.38	0.07	0.30	0.30	0.36
Education and Humanities	0.08	0.08	0.10	0.11	0.08	0.09	0.06	0.08
Health	0.16	0.16	0.16	0.24	0.19	0.17	0.17	0.17

Notes: This table shows market shares across majors and broad majors in periods 1 and 2 across four demographic groups. Market shares represent the proportion of students enrolled in each major category. Although the model tends to overpredict non-enrollment, we hold the outside option value flow fixed in counterfactuals, which attenuates issues with our welfare comparisons.

Table G8—Goodness of Fit: Additional Application and Assignment Characteristics

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
Share unassigned in 1st period	0.292	0.414	0.309	0.418	0.496	0.522	0.134	0.296
Share assigned to 1st preference	0.526	0.297	0.522	0.293	0.385	0.243	0.681	0.353
Share assigned to 2nd preference	0.100	0.145	0.090	0.142	0.067	0.109	0.100	0.176
Share assigned to 3rd preference	0.033	0.079	0.033	0.079	0.021	0.062	0.033	0.095
Student observed ability score (top reported pref.)	0.963	1.111	0.900	1.108	0.763	0.948	1.413	1.216
Program observed ability score (top reported pref.)	1.162	1.467	1.091	1.464	1.091	1.349	1.295	1.543
Tuition (top reported pref.)	3.601	3.780	3.432	3.708	3.000	3.345	3.581	3.821
Distance (top reported pref.)	12.126	11.714	12.187	11.684	13.271	12.416	11.462	11.563
Relative position (top reported pref.)	-0.352	-1.849	-0.388	-1.928	-0.883	-2.099	0.780	-1.750
Correlation: Normalized major change and grades	-0.203	-0.098	-0.133	-0.089	-0.169	-0.109	-0.064	-0.106
Correlation: Normalized broad major change and grades	-0.116	-0.063	0.030	-0.093	-0.089	-0.116	-0.087	-0.064
Correlation: Normalized math type change and grades	-0.038	0.006	-0.089	0.010	-0.028	0.007	-0.094	-0.010
Risk in ROL 1 (portfolio diversity)	0.140	0.325	0.133	0.349	0.219	0.363	0.050	0.303
Share of broad majors within ROL 1 (enrolled)	0.949	0.858	0.958	0.850	0.944	0.854	0.959	0.857
Share assigned to top true preference	0.116	0.230	0.152	0.046	0.127	0.168	0.152	0.282
Share of reapplicants assigned to top true pref.	0.046	0.090	0.080	0.046	0.055	0.069	0.068	0.107
Share with top true preference changed	0.462	0.665	0.607	0.617	0.509	0.718	0.448	0.582

Note: This table shows additional application and assignment characteristics from the model and data across four demographic groups. The moments include assignment status, characteristics of top reported preferences, correlations between preference changes and academic performance, portfolio risk measures, and top true preference assignment outcomes.

Table G9—Goodness of Fit: Residual Variances and Expected Wages

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
SSR: Grades equation (Year 1)	0.672	0.713	0.570	0.638	0.744	0.785	0.568	0.698
SSR: Grades equation (Year 2)	0.608	0.646	0.541	0.588	0.633	0.681	0.517	0.626
Variance of residuals: Stayers (Year 1)	0.603	0.401	0.529	0.364	0.632	0.407	0.524	0.396
Variance of residuals: Stayers (Year 2)	0.607	0.636	0.538	0.575	0.647	0.675	0.521	0.613
Expected log wages: Top true preference	2.784	2.775	2.722	2.765	2.710	2.692	2.830	2.863
Expected log wages: Random assignment	2.521	2.548	2.491	2.540	2.478	2.472	2.567	2.638

Note: This table shows the sum of squared residuals (SSR) from grade equations, variance of residuals for students who stay in their initial assignment, and expected log wages under different assignment scenarios across four demographic groups. Expected wages compare outcomes when students are assigned to their top true preference versus random assignment in the surveys.

Table G10—Goodness of Fit: Correlations between Signals and Outcomes

Moment	Overall		Female		Low-income		High-ability	
	Model	Data	Model	Data	Model	Data	Model	Data
Correlation: Signal and graduation from 1st enrollment	0.163	0.318	0.140	0.283	0.222	0.338	0.158	0.307
Correlation: Signal and dropout	-0.220	-0.281	-0.213	-0.301	-0.273	-0.344	-0.225	-0.259
Correlation: Signal and program switch	-0.158	-0.218	-0.125	-0.182	-0.196	-0.246	-0.111	-0.216
Correlation: Signal and major switch	-0.166	-0.213	-0.132	-0.181	-0.193	-0.227	-0.111	-0.208
Correlation: Signal and broad major switch	-0.134	-0.158	-0.076	-0.138	-0.157	-0.208	-0.114	-0.159
Correlation: Signal and math type switch	-0.117	-0.139	-0.069	-0.140	-0.138	-0.121	-0.117	-0.152
Correlation: Signal and major switch within math type	-0.120	-0.171	-0.117	-0.134	-0.139	-0.206	-0.057	-0.157
Correlation: Signal and math type switch within major	-0.036	-0.099	-0.022	-0.097	-0.050	-0.066	-0.053	-0.111
Correlation: Signal and switch up (higher preference)	-0.027	0.065	-0.016	0.062	-0.039	0.016	-0.018	0.069
Correlation: Signal and switch down (lower preference)	-0.048	-0.133	-0.050	-0.122	-0.043	-0.129	-0.019	-0.130
Correlation: Signal and switch out (feasible)	-0.153	-0.225	-0.121	-0.211	-0.181	-0.216	-0.101	-0.239
Correlation: Signal and switch out (unfeasible)	-0.032	-0.011	-0.025	-0.011	-0.044	-0.044	-0.045	-0.011

Note: This table shows correlations between students' ability signals and various outcomes across four demographic groups.

Table G11—Goodness of Fit: PSU Score Evolution for Retakers

Moment	Model	Data
Mean log ratio (future/actual): Language	0.026	0.040
Variance of log ratio: Language	0.006	0.010
Mean log ratio (future/actual): Math	0.024	0.043
Variance of log ratio: Math	0.006	0.007
Mean log ratio (future/actual): History (takers)	0.034	0.055
Variance of log ratio: History (takers)	0.005	0.010
Mean log ratio (future/actual): Science (takers)	0.016	0.070
Variance of log ratio: Science (takers)	0.006	0.012
Mean log ratio (future/actual): History (non-takers)	0.045	0.067
Variance of log ratio: History (non-takers)	0.005	0.011
Mean log ratio (future/actual): Science (non-takers)	0.014	0.023
Variance of log ratio: Science (non-takers)	0.006	0.013

Note: This table shows the mean and variance of the logarithm of the ratio between future and actual PSU (Prueba de Selección Universitaria) scores for students who retake the exam. For each test, we compute $\log(\text{future score}/\text{actual score})$ and report the mean and variance of these log ratios. Language and Math are mandatory tests taken by all retakers. For History and Science, which are optional, we separate students into “takers” (those who had a score > 0 in the initial attempt) and “non-takers” (those who had a score of 0 initially). These moments capture the distribution of score improvements when retaking the PSU, which is important for understanding reapplication and admission dynamics.

Table G12—Goodness of Fit: Wage Regression Moments

Moment	Model	Data
<i>Wage Regression Coefficients</i>		
Intercept (log avg wage)	0.574	0.647
Science (broad major)	0.714	0.610
Social Science (broad major)	0.793	0.634
Educ. & Humanities (broad major)	0.436	0.509
Health (broad major)	0.747	0.629
Dummy: Top reported preference	0.001	0.140
Grades	0.050	0.026
Observed ability	0.004	0.067
Female	-0.005	-0.000
Low-income	-0.030	-0.057
<i>Regression Residuals</i>		
Sum of squared residuals (SSR)	0.414	0.404
Average product of residuals	0.145	0.314
<i>Expected Log Wages Regression</i>		
Coeff: Top true vs. Random (intercept)	0.051	0.063
<i>Beliefs over Wage Uncertainty</i>		
Dep. mass within quintile (top rep. pref.)	0.493	0.496

Note: This table shows wage regression moments. The wage regression estimates the relationship between log individual wages and program characteristics (broad major), student characteristics (grades, observed ability, demographics), and an indicator for being assigned to the top reported preference. The regression pools observations from students assigned to their top true preference and random programs. SSR (Sum of Squared Residuals) and average product of residuals assess regression fit quality. The expected log wages regression compares predicted wages between top true preference and random assignments, with the intercept capturing the average wage difference. The dependent mass within quintile measures the subjective probability that students report of their wage falling within $\pm 10\%$ of the expected wage reported in the survey.

Table G13—Goodness of Fit: Learning and Subjective Beliefs

Moment	Model	Data
<i>Panel A: Difference in Grade Expectations (Subjective - Realized)</i>		
Science (broad major)	0.630	0.630
Social Science (broad major)	0.664	0.664
Educ. & Humanities (broad major)	0.361	0.361
Health (broad major)	0.843	0.843
Female	-0.185	-0.185
Low-income	0.084	0.084
Avg. obs. ability	-0.207	-0.207
Verbal type	-0.045	-0.045
<i>Panel B: Log Variance of Ability Beliefs (Realized)</i>		
Science (broad major)	-0.252	-0.191
Social Science (broad major)	-0.227	-0.139
Educ. & Humanities (broad major)	-0.480	-0.548
Health (broad major)	-0.957	-0.825
Female	-0.249	-0.245
Low-income	0.081	0.068
Avg. obs. ability	-0.524	-0.503
Verbal type	0.092	0.351
<i>Panel C: Log Variance of Ability Beliefs (Subjective)</i>		
Science (broad major)	-1.602	-1.768
Social Science (broad major)	-1.709	-1.746
Educ. & Humanities (broad major)	-1.522	-1.770
Health (broad major)	-1.836	-1.749
Female	-0.031	-0.070
Low-income	0.001	0.003
Avg. obs. ability	-0.307	-0.302
Verbal type	-0.069	-0.175
<i>Panel D: Change in Norm of Broad Majors (Top Quartile)</i>		
Science (broad major)	0.117	0.089
Social Science (broad major)	0.053	0.029
Educ. & Humanities (broad major)	0.080	0.048
Health (broad major)	0.058	0.070
Signal	-0.082	-0.120

Note: This table shows moments related to learning and subjective beliefs. Panel A reports the difference between subjective grade expectations and realized grades at the top reported preference; these moments are matched exactly by construction, as the corresponding mean-bias parameters are just-identified by them. Panels B and C show how the log variance of ability beliefs varies by demographics and program characteristics, comparing realized beliefs (based on actual enrollment) and subjective beliefs (stated expectations). Panel D shows the regression of the change in the norm of broad majors for students in the top quartile, with coefficients on broad major dummies and the signal about ability.

ENDOGENOUS ENROLLMENT

To assess the robustness of our results to the exogenous enrollment assumption, we develop a variant of the model in which take-up endogenously responds to the value of the assignment relative to the student’s outside option and current program. The implementation preserves the closed-form inclusive values that make the estimator feasible, while separating assignment from enrollment in the forward simulation process. In summary, the endogenous take-up model leads to a similar fit and leaves the counterfactual conclusions qualitatively unchanged. Detailed estimation, fit, and counterfactual results are available upon request.

H1. Model

Continuation values. Let $\bar{v}_{i,j,t}$ denote the expectation of the continuation value $v_{i,j,t}$ taken over all sources of randomness except the preference shocks, and let

$$\mathcal{J}_{i,j,t}^+(\omega) := \{k \in \mathcal{J} \setminus \{j\} : s_{i,k,t}(\omega) \geq \bar{s}_{k,t}(\omega)\}$$

be the set of new programs that are ex post feasible for student i in state ω . For a student who begins period $t = 2$ enrolled in program j , we now treat the enrollment options—each ex post feasible new program, the current program, and the outside option—as a single terminal choice set. Leveraging the i.i.d. type-I extreme value preference shocks (with scale σ), the fixed-cutoff inclusive value is

$$(H1) \quad IV_{i,j,t}^E(\omega) = \sigma \log \left[\sum_{k \in \mathcal{J}_{i,j,t}^+(\omega)} \exp\left(\frac{\bar{v}_{i,k,t} - C_{i,t}^S}{\sigma}\right) + \exp\left(\frac{\bar{v}_{i,j,t}}{\sigma}\right) + \exp\left(\frac{\bar{v}_{i,\emptyset,t}}{\sigma}\right) \right] + \sigma\gamma,$$

where $C_{i,t}^S$ is the cost of starting a new program and γ is Euler’s constant. The period-1 problem and the period-2 problem of a student who begins from the outside option are analogous, with the current-program term $\exp(\bar{v}_{i,j,t}/\sigma)$ omitted and the feasible set adjusted accordingly. As in the baseline, the expectation over cutoffs is approximated by averaging (H1) over the precomputed bootstrap cutoff draws, and the same Gauss-Hermite step is applied over future scores when the student may retake the exam.

Equation (H1) has a simple interpretation: when forming her application list, the student forecasts the value of a future reapplication as the value of picking the best option available ex post—enrolling in a feasible new program (net of the switching cost), staying in her current program, or taking the outside option. Formally, it is the baseline continuation value evaluated at $P_{i,t}^E \rightarrow 1$. The take-up friction introduced below—the possibility that an assignment is rejected at the enrollment stage—is deliberately left out of this forecast: incorporating it would break the log-sum structure, while leaving it out keeps (H1) in closed form, exactly as in the baseline model.

Post-assignment take-up. The endogenous model separates assignment from enrollment in the forward simulation process (realized path). Let $\Delta_{i,k,t}$ be the

realized surplus from accepting assigned program k relative to the fallback available upon rejection.³¹ Then, enrollment in the newly assigned program occurs with probability

$$(H2) \quad q_{i,k,t} = \frac{\exp\left(\frac{\Delta_{i,k,t} + \kappa_E}{\sigma}\right)}{1 + \exp\left(\frac{\Delta_{i,k,t} + \kappa_E}{\sigma}\right)},$$

where σ is again the Gumbel scale and κ_E is a take-up intercept that affects the sensitivity of enrollment to the assignment surplus.³²

ROL formation and realization. When deciding where to apply, students in the endogenous model attach the probability-weighted surplus $q_{i,k,t} \Delta_{i,k,t}$ to each program k , which is the natural analog of the baseline (i.e., $P_{i,t}^E \Delta_{i,k,t}$) but now responds to the assignment surplus. In the forward simulation, the ROL is constructed and the mechanism is run exactly as in the baseline; the only change is that, conditional on being assigned to program k , the student enrolls if $\xi_{i,t} < q_{i,k,t}$ rather than $\xi_{i,t} < P_{i,t}^E$, where $\xi_{i,t} \sim U[0, 1]$ is the existing uniform draw. If the assignment is rejected, the student falls back to the outside option in period 1 and to $\max\{\text{current}, \text{outside}\}$ in period 2.

Remark. The aforementioned model changes only affect the terminal log-sum and the assignment and enrollment objects; however, the current integration over cutoffs, future scores, ability, and preference learning remains unchanged, keeping the model computationally feasible.³³

H2. Results

Estimation. The model fit that results from endogenizing enrollment is similar to that of the baseline model. The main difference lies in the regression-discontinuity estimate for re-applications, where the endogenous model overestimates the effect of being just above the cutoff. The reason is that the baseline model uses a student-specific take-up probability $P_{i,t}^E$ that is constant across assigned programs, dampening differences across programs in the ROL. In the en-

³¹Note that the fallback defining $\Delta_{i,k,t}$ depends on the student's state. In period 1, it is the outside option. In period 2, for a student currently enrolled in program j , it is $\max\{U_{i,j,t}^{cur}, U_{i,t}^{out}\}$; for a student in the outside option, it remains the outside value. Here, U^{new} , U^{cur} , and U^{out} denote realized (shock-inclusive) values of the new, current, and outside alternatives. Thus, for example,

$$\Delta_{i,j,k,t} = U_{i,k,t}^{new} - \max\{U_{i,j,t}^{cur}, U_{i,t}^{out}\}.$$

For notational simplicity, we suppress the dependence of $\Delta_{i,k,t}$ on the current state j .

³²We use the same scale σ as in the preference shocks for parsimony, so the additional parameter shifts take-up levels rather than separately identifying the slope of take-up in surplus. Equivalently, the student enrolls in the new program whenever $\Delta_{i,k,t} + \kappa_E + \zeta_{i,k,t}^E \geq 0$ for a logistic shock $\zeta_{i,k,t}^E \sim \text{Logistic}(0, \sigma)$, so take-up given assignment is generally strictly below one.

³³Richer sequential or program-specific enrollment models—in which enrollment shocks are integrated in a separate lower-level decision, or take-up probabilities multiply utilities program by program—generally break the inclusive-value shortcut and would require numerical integration over the preference shocks, which would make the model computationally unfeasible.

ogenous model, take-up is increasing in the assignment surplus, so higher-value assignments receive a larger take-up weight while low-surplus assignments are more likely to be rejected. This makes the value of assignment vary more sharply around the cutoff, amplifying the re-application discontinuity.

Counterfactuals. To isolate the role of endogenous take-up, we compare the endogenous and exogenous models at the baseline estimated parameters, holding $\kappa_E = 0$ across counterfactuals.³⁴ The main takeaways of the paper remain unchanged: every counterfactual moves outcomes in the same direction under both models. Endogenizing take-up produces moderate level shifts that are consistent across scenarios—take-up and the share of wasted (assigned-but-unfilled) seats fall, the unassigned share rises, graduation falls slightly, and welfare rises—because low-surplus assignments are now more likely to be rejected, so fewer seats are wasted while more students remain unassigned.

IDENTIFICATION IN THE STYLIZED MODEL

We formalize the identification arguments of the stylized model. We show that the directional switching decisions of students around admission cutoffs, and how strongly each responds to its match-quality signal, identify the switching cost and the relative scale of learning, while a persistent preference shifter—distance, in the structural model—pins down their levels.

To do so, we extend the model discussed in Section II in three important ways. First, we assume that students can be partitioned based on their vector of observable characteristics $X \in \mathcal{X}$, where $\mathcal{X}$ is a set of types that collects student characteristics (e.g., $\{\text{low-income, high-income}\} \times \{\text{male, female}\}$) and observable student-program characteristics (e.g., the student’s distance to each program). Furthermore, we assume that student types determine the observable part of the expected utility from each option, which we denote by $u^a(X)$ for $a \in \{h, l\}$. Second, we allow for match-quality learning in both programs (i.e., not only in the top choice). Formally, let $\xi^a \sim F_\xi$ be the match-quality shock learned by a student in program $a \in \{h, l\}$. Finally, we assume that this shock enters linearly in students’ utilities, modulated by the constants λ_ξ^a . As a result, a student assigned to their second choice gets an ex post utility of $u^l(X) - \lambda_\xi^l \xi^l$, while a student assigned to their top choice gets an ex post utility of $u^h(X) + \rho - \lambda_\xi^h \xi^h$, where we orient ξ^a so that larger realizations are worse news about program a .³⁵ Note that the model in Section II is a special case of this more general model, where $u^h(X) = u^l(X) = 1$ for all $X \in \mathcal{X}$, $F_\xi \sim U[-1, 1]$, $\lambda_\xi^h = 1$ and $\lambda_\xi^l = 0$.

Based on these primitives, students may decide to switch up or down based on the intensity of their preferences and the realized match-quality shocks. In particular, a student assigned to their top choice will switch down if the expected

³⁴At $\kappa_E = 0$, the model reproduces an aggregate take-up rate close to that in the data.

³⁵Throughout, ρ denotes preference intensity, as in Section II, and is distinct from the weak-truth-teller probability ρ used elsewhere in the paper.

gains from doing so exceeds the switching costs, i.e.,

$$\lambda_\xi^h \xi^h > \tau + \Delta(X) + \rho,$$

where $\Delta(X) := u^h(X) - u^l(X)$ is the difference in the observable part of the expected utility between the top and fallback programs. Similarly, a student assigned to their fallback will switch up if

$$\lambda_\xi^l \xi^l > \tau - \Delta(X) - \rho.$$

RDD moments. We exploit the discontinuity in initial assignment at the admission cutoff for h , restricting attention to students who report truthfully and clear the cutoff on their fallback option l , and conditioning on their observable type X . Consequently, students only differ in their assignment, with students just above the cutoff being assigned to h , while those just below being assigned to l . Let $H_X(t)$ be the distribution of ρ conditional on X among students in the selected sample and around the cutoff, i.e.,

$$H_X(t) = \Pr(\rho \leq t \mid \{h, l\} \text{ reported, } s_h^1 = \bar{s}_h^1, s_l^1 \geq \bar{s}_l^1, X).$$

Note that H_X is not necessarily equal to the unconditional distribution of ρ , since some students may decide to truncate their lists and only report $\{h\}$ (see Lemma 1). Furthermore, let $D_X(\xi^h)$ and $U_X(\xi^l)$ denote the limiting downward- and upward-switching probabilities, respectively, as the cutoff for the top choice is approached from either side, conditional on X and the match-quality shocks ξ^h and ξ^l , i.e.,

$$D_X(\xi^h) = \lim_{s \downarrow \bar{s}_h^1} \Pr(\text{switch down} \mid s_h^1 = s, X, \xi^h),$$

$$U_X(\xi^l) = \lim_{s \uparrow \bar{s}_h^1} \Pr(\text{switch up} \mid s_h^1 = s, X, \xi^l).$$

In the remainder of this section, we make three assumptions that allow us to identify the primitives of interest. First, we assume that the conditional distribution of (ρ, ξ^h, ξ^l) is continuous around the cutoff, so that the two sides share H_X (cutoff continuity). Second, we assume that the signals are orthogonal to ρ given the type X , so that conditioning on a signal does not reshape H_X (signal orthogonality). Finally, we assume that the relevant thresholds are interior, where H_X is strictly increasing (interior support). Based on these assumptions, in Propositions 7–9 we show that the directional switching surfaces $D_X(\xi^h)$ and $U_X(\xi^l)$ identify the primitives of interest up to a common scale; we then show that variation in a persistent special regressor—the role played by distance in the structural model—delivers their levels.³⁶

³⁶See Kelchtermans and Verboven (2010) for evidence on the importance of distance as a driver of

Proposition 7 (Cutoff surfaces). *Under cutoff continuity, signal orthogonality, and interior support,*

$$(I1) \quad D_X(\xi^h) = H_X\left(\lambda_\xi^h \xi^h - \tau - \Delta(X)\right),$$

$$(I2) \quad U_X(\xi^l) = 1 - H_X\left(\tau - \Delta(X) - \lambda_\xi^l \xi^l\right).$$

Proof. A top holder near the cutoff switches down iff $\rho < \lambda_\xi^h \xi^h - \tau - \Delta(X)$; by signal orthogonality the conditional law of ρ is the marginal H_X , giving (I1). A fallback holder switches up iff $\rho > \tau - \Delta(X) - \lambda_\xi^l \xi^l$, giving (I2). $\square$

Proposition 8 (Differential-learning cancellation). *For any X and any values on a common interior level set,*

$$(I3) \quad D_X(\xi^h) = 1 - U_X(\xi^l) \iff \lambda_\xi^h \xi^h + \lambda_\xi^l \xi^l = 2\tau,$$

equivalently the affine level set

$$(I4) \quad \xi^h = -\frac{\lambda_\xi^l}{\lambda_\xi^h} \xi^l + \frac{2\tau}{\lambda_\xi^h}.$$

Both the selected distribution H_X and the observable gap $\Delta(X)$ cancel, so the matched level-set slope identifies $\lambda_\xi^l/\lambda_\xi^h$ and the intercept identifies τ/λ_ξ^h .³⁷

Proof. By (I1) and (I2), $D_X(\xi^h) = 1 - U_X(\xi^l)$ equates the arguments of the strictly increasing H_X , $\lambda_\xi^h \xi^h - \tau - \Delta(X) = \tau - \Delta(X) - \lambda_\xi^l \xi^l$, which is (I3); the gap $\Delta(X)$ cancels. The converse follows by substitution. $\square$

Proposition 9 (Signal derivatives). *If H_X is differentiable at the relevant thresholds with density h_X , and $\lambda_\xi^h, \lambda_\xi^l > 0$, then*

$$(I5) \quad \frac{\partial D_X(\xi^h)}{\partial \xi^h} = \lambda_\xi^h h_X\left(\lambda_\xi^h \xi^h - \tau - \Delta(X)\right), \quad \frac{\partial U_X(\xi^l)}{\partial \xi^l} = \lambda_\xi^l h_X\left(\tau - \Delta(X) - \lambda_\xi^l \xi^l\right).$$

On a matched level set the two densities coincide, so

$$(I6) \quad \frac{\partial D_X(\xi^h)/\partial \xi^h}{\partial U_X(\xi^l)/\partial \xi^l} = \frac{\lambda_\xi^h}{\lambda_\xi^l}.$$

students' choices in higher education.

³⁷The argument parallels Narita (2018), where the asymmetry in reapplication between randomly assigned first- and lower-choice holders identifies the switching cost; here the score cutoff plays the role of his admissions lotteries.

Identification. The two directional surfaces identify the primitives up to a common scale. The matched level set (I4) identifies the relative learning scale $\lambda_\xi^l/\lambda_\xi^h$ from its slope and the switching cost in top-learning units τ/λ_ξ^h from its intercept; the derivative ratio (I6) identifies $\lambda_\xi^h/\lambda_\xi^l$, an overidentifying check on the same relative scale. Both statements are robust to the selection generating H_X and to the observable gap $\Delta(X)$, which cancel.³⁸ The two surfaces alone, however, pin down only these ratios: a common rescaling of $(\rho, \lambda_\xi^h, \lambda_\xi^l, \tau, \Delta(X))$ leaves D_X and U_X unchanged, so recovering the levels requires an anchor in known units.

Levels. We anchor the scale with the observable gap. Suppose that the observable gap $\Delta(X)$ can be additively decomposed into a function of the observables $\tilde{X}$ and a persistent special regressor w that enters linearly with a unit coefficient by the scale normalization, i.e., $\Delta(X) = \Delta(\tilde{X}) + w$, where $\tilde{X}$ collects the remaining observables. Specifically, $w \equiv d^l - d^h$, where d^a denotes the student's distance to program a : since distance is a disutility, the distance component of the observable gap is exactly $d^l - d^h$, so a larger w —a relatively farther fallback—raises the relative value of the top choice. It is observed and continuously distributed. We assume w is independent of (ρ, ξ^h, ξ^l) conditional on $\tilde{X}$: distance shifts payoffs, not the distribution of intensities or signals. Because w is persistent, it also shifts first-period behavior, and a derivative of the surfaces with respect to w alone would confound the payoff index with the selection response of H_X . We interpret ρ as the unobserved component of the top-minus-fallback gap, so that a student ranks h first when $\Delta(X) + \rho > 0$. As in Section II, a student lists l second unless the gap exceeds a truncation threshold $\bar{a}(\tilde{X}, w)$: above it, students report h alone to preserve the option of reapplying rather than lock in the fallback, so the threshold embeds this option value and may depend on w without restriction. The selection is therefore a double truncation of the population distribution $F \equiv F_{\rho|\tilde{X}}$. The listing shares undo it. Formally, let $S_l(\tilde{X}, w)$ be the observed share ranking l first and $S(\tilde{X}, w)$ be the observed share reporting (h, l) . In the remainder, we omit the arguments of S_l and S for notational convenience when clear from the context. Then, we have $S_l = F(-\Delta(\tilde{X}) - w)$ and $S_l + S H_X(t) = F(t)$ on the interior of the selected support, so the share-corrected surfaces satisfy

$$(I7) \quad \begin{aligned} S_l + S D_X(\xi^h) &= F(\lambda_\xi^h \xi^h - \tau - \Delta(\tilde{X}) - w), \\ S_l + S [1 - U_X(\xi^l)] &= F(\tau - \Delta(\tilde{X}) - w - \lambda_\xi^l \xi^l). \end{aligned}$$

The left-hand sides are observed, and the right-hand sides are free of the selection: w may move the ranking and the truncation arbitrarily because the shares absorb them. Since w enters both indices with a unit coefficient and is excluded

³⁸The cancellation is likewise unaffected by changes in preferences between the two periods that shift the top-fallback gap on both margins alike since such changes enter the switch conditions exactly as ρ does and are absorbed into H_X .

from F , the levels follow from

$$(I8) \quad \begin{aligned} \lambda_\xi^h &= -\frac{\partial(S_l + S D_X)/\partial \xi^h}{\partial(S_l + S D_X)/\partial w}, & \lambda_\xi^l &= \frac{\partial(S_l + S [1 - U_X])/\partial \xi^l}{\partial(S_l + S [1 - U_X])/\partial w}, \\ \tau &= \frac{\lambda_\xi^h}{2} \times \text{intercept of (I4)}, \end{aligned}$$

which separately identifies $(\lambda_\xi^h, \lambda_\xi^l, \tau)$. The slope of (I4) and the derivative ratio (I6) become overidentifying restrictions, and, given λ_ξ^h , the corrected surfaces in (I7) trace F itself on the covered support, up to the location shift $\Delta(\tilde{X})$.